Compare Report Summary

Compare Tool Version: 8.12.00
Left Content: C:\Users\Dune Admin\Documents\Studio 5000\Projects\DUNEW2PLC1_oye.ACD
Left Content Software Revision: 33.00
Right Content: C:\Users\Dune Admin\Desktop\PLC_UK\DUNEWINDER1_v4_2.ACD
Right Content Software Revision: 33.00

Options

Include tags: Yes
Include tag data values in compare: No
Include constant tag data values in compare: No
Include descriptions in compare: Yes
Compare tracked components only: No
Compare logix definition only: No

Left Content	Right Content
Controller	Controller
53 Controller Tag Differences	100 Controller Tag Differences
1 Controller Property Difference	2 Controller Property Differences
Program	Program
0 Property Difference	0 Property Difference
0 Tag Difference	0 Tag Difference
0 Ladder Difference	0 Ladder Difference
0 Function Block Difference	0 Function Block Difference
0 SFC Difference	0 SFC Difference
0 Structured Text Difference	0 Structured Text Difference
0 SEQ Difference	0 SEQ Difference
0 Parameter Connection Difference	0 Parameter Connection Difference
Task	Task
1 Task Property Difference	4 Task Property Differences
Program	Program
0 Property Difference	0 Property Difference
71 Tag Differences	90 Tag Differences
90 Ladder Differences	133 Ladder Differences
0 Function Block Difference	0 Function Block Difference
0 SFC Difference	0 SFC Difference
0 Structured Text Difference	0 Structured Text Difference
0 SEQ Difference	0 SEQ Difference
0 Parameter Connection Difference	0 Parameter Connection Difference
11 Module Differences	11 Module Differences
0 Data Type Difference	0 Data Type Difference
0 Configuration Difference	0 Configuration Difference
0 Add-On Instruction Difference	0 Add-On Instruction Difference
0 AlarmDefinition Difference	0 AlarmDefinitions Difference
0 AlarmCondition Difference	1 AlarmCondition Difference
4 Quick Watch Differences	0 Quick Watch Difference
Library Object	Library Object
0 Library Object Difference	0 Library Object Difference

Controller Properties

Property	Left Content	Right Content
ProjectSN	16#d028_27b7	16#d01c_9992
CommPath	---	AB_ETHIP-1\192.168.140.13

Controller Tags

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
a_actual		REAL		700.0062	---	---	---	---
a_signed		REAL		-243.85713	---	---	---	---
accel_xy		REAL	Tangential acceleration of the winding head in xy-plane	0.0	---	---	---	---
ACTUATOR_POSITION_0	---	---	---	---		BOOL		0
ACTUATOR_POSITION_1	---	---	---	---		BOOL		0
ACTUATOR_POSITION_2	---	---	---	---		BOOL		0
Adjust	---	---	---	---		REAL		0.0
Adjust_Hi_Lim	---	---	---	---		REAL		10.0
Adjust_Lo_Lim	---	---	---	---		REAL		0.0
Adjusted	---	---	---	---		DINT		0
adjusted_tension	---	---	---	---		REAL		0.0
always_regulate_speed		BOOL		1	---	---	---	---
AnalogVin0	---	---	---	---		REAL		0.0
APA_IS_VERTICAL		BOOL		0	---	---	---	---
Beacon_Am	---	---	---	---		BOOL		0
Beacon_Gn	---	---	---	---		BOOL		0
Beacon_Rd	---	---	---	---		BOOL		0
Beacon_S	---	---	---	---		BOOL		0
blink_off		TIMER			---	---	---	---
blink_on		TIMER			---	---	---	---
BLINKY	---	---	---	---		BOOL		0
bypass_xy_move		BOOL		1	---	---	---	---
check_tension_stable		BOOL	Check if the tension is stable at the setpoint before moving the winding head	1	---	---	---	---
CORNERS_GO	---	---	---	---		BOOL		0
current_command		REAL		0.0	---	---	---	---
current_command_high		REAL		1.5	---	---	---	---
current_command_low		REAL		-0.5	---	---	---	---
disable	---	---	---	---		BOOL		0
ENABLE_LIMIT_SWITCHES	---	---	---	---		BOOL		0
EOT_MINUS_X	---	---	---	---		BOOL	End of Travel Limit Switch - Min	0

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
EOT_MINUS_X	---	---	---	---			us X Direction	
EOT_MINUS_Y	---	---	---	---		BOOL	End of Travel Limit Switch - Minus Y Direction	0
EOT_MINUS_Z	---	---	---	---		BOOL	End of Travel Limit Switch - Minus Z Direction	0
EOT_OK	---	---	---	---		BOOL		0
EOT_PLUS_X	---	---	---	---		BOOL	End of Travel Limit Switch - Plus X Direction	0
EOT_PLUS_Y	---	---	---	---		BOOL	End of Travel Limit Switch - Plus Y Direction	0
EOT_PLUS_Z	---	---	---	---		BOOL	End of Travel Limit Switch - Plus Z Direction	0
EOT_STAT	---	---	---	---		DINT		0
ERROR_NOGO	---	---	---	---		BOOL		0
gamma		REAL		20.0	---	---	---	---
GetTemp	---	---	---	---		BOOL		1
GetTempTMR	---	---	---	---		TIMER		
GO_NOGO_STAT	---	---	---	---		DINT		0
HEAD_POS		DINT		0	---	---	---	---
HOMENOW	---	---	---	---		BOOL		0
J		REAL		2000.0	---	---	---	---
latch_ouput	Local:3:O.Pt01.Data			0	---	---	---	---
LCAP	---	---	---	---		TIMER	Load Cell Amp Power	
MACHINE_SW_STAT2	---	---	---	---		BOOL[32]		
MAS		BOOL	Motion Axis Stop	1		BOOL		1
max_tolerable_tension		REAL		9.0	---	---	---	---
MAX_TOLERABLE_Z		REAL		5.0	---	---	---	---
MCD_X_axis	---	---	---	---		MOTION_INSTRUCTION		
MCD1_X_axis	---	---	---	---		MOTION_INSTRUCTION		
MIDDLE_GO	---	---	---	---		BOOL		0
MIDDLE_NOGO	---	---	---	---		BOOL		0
MOUNT_NOGO	---	---	---	---		BOOL		0
MOUNT_SENSE	---	---	---	---		BOOL	Low = At height of mount	0
neutral_cv		REAL		2.5	---	---	---	---

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
New_Jog_Speed_Neg	---	---	---	---		DINT		-50
New_Jog_Speed_Pos	---	---	---	---		REAL		50.0
New_Speed	---	---	---	---		DINT		25
NEXTSTATE		DINT		1		DINT	NEXT State in State Machine	1
OT_OR	---	---	---	---		BOOL		0
Overtension_Reset	---	---	---	---		BOOL		1
overtension_timer		TIMER			---	---	---	---
PARK_AREA	---	---	---	---		BOOL		0
PARK_SENSE	---	---	---	---		BOOL	Low = In Park Area, No Z Transfer	0
pid_cv_high_limit		REAL		10.0	---	---	---	---
pid_cv_low_limit		REAL		0.0	---	---	---	---
pid_cv_out		REAL		2.5	---	---	---	---
PID_LOOP_TIMER		TIMER			---	---	---	---
pid_loop_timer_bypass		BOOL		0	---	---	---	---
pid_start_bias		BOOL		0	---	---	---	---
PID_TMR	---	---	---	---		TIMER		
Read_X_MtrTemp	---	---	---	---		MESSAGE		
Read_Y_MtrTemp	---	---	---	---		MESSAGE		
Read_Z_MtrTemp	---	---	---	---		MESSAGE		
Res1	---	---	---	---		TIMER		
ROT_LOCK_KEY	---	---	---	---		BOOL		0
SetPoint	---	---	---	---		REAL		6.25
SetPoint_PID	---	---	---	---		PID		
SetPoint_PV		REAL	wire tension from load cell (N)	2.3567553		REAL		0.026544806
SIDES_GO	---	---	---	---		BOOL		0
SLS_Active	---	---	---	---		BOOL		0
speed_regulator_switch		BOOL	This is our "enable safe speed" switch, repurposed to switch the speed regulator	1	---	---	---	---
Startup_Timer	---	---	---	---		TIMER		
STATE7_IND1	---	---	---	---		BOOL		0
sum	---	---	---	---		DINT		3
T02	---	---	---	---		TIMER		

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
T03	---	---	---	---		TIMER		
t04	---	---	---	---		TIMER		
tension	Local:4:I.Ch01.Data		wire tension (N)	2.357425	---	---	---	---
Tension_10N_test	---	---	---	---		BOOL		0
Tension_Above_10N	---	---	---	---		BOOL		0
Tension_Above_10N_Light	---	---	---	---		BOOL		0
TENSION_CONTROL_OK		BOOL		0	---	---	---	---
Tension_Increase	---	---	---	---		BOOL		0
tension_limited_speed		REAL		0.0	---	---	---	---
tension_motor_out	Local:5:O.Ch00.Data			2.5	---	---	---	---
TENSION_ON_SW	---	---	---	---		BOOL		0
TENSION_ON_SWITCH		BOOL		0	---	---	---	---
tension_on_switch_delay_on_start		TIMER			---	---	---	---
Tension_Reduction	---	---	---	---		BOOL		0
tension_setpoint		REAL		6.25	---	---	---	---
tension_stabilizing		BOOL		1	---	---	---	---
tension_stable_timer		TIMER			---	---	---	---
Tension_T1	---	---	---	---		TIMER		
test	---	---	---	---		BOOL		0
Test_Latch	---	---	---	---		DINT		0
Test250324	---	---	---	---		BOOL		0
TIMEDELAY	---	---	---	---		COUNTER		
TSD	---	---	---	---		TIMER		
TWB	---	---	---	---		TIMER	debounce on switch	
v_0		REAL		0.049999997	---	---	---	---
v_xy		REAL	velocity of winding head in xy plane	0.0	---	---	---	---
v_xyz		REAL	Velocity of winding head in xyz space	0.0	---	---	---	---
winding_head_pid		PID			---	---	---	---
wire_break_proxy		BOOL		0	---	---	---	---
WIRE_BREAK_SW	---	---	---	---		BOOL		0
wire_break_tension		REAL		2.5	---	---	---	---
X_Axis_SS1	---	---	---	---		SAFE_STOP_1		
X_DIR	---	---	---	---		DINT		0

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
X_MtrTemp	---	---	---	---		REAL		21.297115
X_Otemp	---	---	---	---		BOOL		0
X_PARK_OK	---	---	---	---		BOOL		1
X_PARKED		BOOL		0	---	---	---	---
X_SENSE	---	---	---	---		BOOL	Low = Inside boundary of frame	0
X_XFER_1	---	---	---	---		BOOL		0
X_XFER_2	---	---	---	---		BOOL		0
X_XFER_OK		BOOL	If at the head, check head support arms. Same for foot	0		BOOL		0
X_Y		COORDINATE_SYSTEM				COORDINATE_SYSTEM		
XY_DECEL_REQ		REAL		0.0	---	---	---	---
XY_DECELERATION_SLS	---	---	---	---		REAL		200.0
xy_regulated_acceleration		REAL		2000.0	---	---	---	---
xy_regulated_deceleration		REAL		2000.0	---	---	---	---
XYZ_Inhibit_Timer	---	---	---	---		TIMER		
Y_Axis_SFX	---	---	---	---		SAFETY_FEEDBACK_INTERFACE		
Y_Axis_SLS	---	---	---	---		SAFELY_LIMITED_SPEED		
Y_Axis_SS1	---	---	---	---		SAFE_STOP_1		
Y_DIR	---	---	---	---		DINT		0
Y_MtrTemp	---	---	---	---		REAL		21.2901
Y_Otemp	---	---	---	---		BOOL		0
Y_SENSE	---	---	---	---		BOOL	Low = Inside boundary of frame	0
Z_Arm_Latched	---	---	---	---		BOOL		0
Z_DIR	---	---	---	---		DINT		0
Z_MtrTemp	---	---	---	---		REAL		21.376606
Z_Otemp	---	---	---	---		BOOL		0
Z_SAFE_TO_WITHDRAW		BOOL		0	---	---	---	---
Z_SPRING_COMP	---	---	---	---		BOOL		0
Z_STAGE_PRESENT		BOOL	Winding head is on the Z-arm	1		BOOL		0

Modules

Module/Property Name	Left Content	Right Content
[1] 5069-IB16/A		
Communications		
Connections		
Connection		
InputTag		
Comments		
Comment.PT08.DATA	Latch top	Latch Top
Comment.PT09.DATA	Latch middle	---
Comment.PT15.DATA	The wire break switch was causing electrical problems, so we use the load-cell as a proxy for wire break	---
Data	[1,0,0,[1,0],[0,0],[1,0],[0,0],[0,0],[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[0,0],[0,0],[0,0]]	[1,0,0,[0,0],[1,0],[0,0],[1,0],[0,0],[1,0],[1,0],[1,0],[0,0],[0,0],[0,0],[0,0],[1,0],[1,0],[0,0],[0,0]]
[2] 5069-IB16/A		
Communications		
Connections		
Connection		
InputTag		
Comments		---
Comment.PT06.DATA	TENSION ON/OFF	---
Comment.PT07.DATA	O/T SWITCH OVERTRAVEL SWITCH	---
Comment.PT13.DATA	Cabinet lighted push button	---
Data	[1,0,0,[0,0],[0,0],[1,0],[0,0],[0,0],[0,0],[0,0],[1,0],[1,0],[0,0],[1,0],[0,0],[1,0],[0,0],[0,0],[1,0]]	[1,0,0,[0,0],[0,0],[1,0],[0,0],[1,0],[0,0],[0,0],[1,0],[1,0],[0,0],[1,0],[0,0],[1,0],[0,0],[0,0],[0,0]]
[3] 5069-OB16/A		
Communications		
Connections		
Connection		
InputTag		
Data	[1,16,0,[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[1,0],[0,0]]	[1,16,0,[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[1,0],[0,0],[1,0],[1,0],[0,0]]
OutputTag		
Comments		
Comment.PT01.DATA	Latching signal output	---
Comment.PT12.DATA	Beacon - Green	Beacon - Blue
Comment.PT13.DATA	Beacon - Amber	Beacon - Green
Comment.PT14.DATA	Beacon - Red	Beacon - Amber
Comment.PT08.DATA	---	Beacon - Red
Data	[[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[1,0,0,0],[0,0,0,0]]	[[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0],[1,0,0,0],[0,0,0,0],[1,0,0,0],[1,0,0,0],[0,0,0,0]]
[4] 5069-IF8/A		
Communications		
ConfigTag		
Data	[388,165, [2,0,5,1,0,0,0.00000000e+000,1.00000000e+001,0.00000000e+000,1.00000000e+001,0.00000000e+000,0.00000000e+000,1.00000000e+002,1.00000000e+002,0.00000000e+000,0.00000000e+000], [2,0,5,1,0,0,0.00000000e+000,1.00000000e+001,0.00000000e+000,1.00000000e+001,0.00000000e+000,0.00000000e+000,1.00000000e+002,1.00000000e+002,0.0	[388,165, [2,0,2,1,0,0,0.00000000e+000,1.00000000e+001,0.00000000e+000,1.00000000e+001,0.00000000e+000,0.00000000e+000,1.00000000e+002,1.00000000e+002,0.00000000e+000,0.00000000e+000], [2,0,2,1,0,0,0.00000000e+000,1.00000000e+001,0.00000000e+000,1.00000000e+001,0.00000000e+000,0.00000000e+000,1.00000000e+002,1.00000000e+002,0.0

Module/Property Name	Left Content	Right Content
Data	0000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000]]	0000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000], [2,0,2,17,0,0,0.00000000e+000,1.000000 00e+001,0.00000000e+000,1.00000000e +001,0.00000000e+000,0.00000000e+00 0,1.00000000e+002,1.00000000e+002,0. 00000000e+000,0.00000000e+000]]
Connections		
Connection		
RPI	12500	100000
InputTag		
Description	Wire tension (N)	---
Data	[1,0,0,[0,0,0,-1.73586123e-002,22319,0], [0,0,0,2.35742497e+000,22316,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0]]	[1,6,0,[0,0,0,-1.45424111e-002,2593,0], [0,0,0,2.65018642e-002,2575,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0], [2,0,0,1.#QNAN000e+000,0,0]]
[5] 5069-OF8/A		
Communications		
Connections		
Connection		
RPI	13100	80000
InputTag		
Data	[1,0,0,[0,0,0,2.50000000e+000,2654,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0]]	[1,0,0,[0,0,0,0.00000000e+000,24009,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0], [2,0,0,0.00000000e+000,0,0]]
OutputTag		
Data	[[0,0,0,2.50000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000],[0,0,0,0.000000	[[0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000], [0,0,0,0.00000000e+000],[0,0,0,0.000000

Module/Property Name	Left Content	Right Content
Data	00e+000]]	00e+000]]
[6] 5069-IB16/A		
Communications		
Connections		
Connection		
InputTag		
Data	[1,0,0,[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0]]	[1,0,0,[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[1,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0],[0,0]]
Comments	---	
Comment.PT07.DATA	---	Fuse 5 OK
192.168.140.10 1783-BMS10CGP DUNEW2NETSW		
Communications		
Connections		
Connection		
InputTag		
Data	[0,31,0,0,0,1,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2]	[0,271,0,0,0,2,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,2]
[1] 1734-IB8S/B		
Communications		
Connections		
Connection		
InputTag		
Comments		
Comment.PT04DATA	EStop Pullcord 2	EStop Operator Station
Comment.PT06DATA	---	XYZ Inhibit Keyswitch
Data	[1,63,-1]	[1,4,-1]
192.168.140.16 2198-D006-ERS4 DUNEW2SERVO1		
Communications		
Connections		
Connection		
OutputTag		
Data	[0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,1]	[0,0,0,0,0,4,0,0,0,0,0,4,0,0,0,0]
Connection		
InputTag		
Data	[1,0,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,0,0,0,17,0,0,0,17]	[1,25503,0.00000000e+000,0,0.00000000e+000,-77084,0.00000000e+000,0,0.00000000e+000,-127,1,0,17,-127,1,0,17]
ExtendedProperties		
public		
PrivateTag	<Data>02 00 0C 00 00 00 56 50 4C 2D 42 30 37 35 32 45 2D 57 00 00 00

Module/Property Name	Left Content	Right Content
PrivateTag	0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 00 0C 00 00 00 56 50 4C 2D 42 30 37 35 32 45 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>	0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 00 0C 00 00 00 56 50 4C 2D 42 30 37 35 32 45 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>
PrivateTag	<Data>1F 00 00 00 07 00 E0 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>	<Data>02 00 0C 00 00 00 56 50 4C 2D 42 30 37 35 32 45 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 00 0C 00 00 00 56 50 4C 2D 42 30 37 35 32 45 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0

Module/Property Name	Left Content	Right Content
PrivateTag	<Data>1F 00 00 00 07 00 E0 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>	0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>
192.168.140.17 2198-D006-ERS4 DUNEW2SERVO2		
Communications		
Connections		
Connection		
OutputTag		
Data	[0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0]	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
Connection		
InputTag		
Data	[1,0,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,0,0,0,17,-127,1,0,16]	[1,-13783,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,0,0.00000000e+000,-127,1,0,17,-127,1,0,16]
ExtendedProperties		
public		
PrivateTag	<Data>02 00 0C 00 00 00 56 50 4C 2D 42 30 36 33 31 54 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 80 3F 0E 00 00 00 50 6F 73 69 74 69 6F 6E 20 55 6E 69 74 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>	<Data>02 00 0C 00 00 00 56 50 4C 2D 42 30 36 33 31 54 2D 57 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 80 3F 0E 00 00 00 50 6F 73 69 74 69 6F 6E 20 55 6E 69 74 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>
PrivateTag	<Data>1F 00 00 00 07 00 E0 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0	<Data>02 00 0C 00 00 00 56 50 4C 2D 42 30 36 33 31 54 2D 57 00 00 00 00 00 00 0

Module/Property Name	Left Content	Right Content
PrivateTag	0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>	0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 00 44 04 00 00 00 52 45 56 53 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 01 00 00 00 00 00 80 3F 00 00 80 3F 00 00 00 00 80 3F 0E 00 00 00 50 6F 73 69 74 69 6F 6E 20 55 6E 69 74 73 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00</Data>
192.168.140.19 In-Sight 8000 Series DUNE2CAM1	LeftOnly	---
192.168.140.19 In-Sight 8000 Series DUNEW2CAM1	---	RightOnly

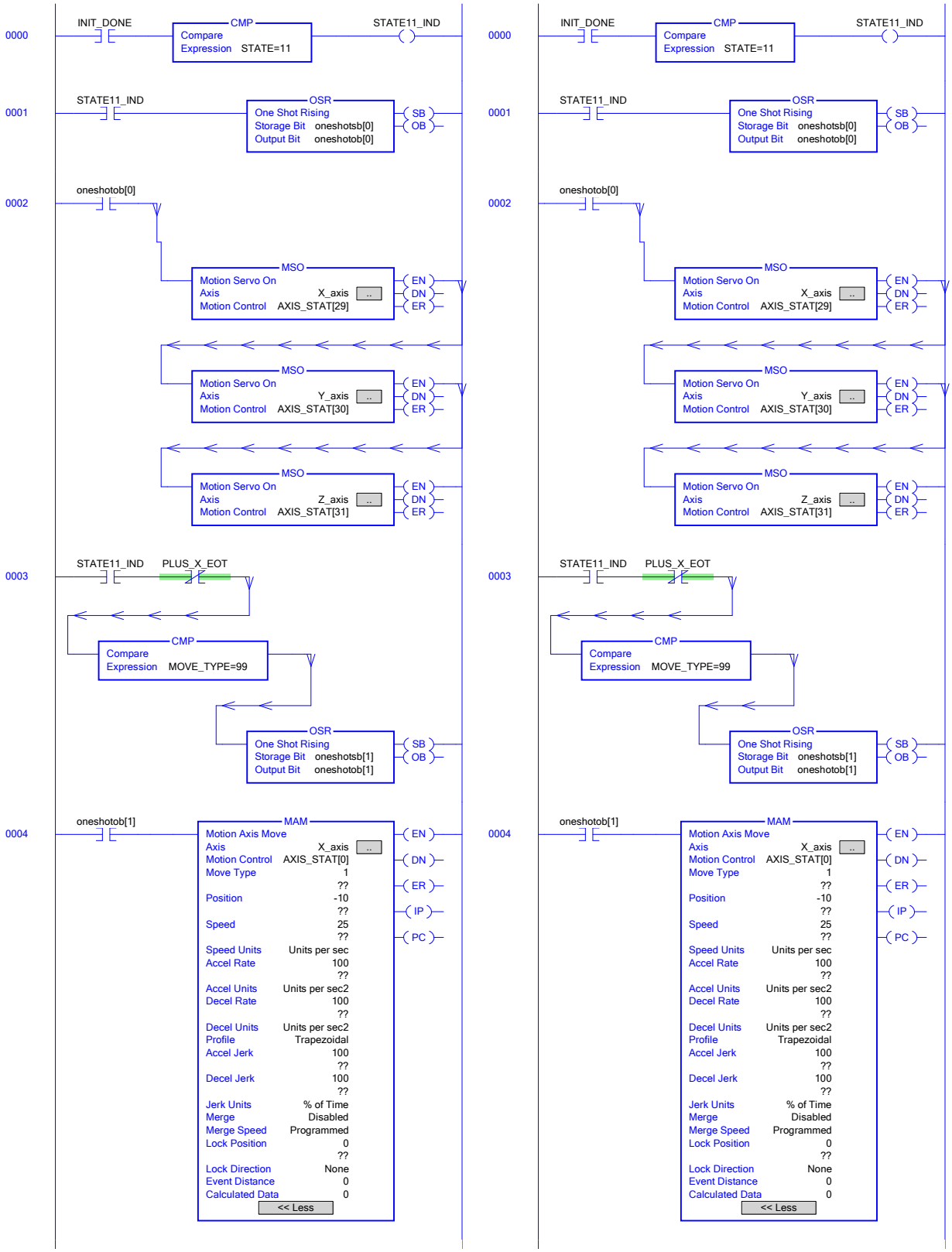
Tasks

Task	Left Content	Right Content
DUNEWINDER2		
Scheduled Programs		
	MainProgram	MainProgram
	UnServo_9	UnServo_9
	EOT_Trip_11	EOT_Trip_11
	Error_State_10	Error_State_10
	Latch_UnLatch_State_6_7_8	Homing
	MoveXY_State_2_3	Latch_UnLatch_State_6_7_8
	MoveZ_State_4_5	MoveXY_State_2_3
	Ready_State_1	MoveZ_State_4_5
	Initialize	Ready_State_1
	PID_Tension_Servo	Initialize
		PID_Tension_Servo
		Motor_Temperature_Monitoring
Unscheduled Programs / Phases		
Scheduled Programs		
	Camera	t
		Tension_Servo

Programs

Program	Left Content	Right Content
EOT_Trip_11		
EOT_TRIP	1 Different Rung	1 Different Rung
Error_State_10		
Parameters and Local Tags	2 Differences	4 Differences
Error	3 Different Rungs	13 Different Rungs
Initialize		
Parameters and Local Tags	0 Difference	6 Differences
Latch_UnLatch_State_6_7_8		
Parameters and Local Tags	14 Differences	17 Differences
Latch_Driver	17 Different Rungs	14 Different Rungs
MainProgram		
Parameters and Local Tags	0 Difference	13 Differences
MainRoutine	24 Different Rungs	21 Different Rungs
MoveXY_State_2_3		
Parameters and Local Tags	43 Differences	28 Differences
MoveXY	20 Different Rungs	39 Different Rungs
xy_speed_regulator	Left Content Only	---
MoveZ_State_4_5		
Parameters and Local Tags	3 Differences	13 Differences
MoveZ	3 Different Rungs	16 Different Rungs
PID_Tension_Servo		
Parameters and Local Tags	9 Differences	3 Differences
PID_Tension	14 Different Rungs	11 Different Rungs
Ready_State_1		
Parameters and Local Tags	0 Difference	6 Differences
Ready	1 Different Rung	1 Different Rung
Safety		
Safety	6 Different Rungs	17 Different Rungs
Camera	Left Only Program	---
Homing	---	Right Only Program
Motor_Temperature_Monitoring	---	Right Only Program
t	---	Right Only Program
Tension_Servo	---	Right Only Program

Program: EOT_Trip_11
Routine: EOT_TRIP



Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
OB

0006 STATE11_IND MINUS_X_EOT

CMP
Compare
Expression MOVE_TYPE=99

OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB

0007 oneshotob[3]

MAM
Motion Axis Move
Axis X_axis ..
Motion Control AXIS_STAT[1]
Move Type 1
??
Position 10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC

0008 AXIS_STAT[1].PC

OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

0009 STATE11_IND PLUS_Y_EOT

CMP
Compare
Expression MOVE_TYPE=99

OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

0010 oneshotob[5]

MAM
Motion Axis Move
Axis Y_axis ..
Motion Control AXIS_STAT[2]
Move Type 1
??
Position -10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
EN
DN
ER
IP
PC

Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
OB

0006 STATE11_IND MINUS_X_EOT

CMP
Compare
Expression MOVE_TYPE=99

OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB

0007 oneshotob[3]

MAM
Motion Axis Move
Axis X_axis ..
Motion Control AXIS_STAT[1]
Move Type 1
??
Position 10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC

0008 AXIS_STAT[1].PC

OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

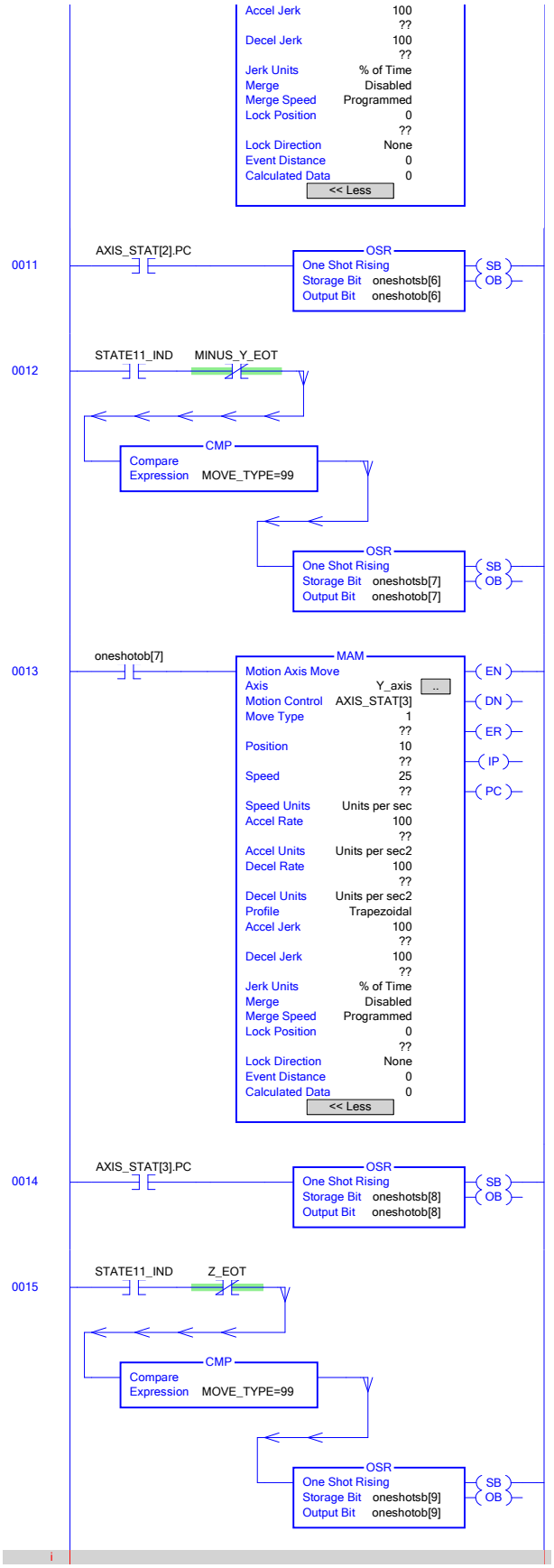
0009 STATE11_IND PLUS_Y_EOT

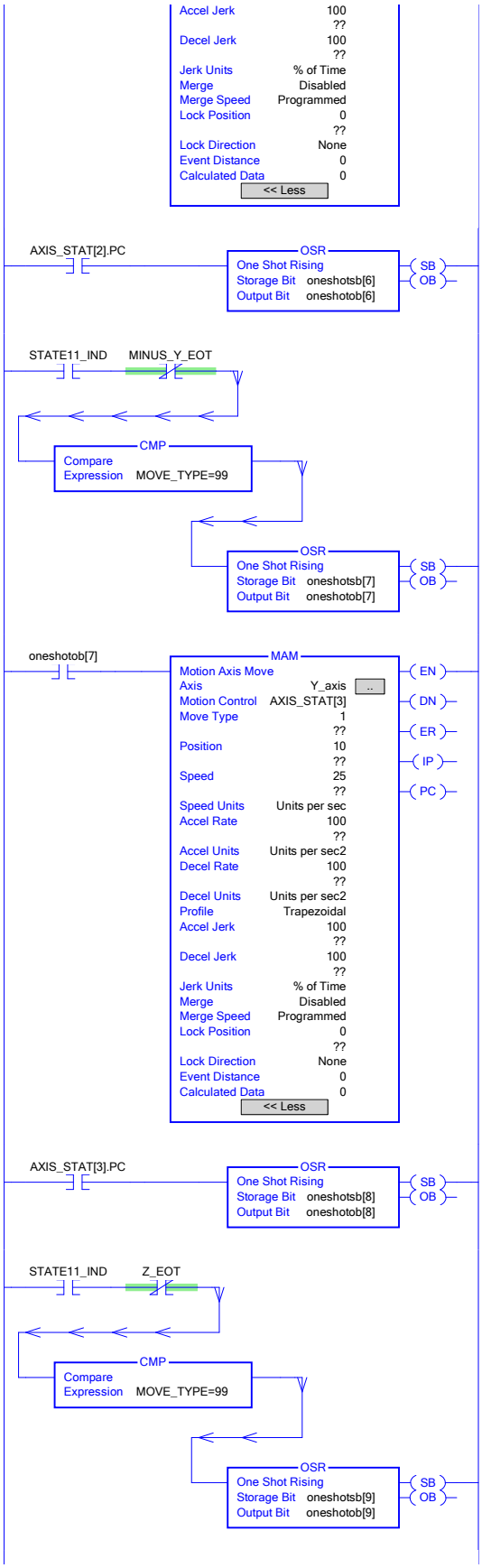
CMP
Compare
Expression MOVE_TYPE=99

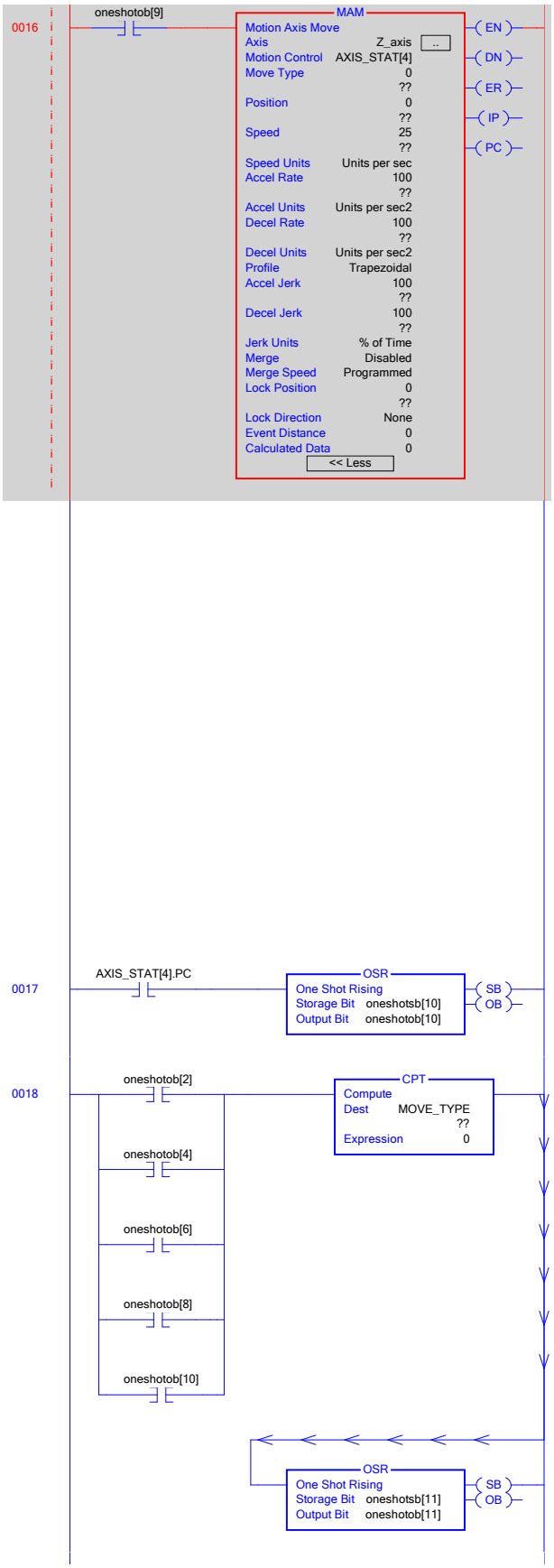
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

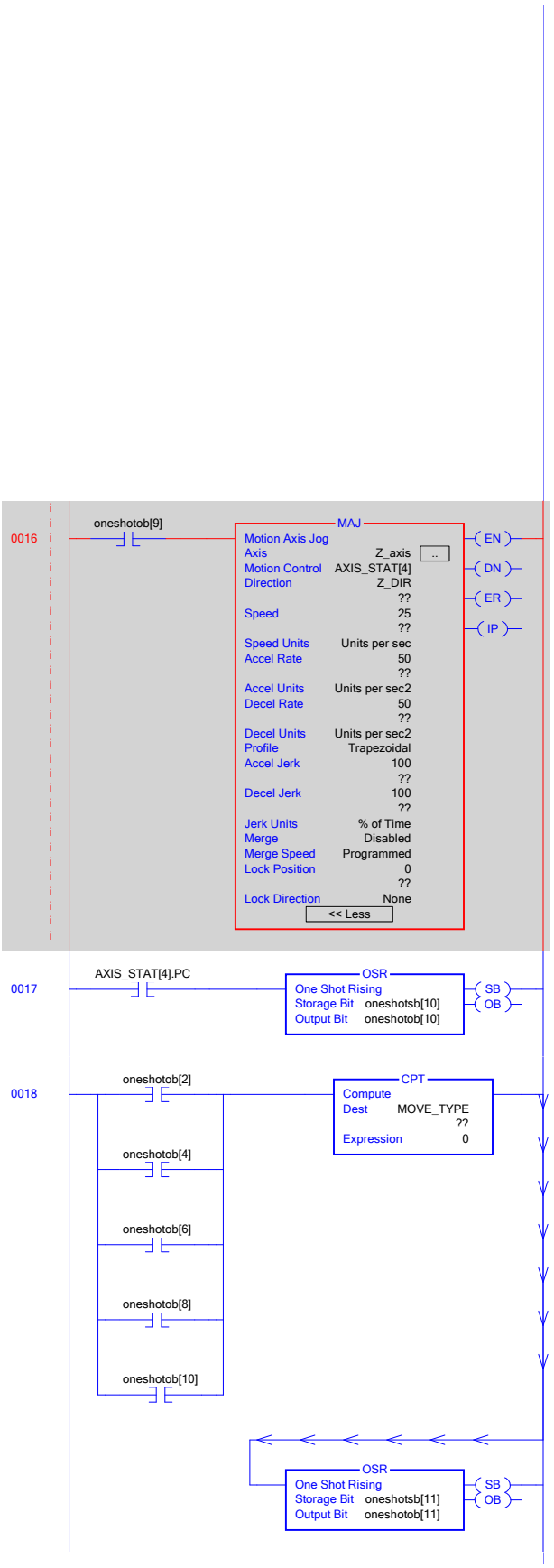
0010 oneshotob[5]

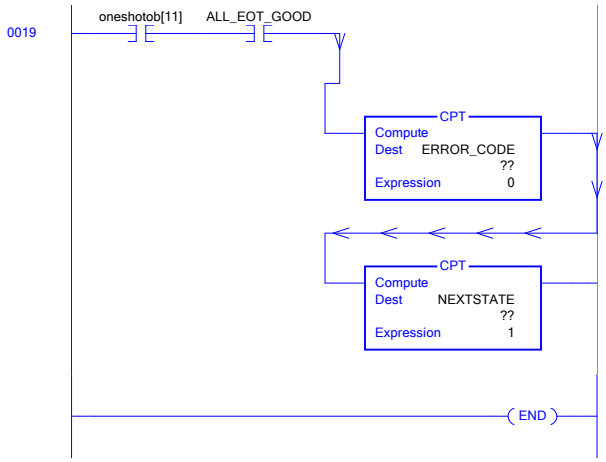
MAM
Motion Axis Move
Axis Y_axis ..
Motion Control AXIS_STAT[2]
Move Type 1
??
Position -10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
EN
DN
ER
IP
PC

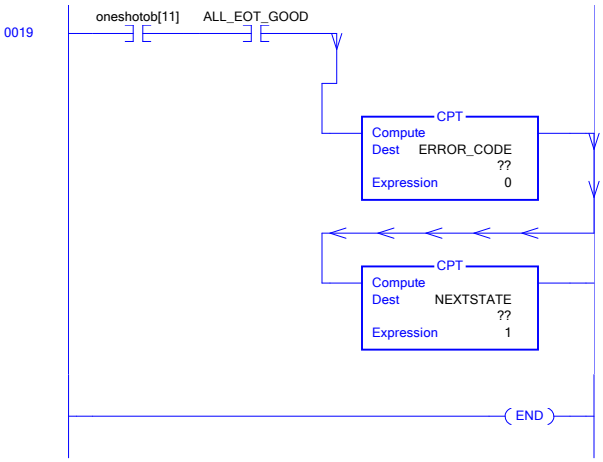

Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
0011
AXIS_STAT[2].PC
OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB
0012
STATE11_IND
MINUS_Y_EOT
CMP
Compare
Expression MOVE_TYPE=99
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
SB
OB
0013
oneshotob[7]
MAM
Motion Axis Move
Axis Y_axis
Motion Control AXIS_STAT[3]
Move Type 1
??
Position 10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC
0014
AXIS_STAT[3].PC
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0015
STATE11_IND
Z_EOT
CMP
Compare
Expression MOVE_TYPE=99
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB


Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
0011
AXIS_STAT[2].PC
OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB
0012
STATE11_IND
MINUS_Y_EOT
CMP
Compare
Expression MOVE_TYPE=99
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
SB
OB
0013
oneshotob[7]
MAM
Motion Axis Move
Axis Y_axis
Motion Control AXIS_STAT[3]
Move Type 1
??
Position 10
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC
0014
AXIS_STAT[3].PC
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0015
STATE11_IND
Z_EOT
CMP
Compare
Expression MOVE_TYPE=99
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB


0016
oneshotob[9]
MAM
Motion Axis Move
Axis Z_axis
Motion Control AXIS_STAT[4]
Move Type 0
??
Position 0
??
Speed 25
??
Speed Units Units per sec
Accel Rate 100
??
Accel Units Units per sec2
Decel Rate 100
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC
0017
AXIS_STAT[4].PC
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB
0018
oneshotob[2]
oneshotob[4]
oneshotob[6]
oneshotob[8]
oneshotob[10]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
OSR
One Shot Rising
Storage Bit oneshotsb[11]
Output Bit oneshotob[11]
SB
OB


0016
oneshotob[9]
MAJ
Motion Axis Jog
Axis Z_axis
Motion Control AXIS_STAT[4]
Direction Z_DIR
??
Speed 25
??
Speed Units Units per sec
Accel Rate 50
??
Accel Units Units per sec2
Decel Rate 50
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
<< Less
EN
DN
ER
IP
0017
AXIS_STAT[4].PC
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB
0018
oneshotob[2]
oneshotob[4]
oneshotob[6]
oneshotob[8]
oneshotob[10]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
OSR
One Shot Rising
Storage Bit oneshotsb[11]
Output Bit oneshotob[11]
SB
OB

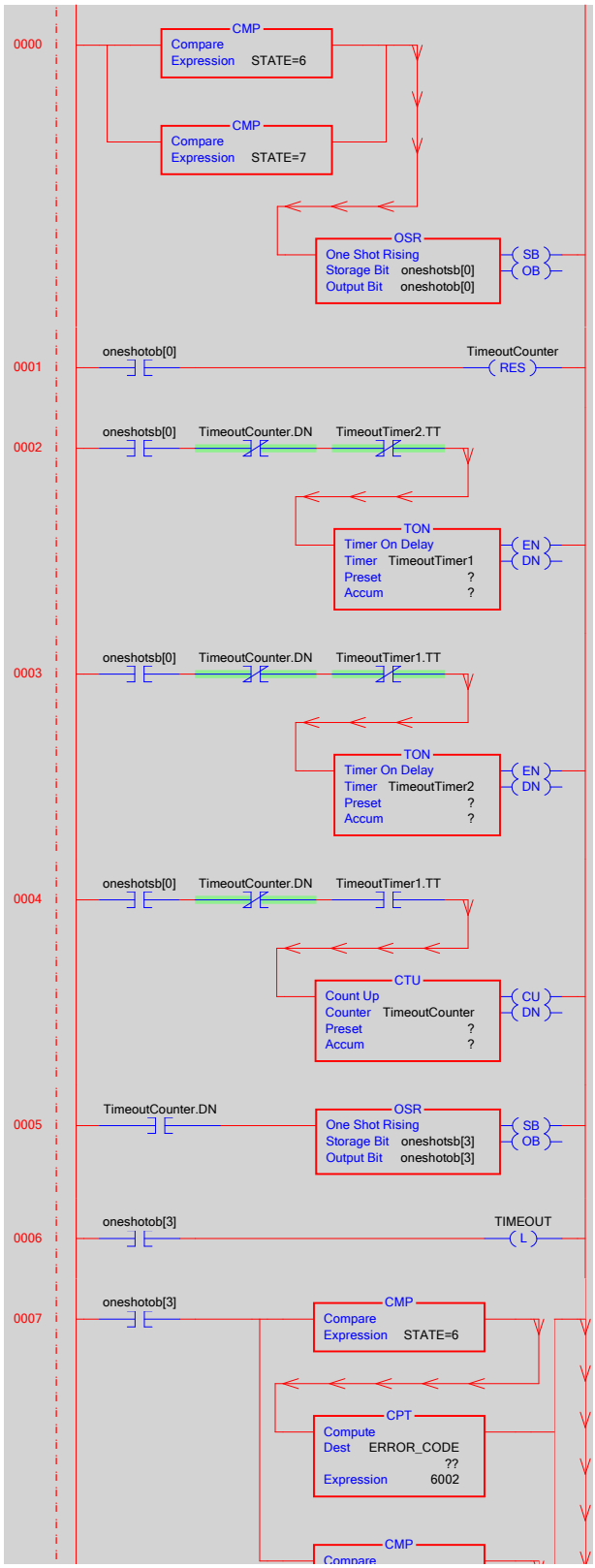

0019
oneshotob[11]
ALL_EOT_GOOD
CPT
Compute
Dest ERROR_CODE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
END

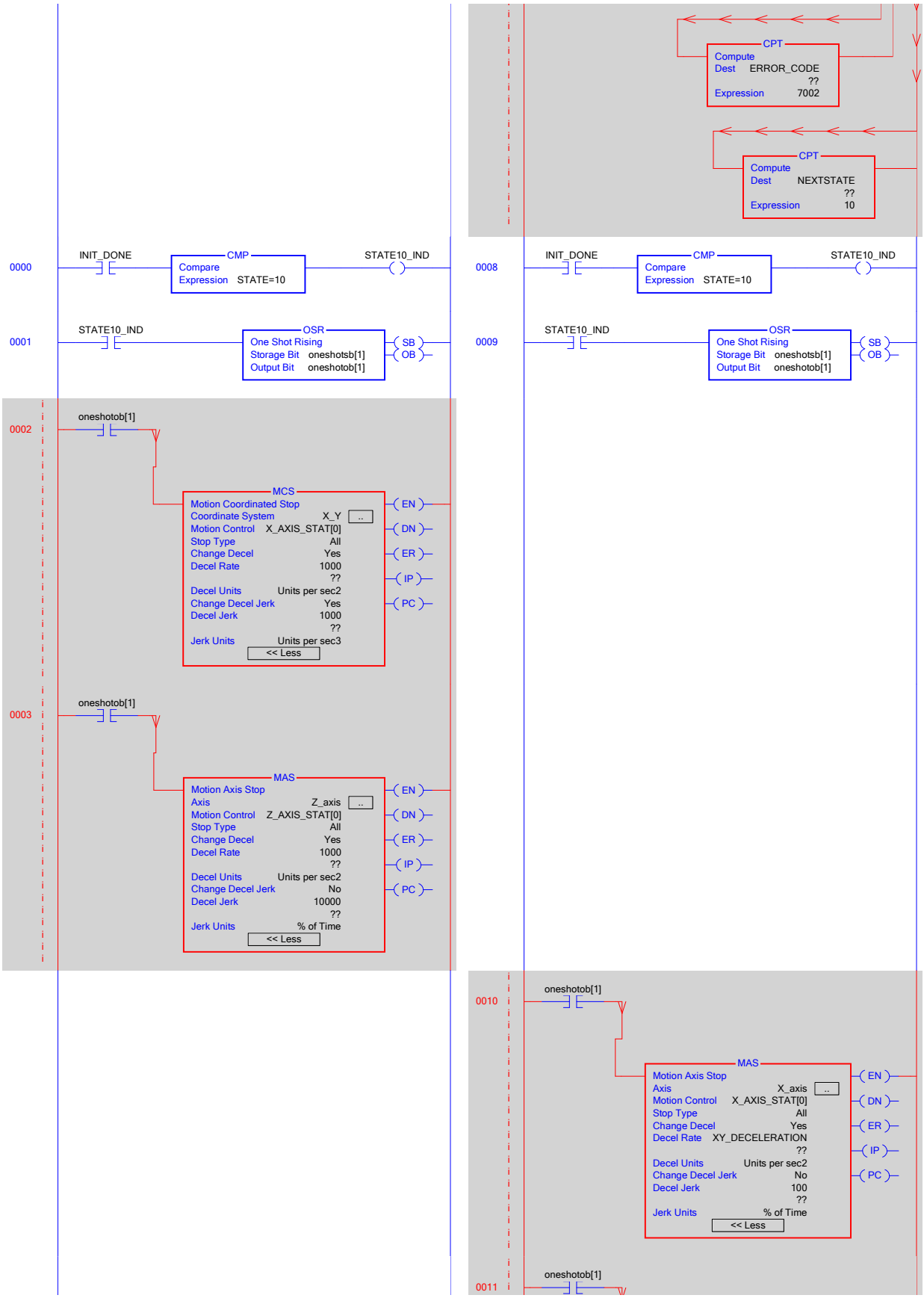

0019
oneshotob[11]
ALL_EOT_GOOD
CPT
Compute
Dest ERROR_CODE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
END

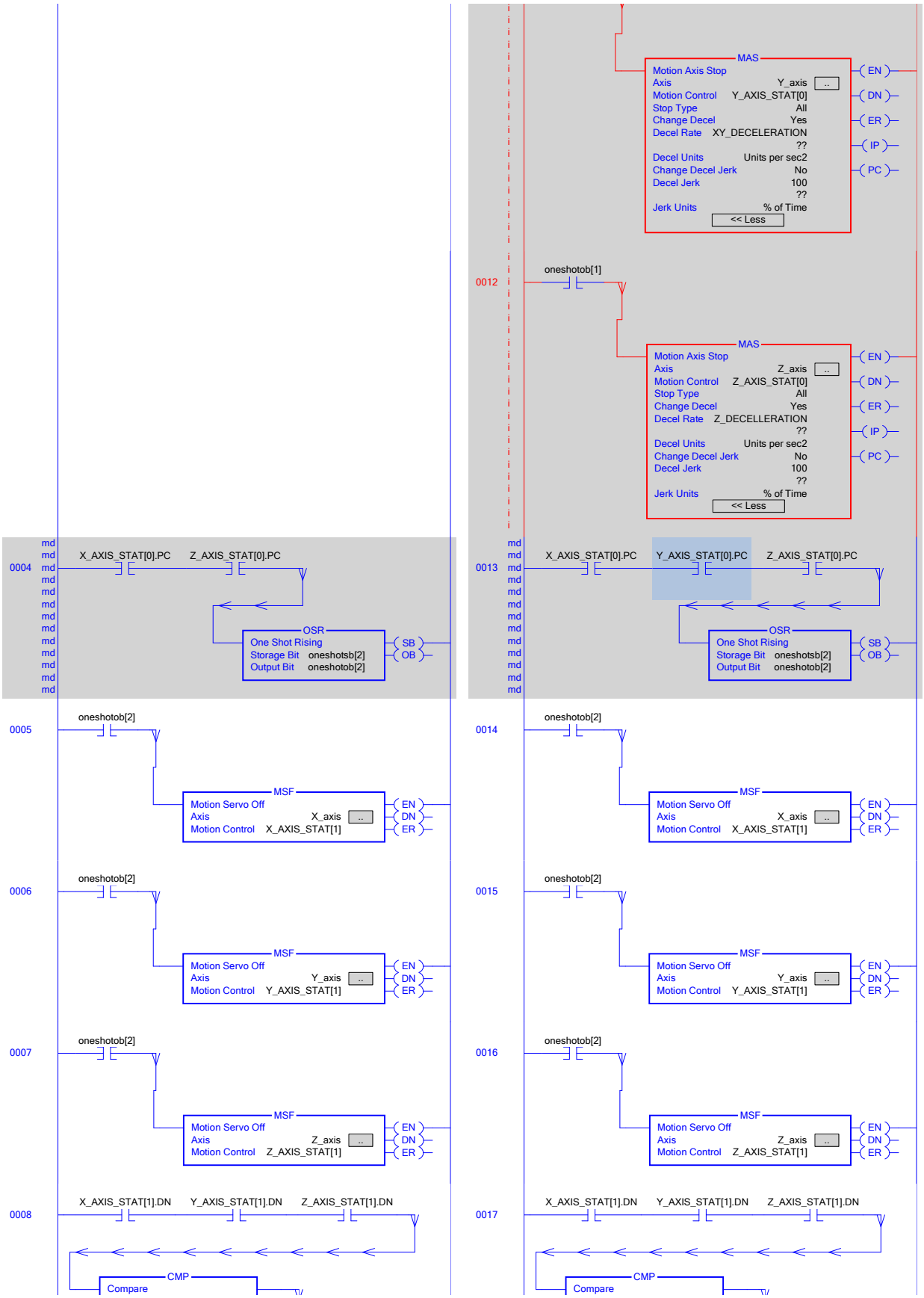
Error_State_10 Parameters and Local Tags

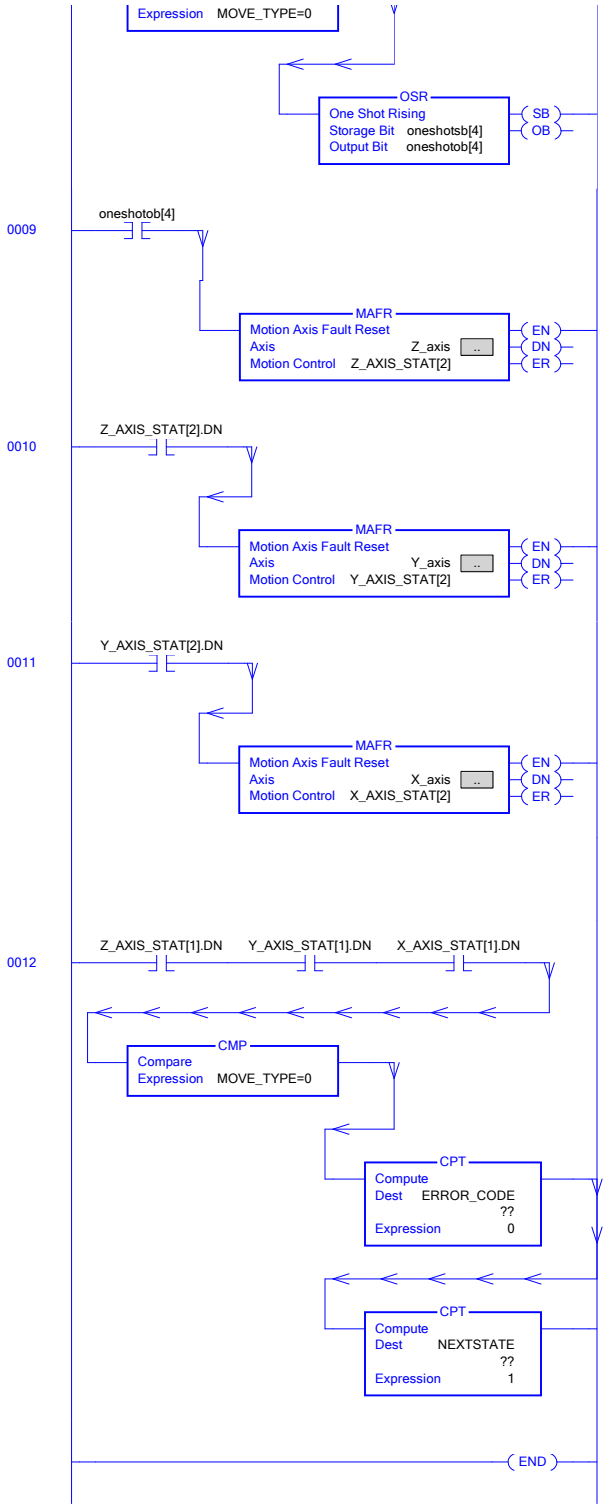
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
MOTION_STOPPED	---	---	---	---		BOOL		0
oneshotob		BOOL[32]				BOOL[32]		
oneshotsb		BOOL[32]				BOOL[32]		
TimeoutCounter	---	---	---	---		COUNTER		
TimeoutTimer1	---	---	---	---		TIMER		
TimeoutTimer2	---	---	---	---		TIMER		

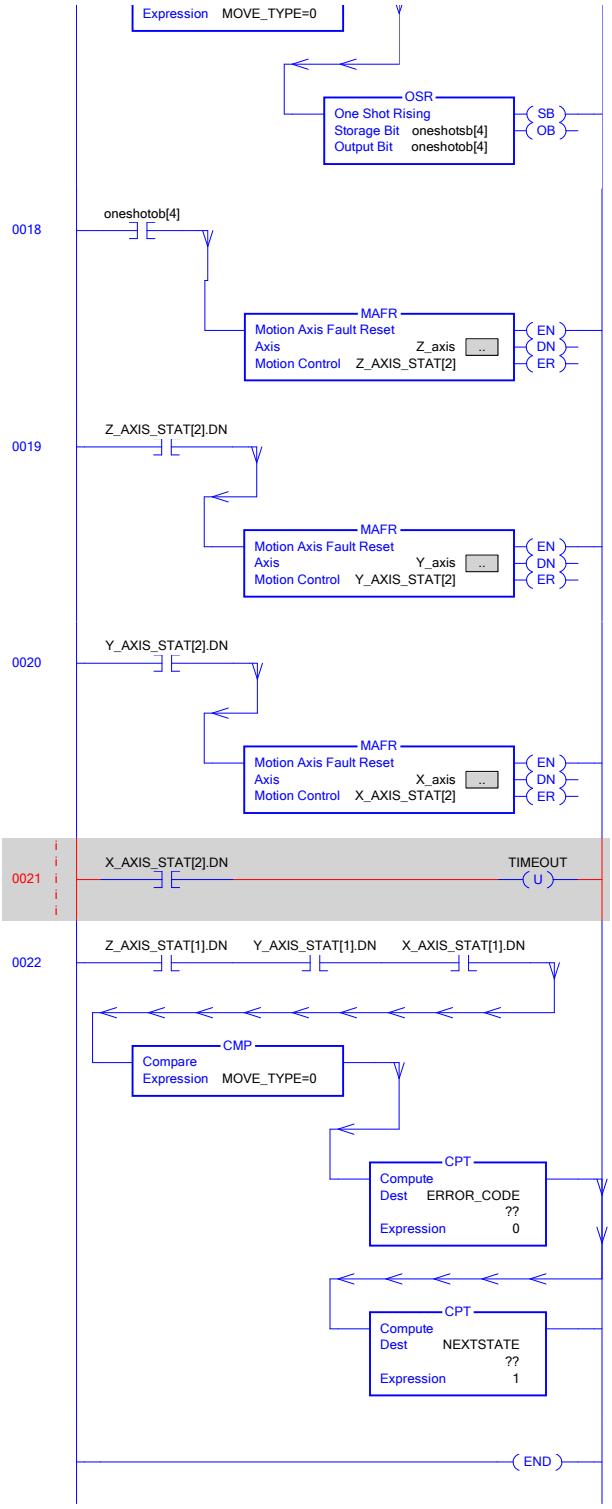
Program: Error_State_10
Routine: Error




0000
INIT_DONE
CMP
Compare
Expression STATE=10
STATE10_IND
0001
STATE10_IND
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB
0002
oneshotob[1]
MCS
Motion Coordinated Stop
Coordinate System X_Y
Motion Control X_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate 1000
??
Decel Units Units per sec2
Change Decel Jerk Yes
Decel Jerk 1000
??
Jerk Units Units per sec3
<< Less
EN
DN
ER
IP
PC
0003
oneshotob[1]
MAS
Motion Axis Stop
Axis Z_axis
Motion Control Z_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate 1000
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 10000
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
CPT
Compute
Dest ERROR_CODE
??
Expression 7002
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0008
INIT_DONE
CMP
Compare
Expression STATE=10
STATE10_IND
0009
STATE10_IND
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB
0010
oneshotob[1]
MAS
Motion Axis Stop
Axis X_axis
Motion Control X_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0011
oneshotob[1]


MAS
Motion Axis Stop
Axis Y_axis
Motion Control Y_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0012
oneshotob[1]
MAS
Motion Axis Stop
Axis Z_axis
Motion Control Z_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
0004
X_AXIS_STAT[0].PC
Z_AXIS_STAT[0].PC
OSR
One Shot Rising
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
SB
OB
0013
X_AXIS_STAT[0].PC
Y_AXIS_STAT[0].PC
Z_AXIS_STAT[0].PC
OSR
One Shot Rising
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
SB
OB
0005
oneshotob[2]
MSF
Motion Servo Off
Axis X_axis
Motion Control X_AXIS_STAT[1]
EN
DN
ER
0014
oneshotob[2]
MSF
Motion Servo Off
Axis X_axis
Motion Control X_AXIS_STAT[1]
EN
DN
ER
0006
oneshotob[2]
MSF
Motion Servo Off
Axis Y_axis
Motion Control Y_AXIS_STAT[1]
EN
DN
ER
0015
oneshotob[2]
MSF
Motion Servo Off
Axis Y_axis
Motion Control Y_AXIS_STAT[1]
EN
DN
ER
0007
oneshotob[2]
MSF
Motion Servo Off
Axis Z_axis
Motion Control Z_AXIS_STAT[1]
EN
DN
ER
0016
oneshotob[2]
MSF
Motion Servo Off
Axis Z_axis
Motion Control Z_AXIS_STAT[1]
EN
DN
ER
0008
X_AXIS_STAT[1].DN
Y_AXIS_STAT[1].DN
Z_AXIS_STAT[1].DN
CMP
Compare
0017
X_AXIS_STAT[1].DN
Y_AXIS_STAT[1].DN
Z_AXIS_STAT[1].DN
CMP
Compare


Expression MOVE_TYPE=0
OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB
0009
oneshotob[4]
MAFR
Motion Axis Fault Reset
Axis Z_axis
Motion Control Z_AXIS_STAT[2]
EN
DN
ER
0010
Z_AXIS_STAT[2].DN
MAFR
Motion Axis Fault Reset
Axis Y_axis
Motion Control Y_AXIS_STAT[2]
EN
DN
ER
0011
Y_AXIS_STAT[2].DN
MAFR
Motion Axis Fault Reset
Axis X_axis
Motion Control X_AXIS_STAT[2]
EN
DN
ER
0012
Z_AXIS_STAT[1].DN
Y_AXIS_STAT[1].DN
X_AXIS_STAT[1].DN
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
END


Expression MOVE_TYPE=0
OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB
0018
oneshotob[4]
MAFR
Motion Axis Fault Reset
Axis Z_axis
Motion Control Z_AXIS_STAT[2]
EN
DN
ER
0019
Z_AXIS_STAT[2].DN
MAFR
Motion Axis Fault Reset
Axis Y_axis
Motion Control Y_AXIS_STAT[2]
EN
DN
ER
0020
Y_AXIS_STAT[2].DN
MAFR
Motion Axis Fault Reset
Axis X_axis
Motion Control X_AXIS_STAT[2]
EN
DN
ER
0021
X_AXIS_STAT[2].DN
TIMEOUT
U
0022
Z_AXIS_STAT[1].DN
Y_AXIS_STAT[1].DN
X_AXIS_STAT[1].DN
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
END

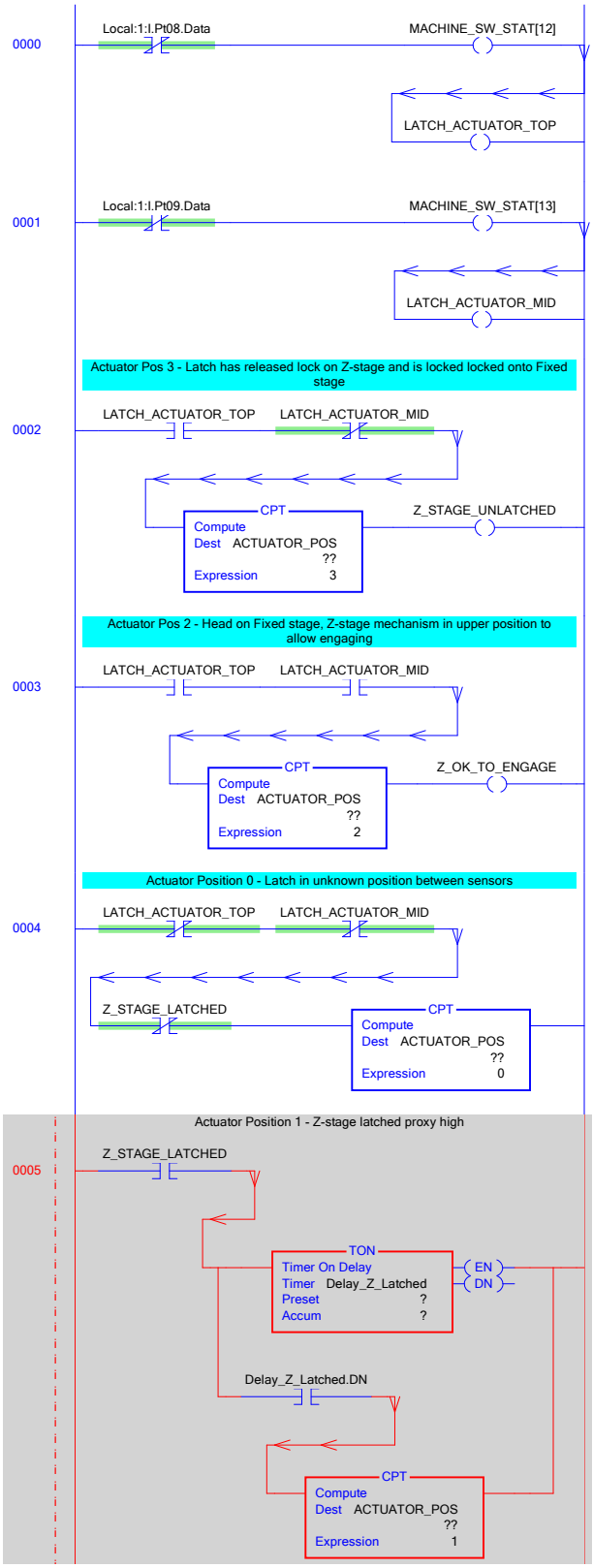
Initialize Parameters and Local Tags

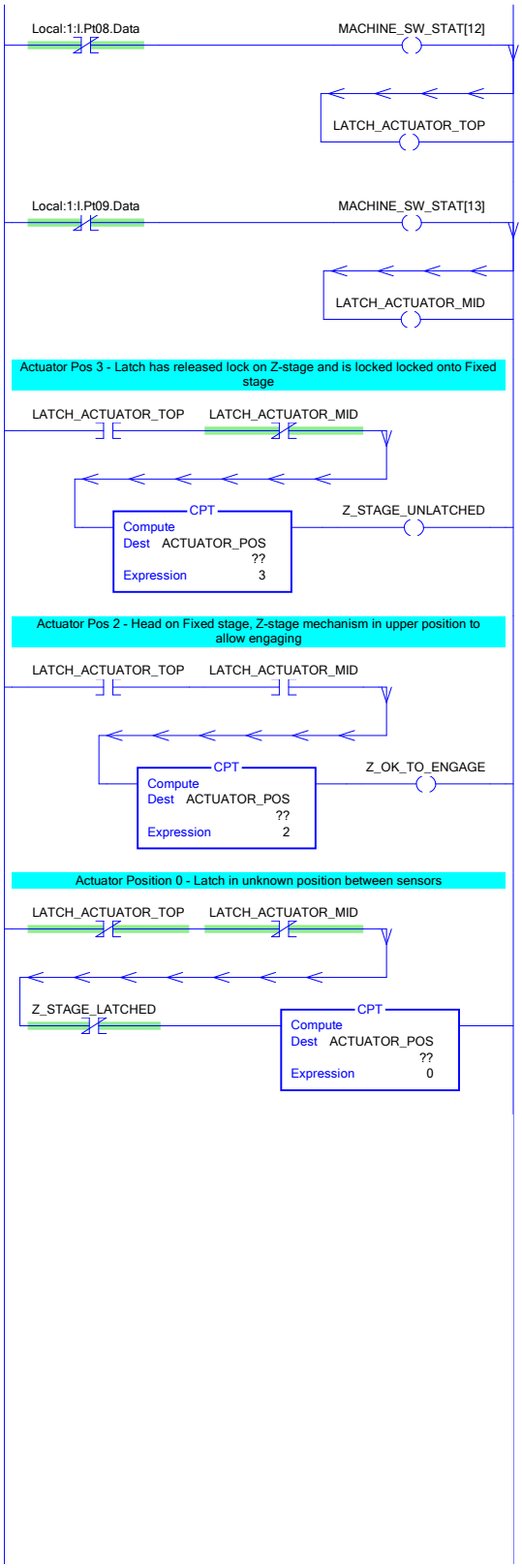
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
STATE_INIT_DN	---	---	---	---		BOOL		0
SW0	---	---	---	---		BOOL		0
SW1	---	---	---	---		BOOL		0
SW2	---	---	---	---		BOOL		0
SW3	---	---	---	---		BOOL		0
SW4	---	---	---	---		BOOL		0

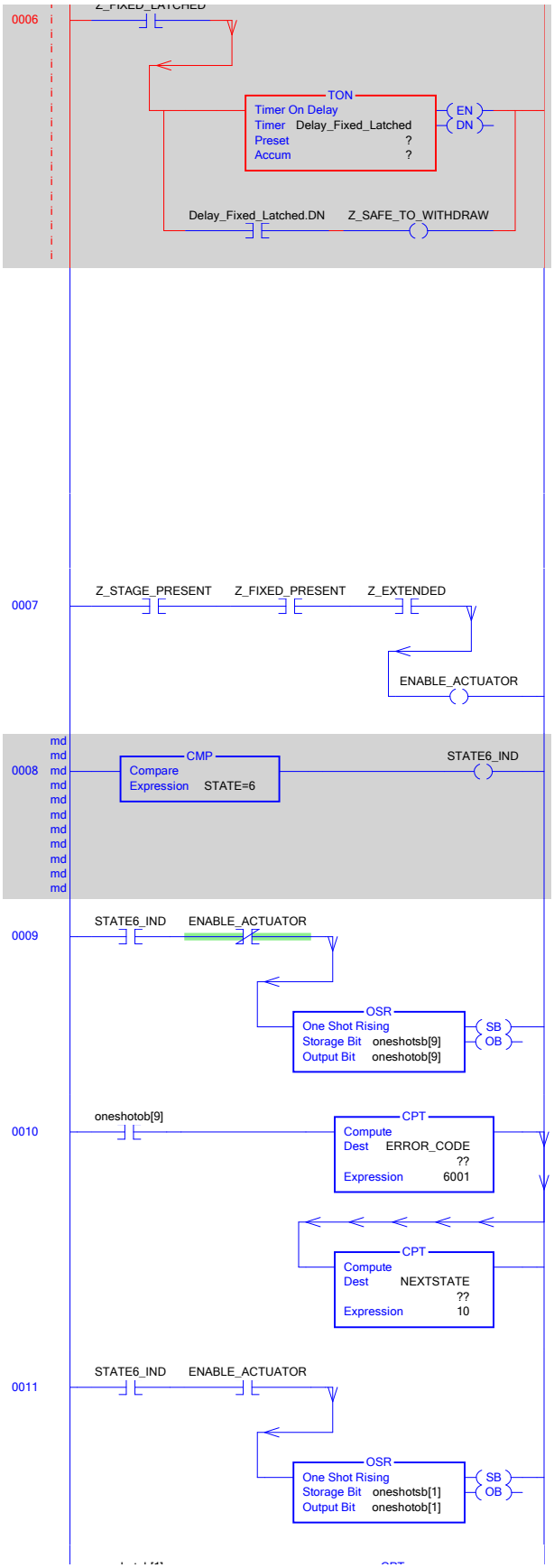
Latch_UnLatch_State_6_7_8 Parameters and Local Tags

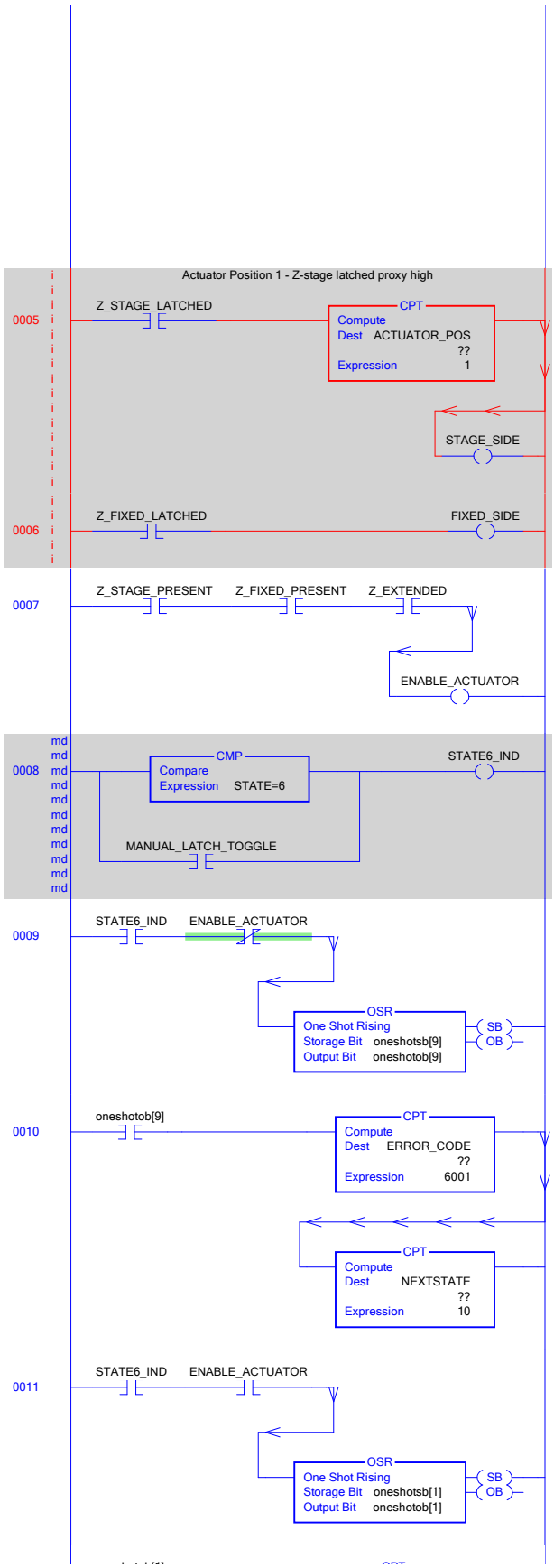
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
BIT0	---	---	---	---		BOOL		0
BIT1	---	---	---	---		BOOL		0
BIT2	---	---	---	---		BOOL		0
bit22	---	---	---	---		BOOL		0
CURRENT_ACT_POS	---	---	---	---		DINT		0
Delay_Fixed_Latched		TIMER			---	---	---	---
Delay_Z_Latched		TIMER			---	---	---	---
FIXED_SIDE	---	---	---	---		BOOL		0
LATCH_ACTUATOR_DOWN	---	---	---	---		BOOL		0
LATCH_EN	---	---	---	---		BOOL		0
LatchCounter	---	---	---	---		COUNTER		
Latching_pulse_duration		TIMER			---	---	---	---
Latching_pulse_interval		TIMER			---	---	---	---
LatchingTimeoutCounter		COUNTER	When you enter state 6 or 7, start counting seconds		---	---	---	---
LatchTimer1	---	---	---	---		TIMER		
LatchTimer2	---	---	---	---		TIMER		
MANUAL_HOME_LATCH	---	---	---	---		BOOL		0
MANUAL_LATCH_TOGGLE	---	---	---	---		BOOL		0
ob4	---	---	---	---		BOOL		0
OFF	---	---	---	---		BOOL		0
oneshotob		BOOL[32]				BOOL[32]		
oneshotsb		BOOL[32]				BOOL[32]		
PREV_ACT_POS		DINT	Store the previous actuator position	2		DINT		3
STAGE_SIDE	---	---	---	---		BOOL		0
test	---	---	---	---		BOOL		1
TimeoutTimer1		TIMER			---	---	---	---
TimeoutTimer2		TIMER			---	---	---	---

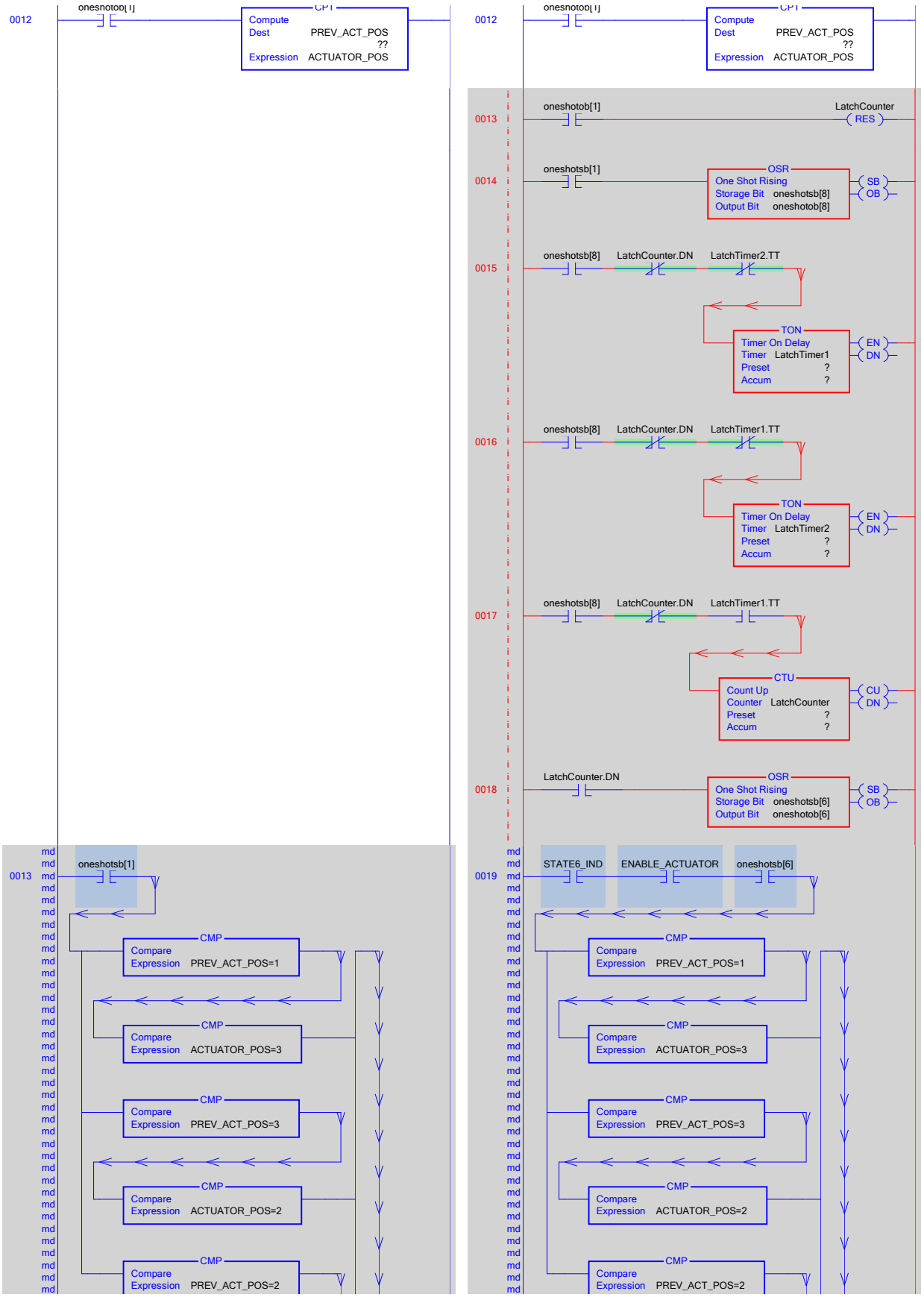
Program: Latch_UnLatch_State_6_7_8
Routine: Latch_Driver

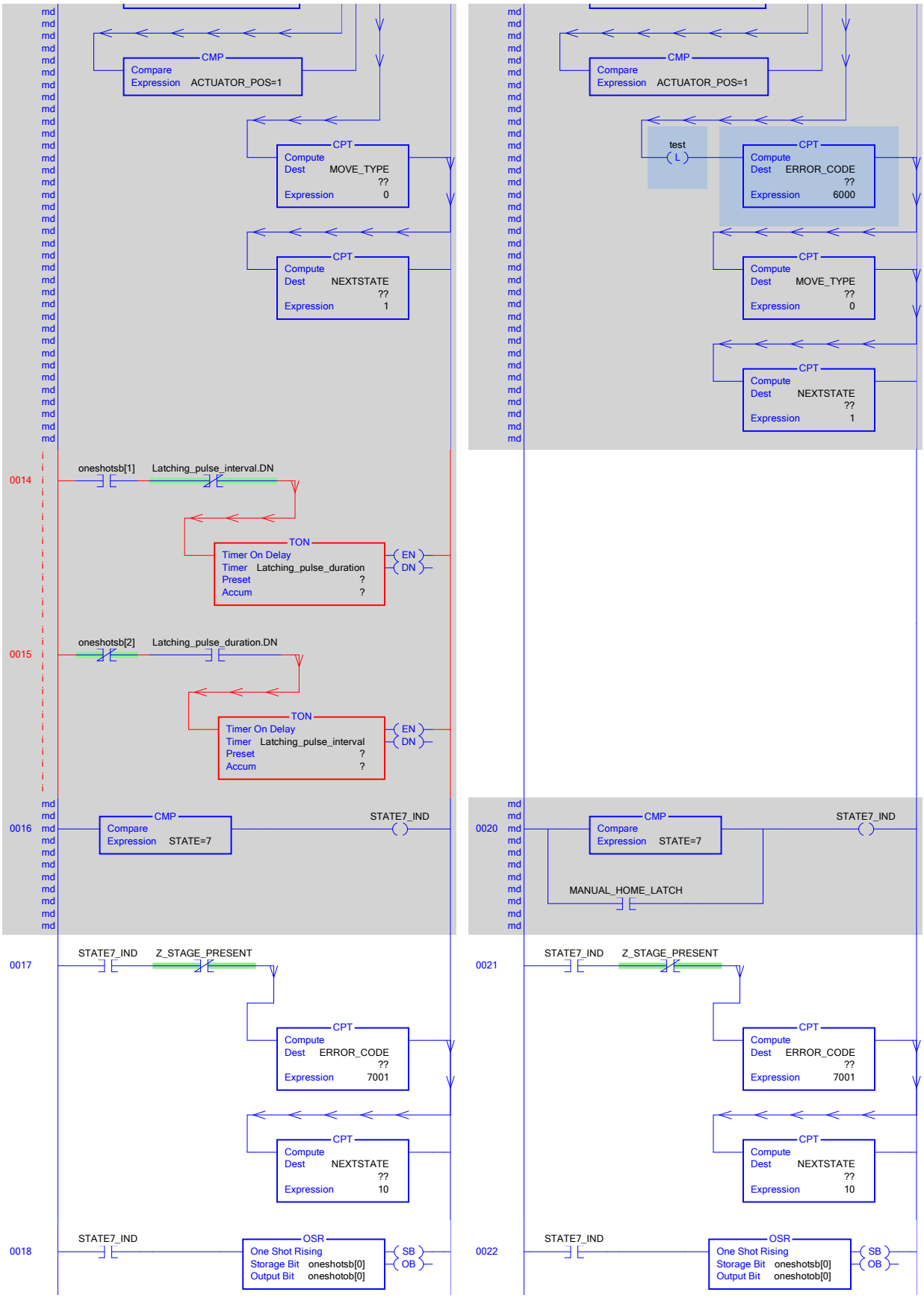





0006
Z_FIXED_LATCHED
TON
Timer On Delay
Timer Delay_Fixed_Latched
Preset ?
Accum ?
EN
DN
Delay_Fixed_Latched.DN
Z_SAFE_TO_WITHDRAW
0007
Z_STAGE_PRESENT
Z_FIXED_PRESENT
Z_EXTENDED
ENABLE_ACTUATOR
0008
CMP
Compare
Expression STATE=6
STATE6_IND
0009
STATE6_IND
ENABLE_ACTUATOR
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB
0010
oneshotob[9]
CPT
Compute
Dest ERROR_CODE
??
Expression 6001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0011
STATE6_IND
ENABLE_ACTUATOR
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]


Actuator Position 1 - Z-stage latched proxy high
0005
Z_STAGE_LATCHED
CPT
Compute
Dest ACTUATOR_POS
??
Expression 1
STAGE_SIDE
0006
Z_FIXED_LATCHED
FIXED_SIDE
0007
Z_STAGE_PRESENT
Z_FIXED_PRESENT
Z_EXTENDED
ENABLE_ACTUATOR
0008
CMP
Compare
Expression STATE=6
STATE6_IND
MANUAL_LATCH_TOGGLE
0009
STATE6_IND
ENABLE_ACTUATOR
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB
0010
oneshotob[9]
CPT
Compute
Dest ERROR_CODE
??
Expression 6001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0011
STATE6_IND
ENABLE_ACTUATOR
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]


0012
oneshotob[1]
CPT
Compute
Dest PREV_ACT_POS
??
Expression ACTUATOR_POS
0013
oneshotsb[1]
CMP
Compare
Expression PREV_ACT_POS=1
CMP
Compare
Expression ACTUATOR_POS=3
CMP
Compare
Expression PREV_ACT_POS=3
CMP
Compare
Expression ACTUATOR_POS=2
CMP
Compare
Expression PREV_ACT_POS=2
0012
oneshotob[1]
CPT
Compute
Dest PREV_ACT_POS
??
Expression ACTUATOR_POS
0013
oneshotob[1]
LatchCounter
RES
0014
oneshotsb[1]
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0015
oneshotsb[8]
LatchCounter.DN
LatchTimer2.TT
TON
Timer On Delay
Timer LatchTimer1
Preset ?
Accum ?
EN
DN
0016
oneshotsb[8]
LatchCounter.DN
LatchTimer1.TT
TON
Timer On Delay
Timer LatchTimer2
Preset ?
Accum ?
EN
DN
0017
oneshotsb[8]
LatchCounter.DN
LatchTimer1.TT
CTU
Count Up
Counter LatchCounter
Preset ?
Accum ?
CU
DN
0018
LatchCounter.DN
OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB
0019
STATE6_IND
ENABLE_ACTUATOR
oneshotsb[6]
CMP
Compare
Expression PREV_ACT_POS=1
CMP
Compare
Expression ACTUATOR_POS=3
CMP
Compare
Expression PREV_ACT_POS=3
CMP
Compare
Expression ACTUATOR_POS=2
CMP
Compare
Expression PREV_ACT_POS=2


CMP
Compare
Expression ACTUATOR_POS=1
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0014
oneshotsb[1]
Latching_pulse_interval.DN
TON
Timer On Delay
Timer Latching_pulse_duration
Preset ?
Accum ?
EN
DN
0015
oneshotsb[2]
Latching_pulse_duration.DN
TON
Timer On Delay
Timer Latching_pulse_interval
Preset ?
Accum ?
EN
DN
0016
CMP
Compare
Expression STATE=7
STATE7_IND
0017
STATE7_IND
Z_STAGE_PRESENT
CPT
Compute
Dest ERROR_CODE
??
Expression 7001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0018
STATE7_IND
OSR
One Shot Rising
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB
CMP
Compare
Expression ACTUATOR_POS=1
test
L
CPT
Compute
Dest ERROR_CODE
??
Expression 6000
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0020
CMP
Compare
Expression STATE=7
STATE7_IND
MANUAL_HOME_LATCH
0021
STATE7_IND
Z_STAGE_PRESENT
CPT
Compute
Dest ERROR_CODE
??
Expression 7001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0022
STATE7_IND
OSR
One Shot Rising
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB

0019 oneshotob[0] HomeCounter (RES)

0020 oneshotsb[0] HomeCounter.DN HomeTimer2.TT
TON
Timer On Delay
Timer HomeTimer1
Preset ?
Accum ?
(EN) (DN)

0021 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
TON
Timer On Delay
Timer HomeTimer2
Preset ?
Accum ?
(EN) (DN)

0022 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
CTU
Count Up
Counter HomeCounter
Preset ?
Accum ?
(CU) (DN)

0023 md HomeCounter.DN AFI Local:3:O.Pt02.Data ( )

0024 HomeCounter.DN
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
(SB) (OB)

0025 oneshotob[7] Z_STAGE_LATCHED LATCH_ACTUATOR_HOMED (L)

0026 oneshotob[7] Z_STAGE_LATCHED
CPT
Compute
Dest ERROR_CODE
??
Expression 7002
LATCH_ACTUATOR_HOMED (L)

0027 LATCH_ACTUATOR_HOMED MACHINE_SW_STAT[0] ( )

0028 LATCH_ACTUATOR_HOMED
OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
(SB) (OB)

oneshotob[4] LatchCounter3

0023 oneshotob[0] HomeCounter (RES)

0024 oneshotsb[0] HomeCounter.DN HomeTimer2.TT
TON
Timer On Delay
Timer HomeTimer1
Preset ?
Accum ?
(EN) (DN)

0025 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
TON
Timer On Delay
Timer HomeTimer2
Preset ?
Accum ?
(EN) (DN)

0026 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
CTU
Count Up
Counter HomeCounter
Preset ?
Accum ?
(CU) (DN)

0027 md HomeCounter.DN AFI Local:3:O.Pt02.Data ( )
BIT0

0028 HomeCounter.DN
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
(SB) (OB)

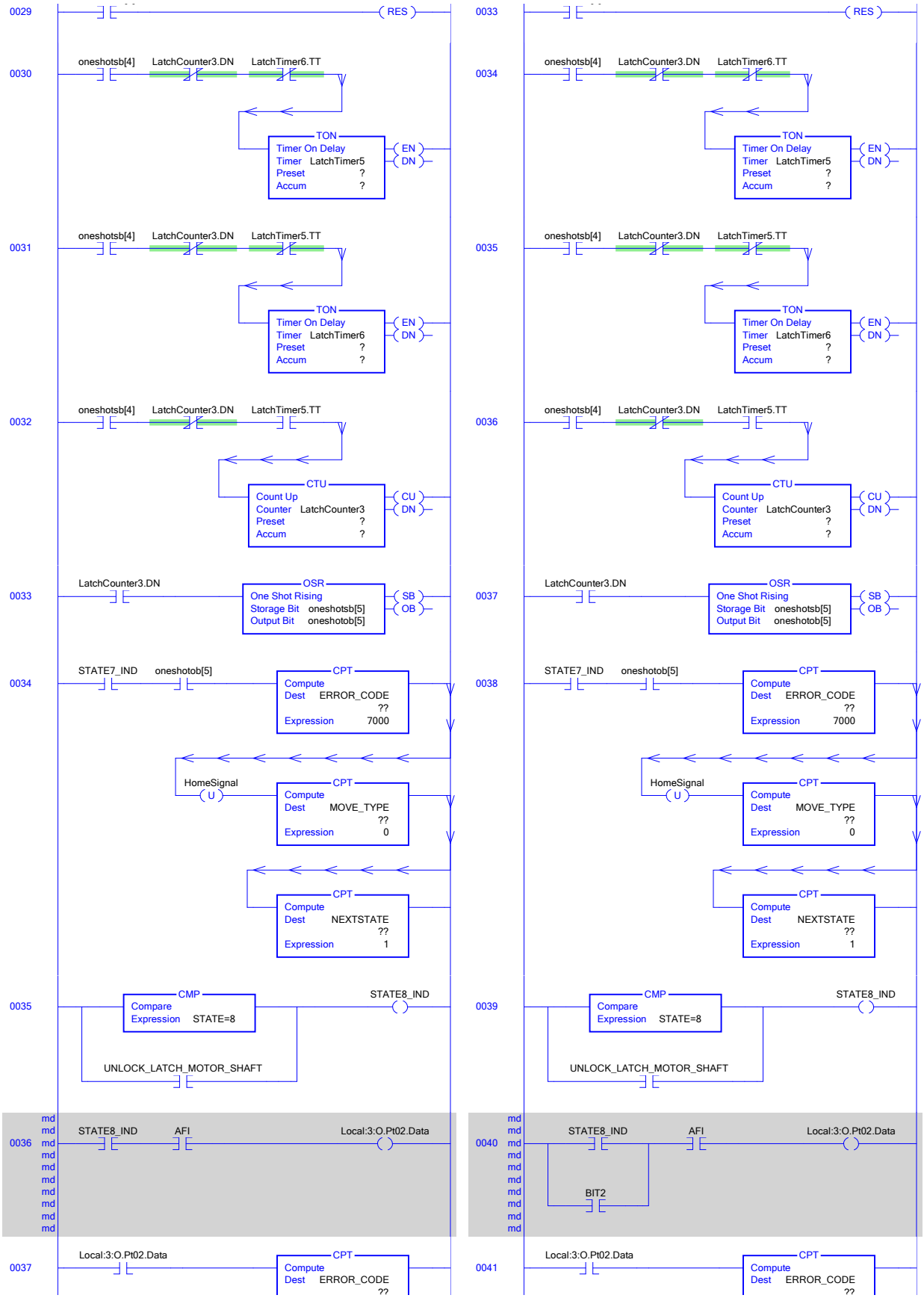
0029 oneshotob[7] Z_STAGE_LATCHED LATCH_ACTUATOR_HOMED (L)

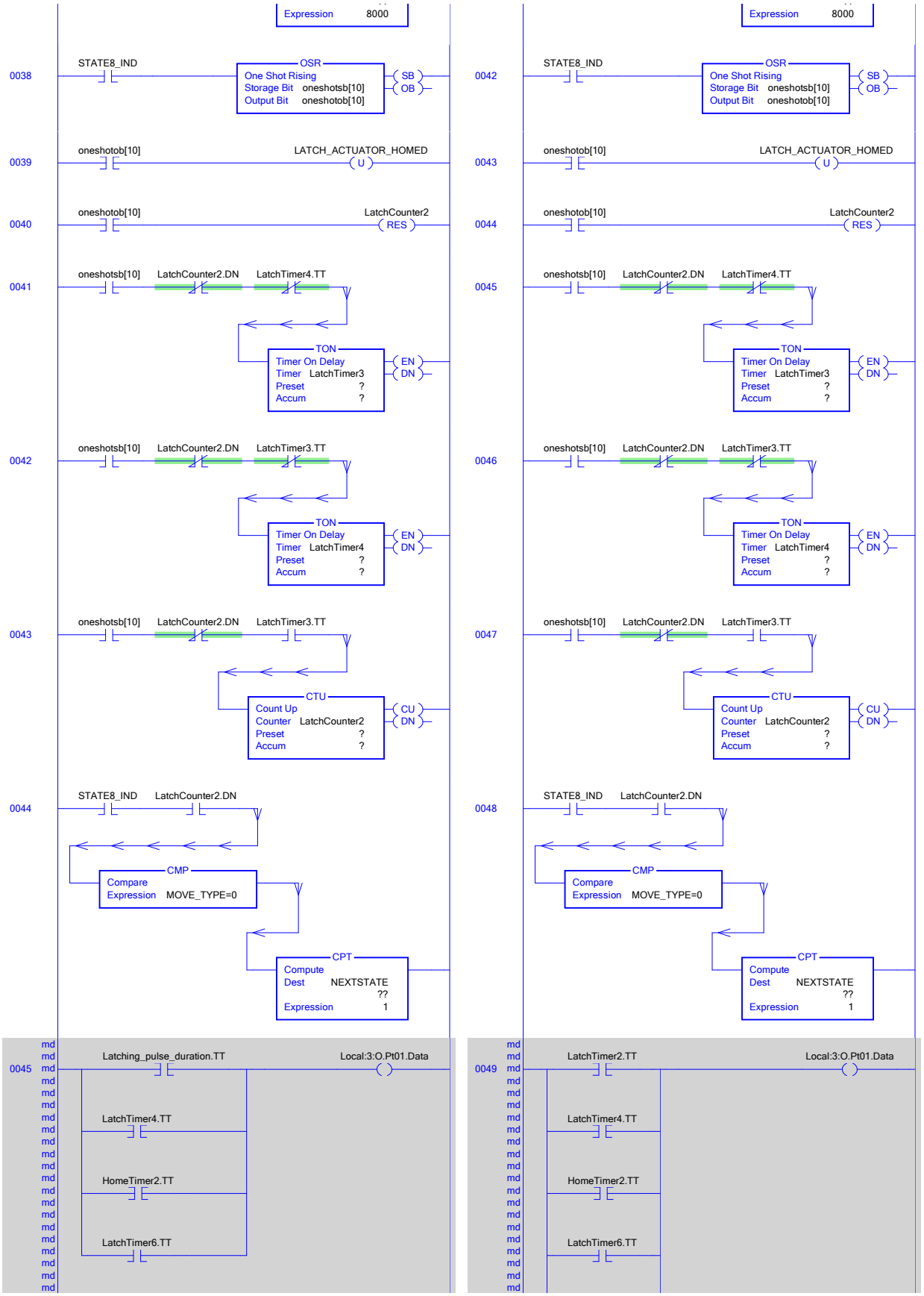
0030 oneshotob[7] Z_STAGE_LATCHED
CPT
Compute
Dest ERROR_CODE
??
Expression 7002
LATCH_ACTUATOR_HOMED (L)

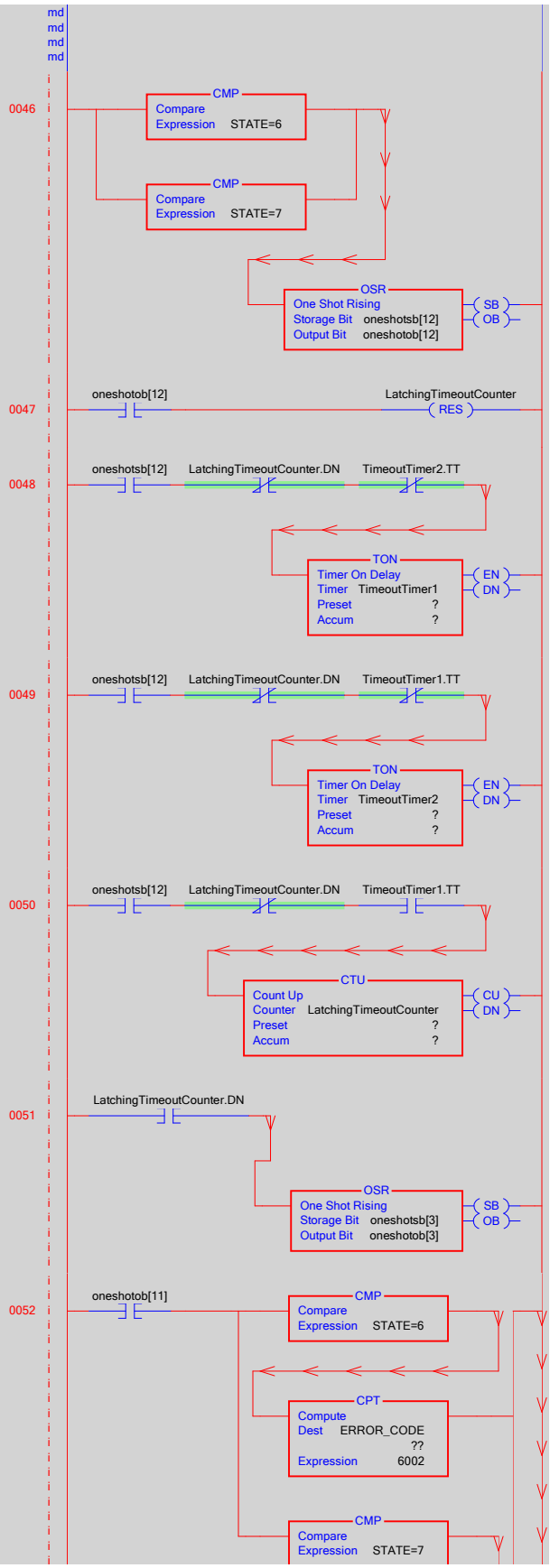
0031 LATCH_ACTUATOR_HOMED MACHINE_SW_STAT[0] ( )

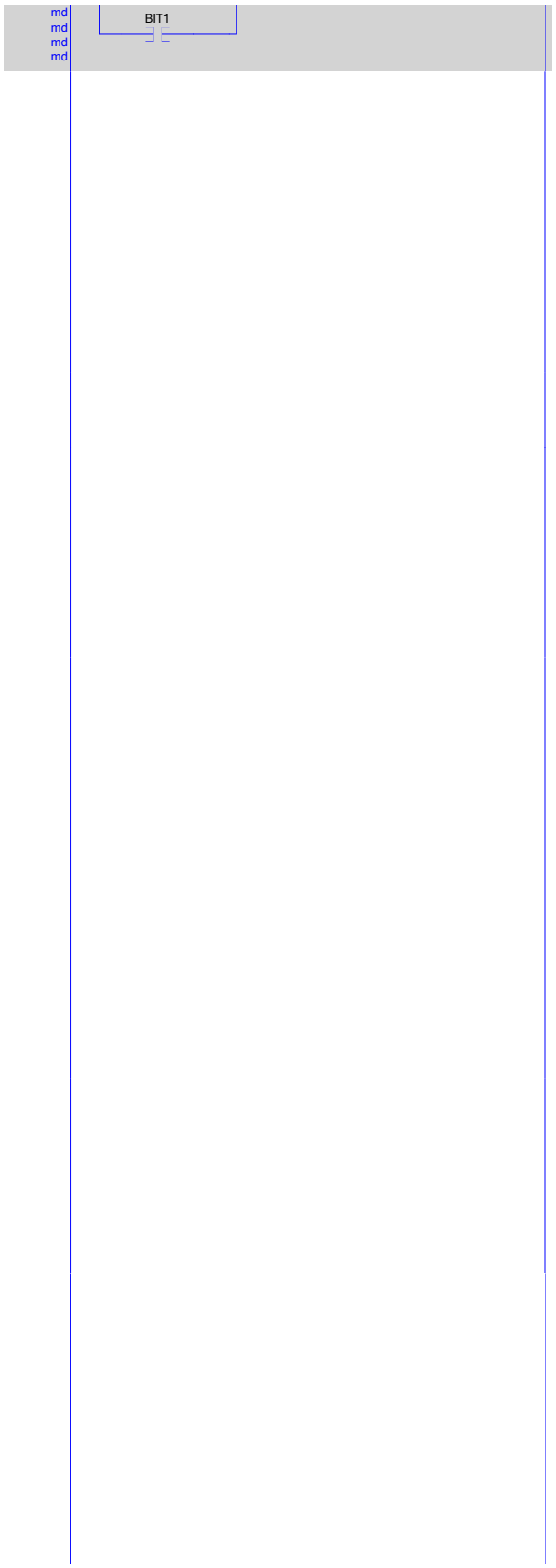
0032 LATCH_ACTUATOR_HOMED
OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
(SB) (OB)

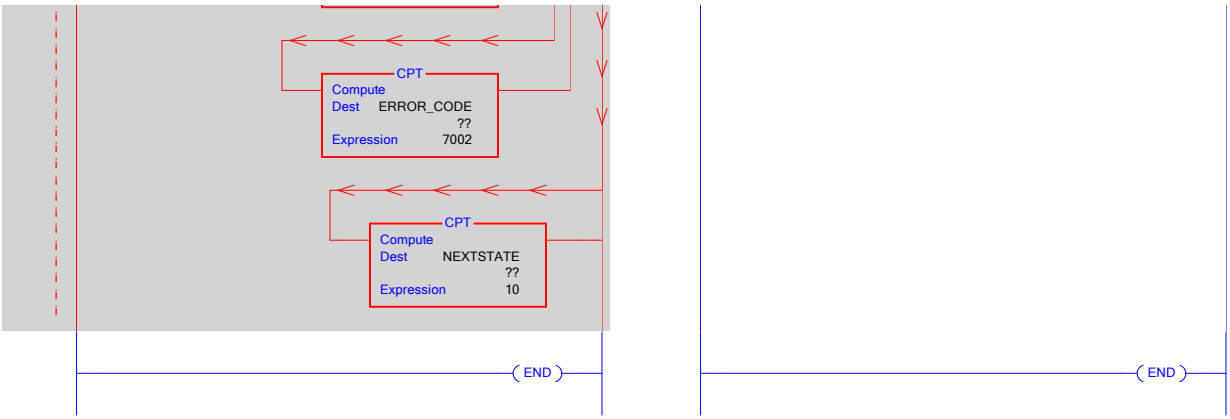
oneshotob[4] LatchCounter3


0029
RES
0030
oneshotsb[4]
LatchCounter3.DN
LatchTimer6.TT
TON
Timer On Delay
Timer LatchTimer5
Preset ?
Accum ?
EN
DN
0031
oneshotsb[4]
LatchCounter3.DN
LatchTimer5.TT
TON
Timer On Delay
Timer LatchTimer6
Preset ?
Accum ?
EN
DN
0032
oneshotsb[4]
LatchCounter3.DN
LatchTimer5.TT
CTU
Count Up
Counter LatchCounter3
Preset ?
Accum ?
CU
DN
0033
LatchCounter3.DN
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
0034
STATE7_IND
oneshotob[5]
CPT
Compute
Dest ERROR_CODE
??
Expression 7000
HomeSignal
U
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0035
CMP
Compare
Expression STATE=8
STATE8_IND
UNLOCK_LATCH_MOTOR_SHAFT
0036
STATE8_IND
AFI
Local:3:O.Pt02.Data
0037
Local:3:O.Pt02.Data
CPT
Compute
Dest ERROR_CODE
??
0033
RES
0034
oneshotsb[4]
LatchCounter3.DN
LatchTimer6.TT
TON
Timer On Delay
Timer LatchTimer5
Preset ?
Accum ?
EN
DN
0035
oneshotsb[4]
LatchCounter3.DN
LatchTimer5.TT
TON
Timer On Delay
Timer LatchTimer6
Preset ?
Accum ?
EN
DN
0036
oneshotsb[4]
LatchCounter3.DN
LatchTimer5.TT
CTU
Count Up
Counter LatchCounter3
Preset ?
Accum ?
CU
DN
0037
LatchCounter3.DN
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
0038
STATE7_IND
oneshotob[5]
CPT
Compute
Dest ERROR_CODE
??
Expression 7000
HomeSignal
U
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0039
CMP
Compare
Expression STATE=8
STATE8_IND
UNLOCK_LATCH_MOTOR_SHAFT
0040
STATE8_IND
AFI
Local:3:O.Pt02.Data
BIT2
0041
Local:3:O.Pt02.Data
CPT
Compute
Dest ERROR_CODE
??


Expression 8000
0038 STATE8_IND
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB
0039 oneshotob[10]
LATCH_ACTUATOR_HOMED
U
0040 oneshotob[10]
LatchCounter2
RES
0041 oneshotsb[10] LatchCounter2.DN LatchTimer4.TT
TON
Timer On Delay
Timer LatchTimer3
Preset ?
Accum ?
EN
DN
0042 oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
TON
Timer On Delay
Timer LatchTimer4
Preset ?
Accum ?
EN
DN
0043 oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
CTU
Count Up
Counter LatchCounter2
Preset ?
Accum ?
CU
DN
0044 STATE8_IND LatchCounter2.DN
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0045 Latching_pulse_duration.TT
Local:3:O.Pt01.Data
LatchTimer4.TT
HomeTimer2.TT
LatchTimer6.TT
Expression 8000
0042 STATE8_IND
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB
0043 oneshotob[10]
LATCH_ACTUATOR_HOMED
U
0044 oneshotob[10]
LatchCounter2
RES
0045 oneshotsb[10] LatchCounter2.DN LatchTimer4.TT
TON
Timer On Delay
Timer LatchTimer3
Preset ?
Accum ?
EN
DN
0046 oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
TON
Timer On Delay
Timer LatchTimer4
Preset ?
Accum ?
EN
DN
0047 oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
CTU
Count Up
Counter LatchCounter2
Preset ?
Accum ?
CU
DN
0048 STATE8_IND LatchCounter2.DN
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0049 LatchTimer2.TT
Local:3:O.Pt01.Data
LatchTimer4.TT
HomeTimer2.TT
LatchTimer6.TT


md
0046
CMP
Compare
Expression STATE=6
CMP
Compare
Expression STATE=7
OSR
One Shot Rising
Storage Bit oneshotsb[12]
Output Bit oneshotob[12]
SB
OB
0047
oneshotob[12]
LatchingTimeoutCounter
RES
0048
oneshotsb[12]
LatchingTimeoutCounter.DN
TimeoutTimer2.TT
TON
Timer On Delay
Timer TimeoutTimer1
Preset ?
Accum ?
EN
DN
0049
oneshotsb[12]
LatchingTimeoutCounter.DN
TimeoutTimer1.TT
TON
Timer On Delay
Timer TimeoutTimer2
Preset ?
Accum ?
EN
DN
0050
oneshotsb[12]
LatchingTimeoutCounter.DN
TimeoutTimer1.TT
CTU
Count Up
Counter LatchingTimeoutCounter
Preset ?
Accum ?
CU
DN
0051
LatchingTimeoutCounter.DN
OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB
0052
oneshotob[11]
CMP
Compare
Expression STATE=6
CPT
Compute
Dest ERROR_CODE
??
Expression 6002
CMP
Compare
Expression STATE=7

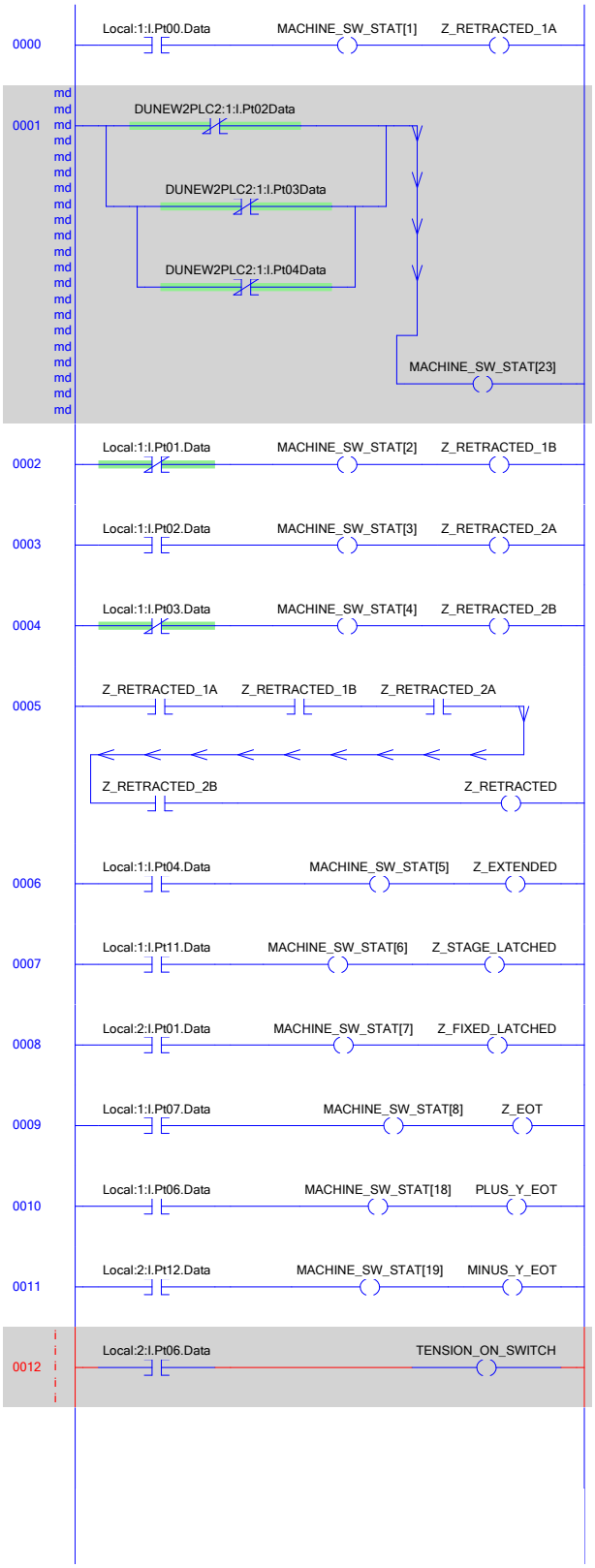

md
BIT1

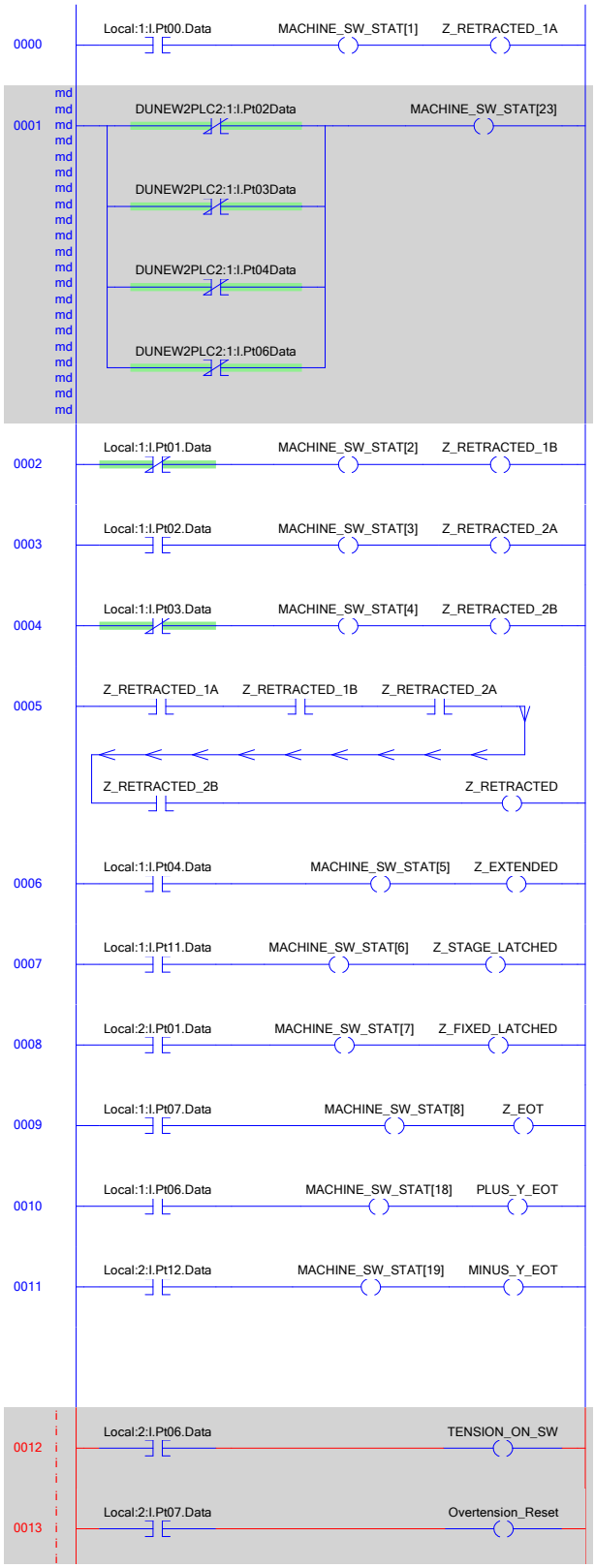

CPT
Compute
Dest ERROR_CODE
??
Expression 7002
CPT
Compute
Dest NEXTSTATE
??
Expression 10
END
END

MainProgram Parameters and Local Tags

	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
CHECKSTATE	---	---	---	---		BOOL		0
INC_STATE NUM1	---	---	---	---		BOOL		0
oneshotob	---	---	---	---		BOOL		0
oneshotsb	---	---	---	---		BOOL		0
SLS_Active	---	---	---	---		BOOL		0
STATE_MACHINE_ERROR	---	---	---	---		BOOL		0
UNKNOWN	---	---	---	---		BOOL		0
X_AXIS_STAT	---	---	---	---		MOTION_INSTRUCTION [10]		
X_estop_active	---	---	---	---		DINT		0
XY_AXIS_STAT	---	---	---	---		MOTION_INSTRUCTION [10]		
Y_AXIS_STAT	---	---	---	---		MOTION_INSTRUCTION [10]		
Y_estop_active	---	---	---	---		DINT		0
Z_estop_active	---	---	---	---		DINT		0

Program: MainProgram
Routine: MainRoutine





0013

0014 Local:2:I.Pt10.Data MACHINE_SW_STAT[21] MINUS_X_EOT

0015 Z_EOT PLUS_Y_EOT MINUS_Y_EOT PLUS_X_EOT
MINUS_X_EOT MAS ALL_EOT_GOOD

0016 Local:1:I.Pt10.Data MACHINE_SW_STAT[9] Z_STAGE_PRESENT

0017 Local:2:I.Pt02.Data MACHINE_SW_STAT[10] Z_FIXED_PRESENT

0018 Local:1:I.Pt12.Data MACHINE_SW_STAT[17] Y_XFER_OK

0019 Local:1:I.Pt13.Data MACHINE_SW_STAT[16] Y_MOUNT_XFER_OK

0020 Local:1:I.Pt15.Data
GRT
Greater Than (A>B)
Source A tension
??
Source B wire_break_tension
??
wire_break_proxy

0021 Local:2:I.Pt00.Data MACHINE_SW_STAT[15] X_XFER_OK

0022 Local:2:I.Pt04.Data MACHINE_SW_STAT[14] X_PARKED

0023 Local:2:I.Pt14.Data MACHINE_SW_STAT[22] APA_IS_VERTICAL

0014

0015 Local:2:I.Pt10.Data MACHINE_SW_STAT[21] MINUS_X_EOT

0016 Z_EOT PLUS_Y_EOT MINUS_Y_EOT
PLUS_X_EOT MINUS_X_EOT
OT_OR
ALL_EOT_GOOD

0017 Local:1:I.Pt10.Data MACHINE_SW_STAT[9] Z_STAGE_PRESENT

0018 Local:2:I.Pt02.Data MACHINE_SW_STAT[10] Z_FIXED_PRESENT

0019 Local:1:I.Pt12.Data MACHINE_SW_STAT[17] Y_XFER_OK

0020 Local:1:I.Pt13.Data MACHINE_SW_STAT[16] Y_MOUNT_XFER_OK

0021 Local:1:I.Pt15.Data WIRE_BREAK_SW

0022 Local:2:I.Pt00.Data MACHINE_SW_STAT[15] X_XFER_OK

0023 Local:2:I.Pt04.Data MACHINE_SW_STAT[14] X_PARK_OK

0024 Local:2:I.Pt14.Data Local:2:I.Pt15.Data MACHINE_SW_STAT[22]
ROT_LOCK_KEY

0024
DUNEW2PLC2:1:I.Pt00Data DUNEW2PLC2:1:I.Pt01Data
MACHINE_SW_STAT[25]

0025
Local:6:I.Pt00.Data
MACHINE_SW_STAT[26]
FRAME_LOC_HD_TOP

0026
Local:6:I.Pt01.Data
MACHINE_SW_STAT[27]
FRAME_LOC_HD_MID

0027
Local:6:I.Pt02.Data
MACHINE_SW_STAT[28]
FRAME_LOC_HD_BTM

0028
Local:6:I.Pt03.Data
MACHINE_SW_STAT[29]
FRAME_LOC_FT_TOP

0029
Local:6:I.Pt04.Data
MACHINE_SW_STAT[30]
FRAME_LOC_FT_MID

0030
Local:6:I.Pt05.Data
MACHINE_SW_STAT[31]
FRAME_LOC_FT_BTM

0031
LIM
Limit Test (CIRC)
Low Limit 80
??
Test Y_axis.ActualPosition
??
High Limit 450
??
Y_Xfer_Window_Btm

0032
LIM
Limit Test (CIRC)
Low Limit 1050
??
Test Y_axis.ActualPosition
??
High Limit 1550
??
Y_Xfer_Window_Mid

0033
LIM
Limit Test (CIRC)
Low Limit 2200
??
Test Y_axis.ActualPosition
??
High Limit 2650
??
Y_Xfer_Window_Top

0034
X_XFER_OK
LIM
Limit Test (CIRC)
Low Limit 400

0025
DUNEW2PLC2:1:I.Pt00Data DUNEW2PLC2:1:I.Pt01Data
MACHINE_SW_STAT[25]

0026
Local:6:I.Pt00.Data
MACHINE_SW_STAT[26]
FRAME_LOC_HD_TOP

0027
Local:6:I.Pt01.Data
MACHINE_SW_STAT[27]
FRAME_LOC_HD_MID

0028
Local:6:I.Pt02.Data
MACHINE_SW_STAT[28]
FRAME_LOC_HD_BTM

0029
Local:6:I.Pt03.Data
MACHINE_SW_STAT[29]
FRAME_LOC_FT_TOP

0030
Local:6:I.Pt04.Data
MACHINE_SW_STAT[30]
FRAME_LOC_FT_MID

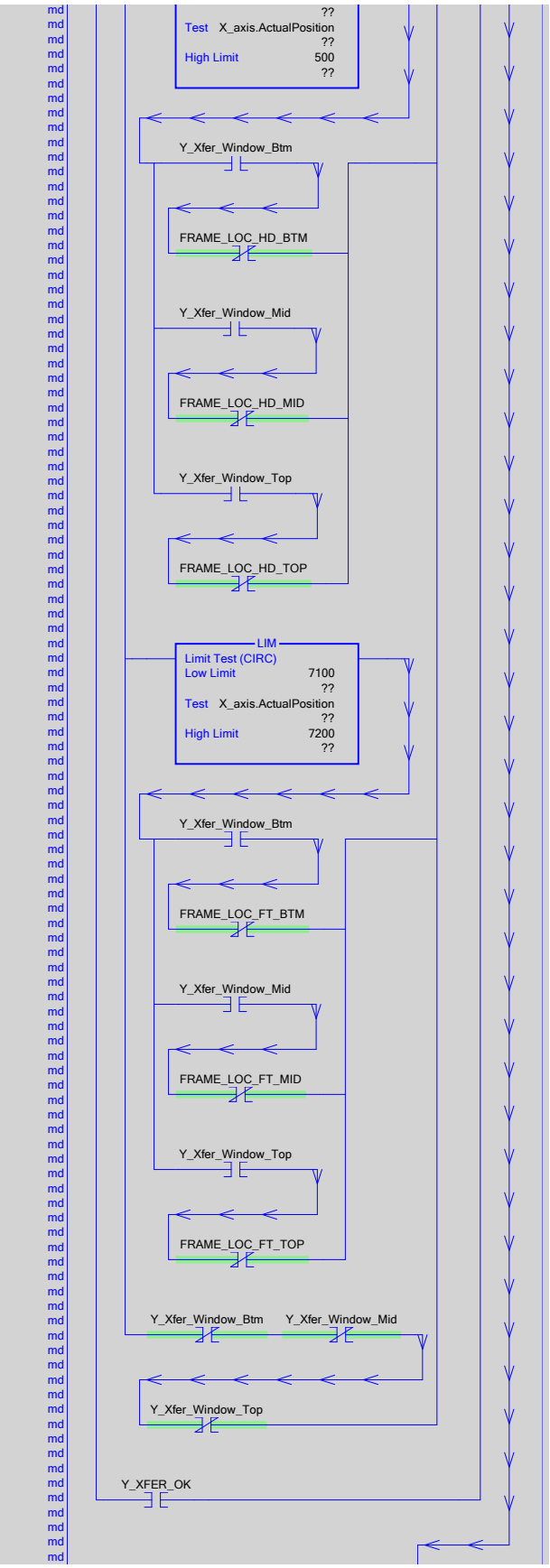
0031
Local:6:I.Pt05.Data
MACHINE_SW_STAT[31]
FRAME_LOC_FT_BTM

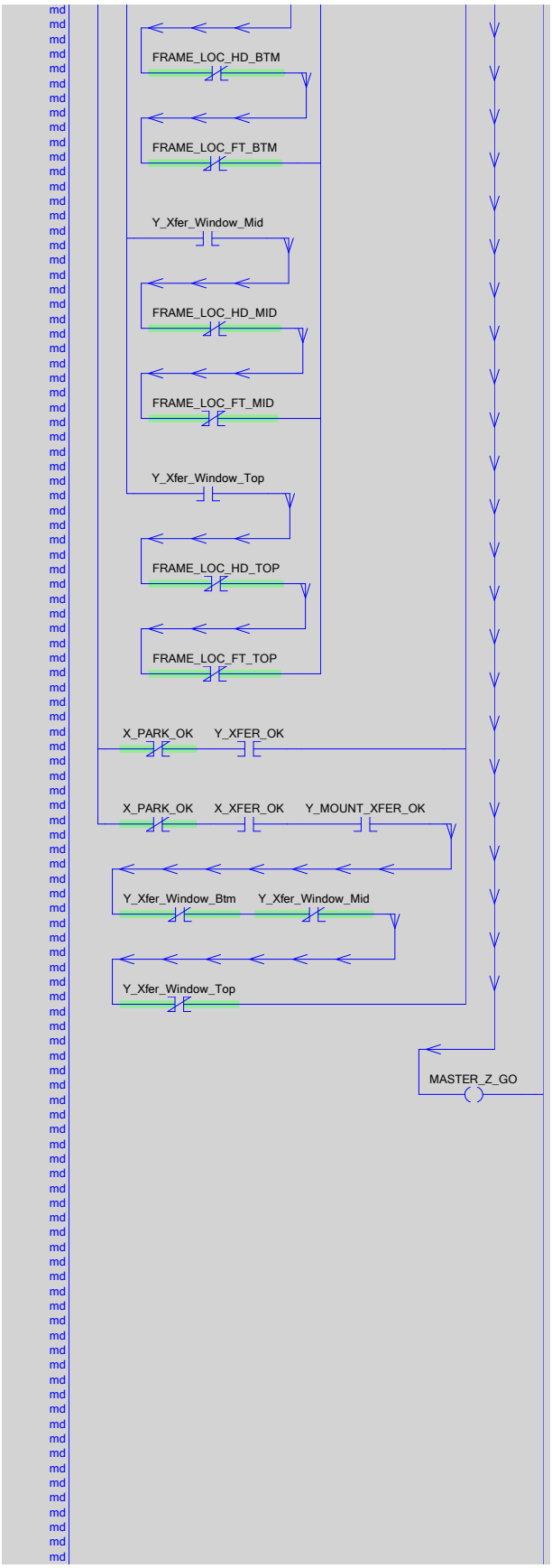
0032
LIM
Limit Test (CIRC)
Low Limit 100
??
Test Y_POSITION
??
High Limit 500
??
Y_Xfer_Window_Btm

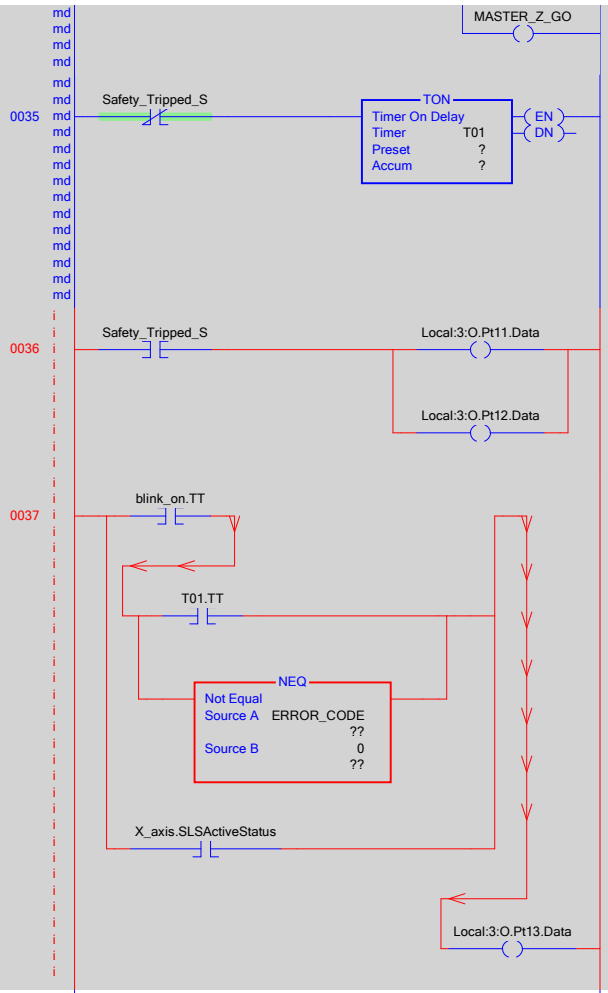
0033
LIM
Limit Test (CIRC)
Low Limit 1180
??
Test Y_POSITION
??
High Limit 1580
??
Y_Xfer_Window_Mid

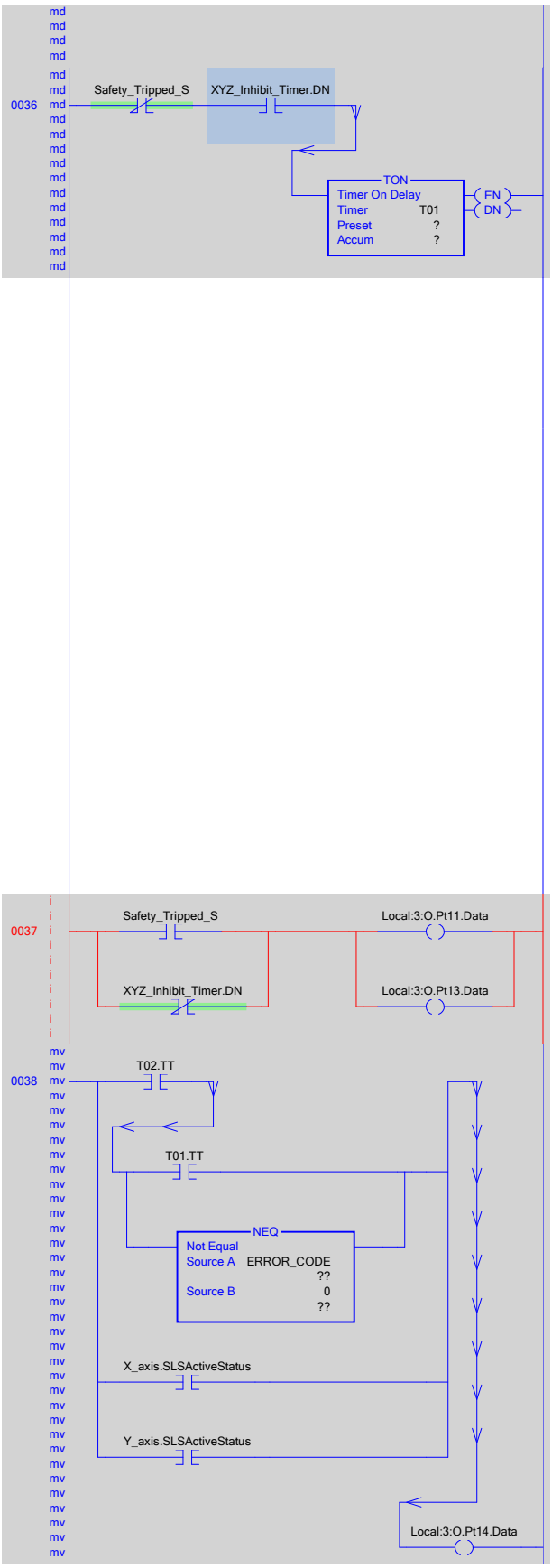
0034
LIM
Limit Test (CIRC)
Low Limit 2260
??
Test Y_POSITION
??
High Limit 2660
??
Y_Xfer_Window_Top

0035
X_PARK_OK X_XFER_OK
Y_Xfer_Window_Btm


??
Test X_axis.ActualPosition
??
High Limit 500
??
Y_Xfer_Window_Btm
FRAME_LOC_HD_BTM
Y_Xfer_Window_Mid
FRAME_LOC_HD_MID
Y_Xfer_Window_Top
FRAME_LOC_HD_TOP
LIM
Limit Test (CIRC)
Low Limit 7100
??
Test X_axis.ActualPosition
??
High Limit 7200
??
Y_Xfer_Window_Btm
FRAME_LOC_FT_BTM
Y_Xfer_Window_Mid
FRAME_LOC_FT_MID
Y_Xfer_Window_Top
FRAME_LOC_FT_TOP
Y_Xfer_Window_Btm
Y_Xfer_Window_Mid
Y_Xfer_Window_Top
Y_XFER_OK


FRAME_LOC_HD_BTM
FRAME_LOC_FT_BTM
Y_Xfer_Window_Mid
FRAME_LOC_HD_MID
FRAME_LOC_FT_MID
Y_Xfer_Window_Top
FRAME_LOC_HD_TOP
FRAME_LOC_FT_TOP
X_PARK_OK
Y_XFER_OK
X_PARK_OK
X_XFER_OK
Y_MOUNT_XFER_OK
Y_Xfer_Window_Btm
Y_Xfer_Window_Mid
Y_Xfer_Window_Top
MASTER_Z_GO


MASTER_Z_GO
0035
Safety_Tripped_S
TON
Timer On Delay
Timer T01
Preset ?
Accum ?
EN
DN
0036
Safety_Tripped_S
Local:3:O.Pt11.Data
Local:3:O.Pt12.Data
0037
blink_on.TT
T01.TT
NEQ
Not Equal
Source A ERROR_CODE
??
Source B 0
??
X_axis.SLSActiveStatus
Local:3:O.Pt13.Data


0036
Safety_Tripped_S
XYZ_Inhibit_Timer.DN
TON
Timer On Delay
Timer T01
Preset ?
Accum ?
EN
DN
0037
Safety_Tripped_S
Local:3:O.Pt11.Data
XYZ_Inhibit_Timer.DN
Local:3:O.Pt13.Data
0038
T02.TT
T01.TT
NEQ
Not Equal
Source A ERROR_CODE
??
Source B 0
??
X_axis.SLSActiveStatus
Y_axis.SLSActiveStatus
Local:3:O.Pt14.Data

0038 blink_on.TT T01.TT Local:3:O.Pt15.Data

0039 T01.TT T01.DN Local:3:O.Pt14.Data

0040 TON Timer On Delay Timer blink_on Preset ? Accum ? EN DN

0041 blink_on.DN TON Timer On Delay Timer blink_off Preset ? Accum ? EN DN

0042 blink_off.DN blink_on RES

0043 INIT_DONE CMP Compare Expression NEXTSTATE=0

CPT Compute Dest STATE ?? Expression 0

0044 INIT_DONE CMP Compare Expression NEXTSTATE=1

CPT Compute Dest STATE

0039 Tension_Above_10N_Light Local:3:O.Pt12.Data

0040 T02.TT T01.TT Tension_T1.TT Local:3:O.Pt15.Data

0041 T01.TT T01.DN Local:3:O.Pt08.Data

0042 TON Timer On Delay Timer T02 Preset ? Accum ? EN DN

0043 T02.DN TON Timer On Delay Timer T03 Preset ? Accum ? EN DN

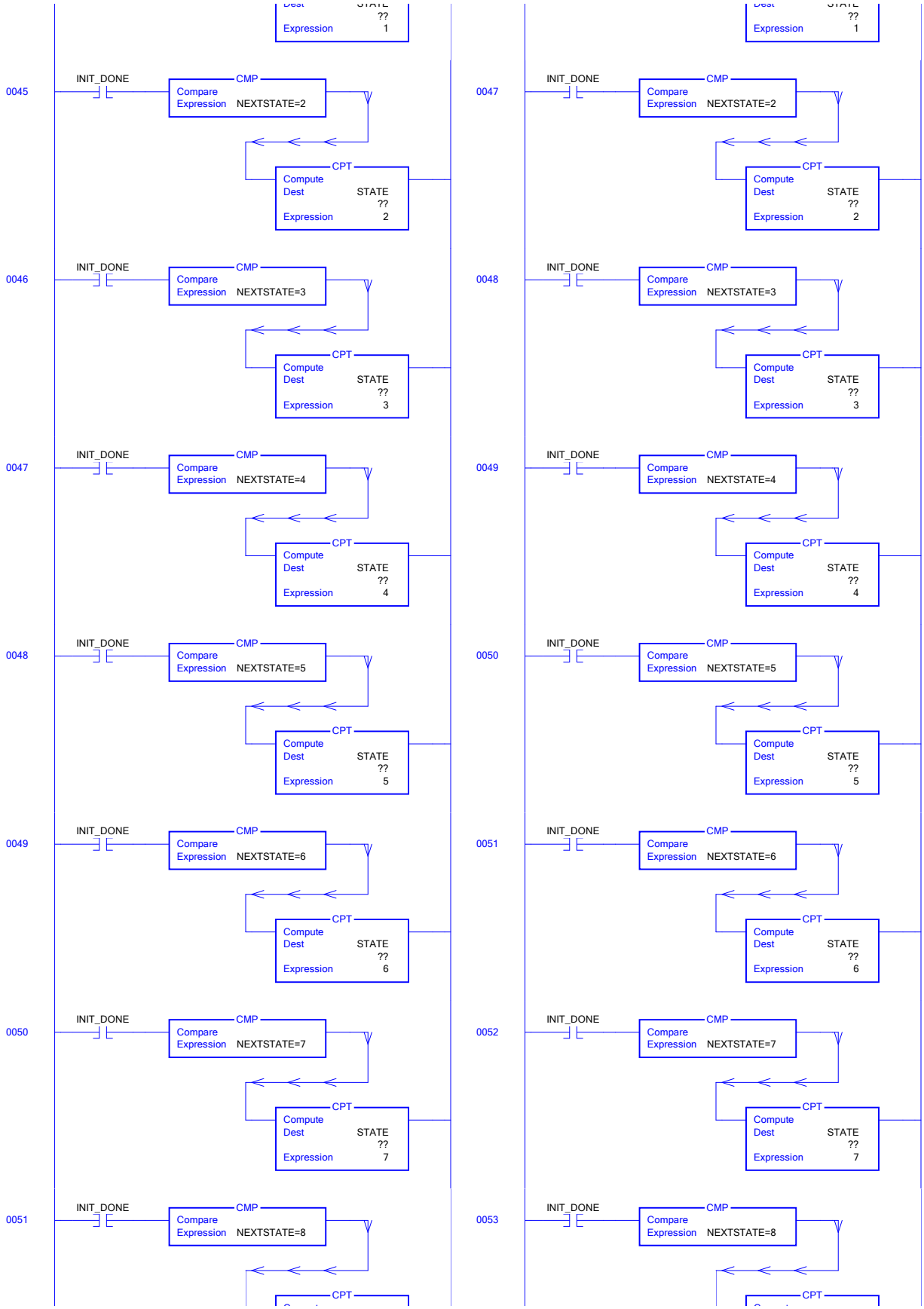
0044 T03.DN T02 RES

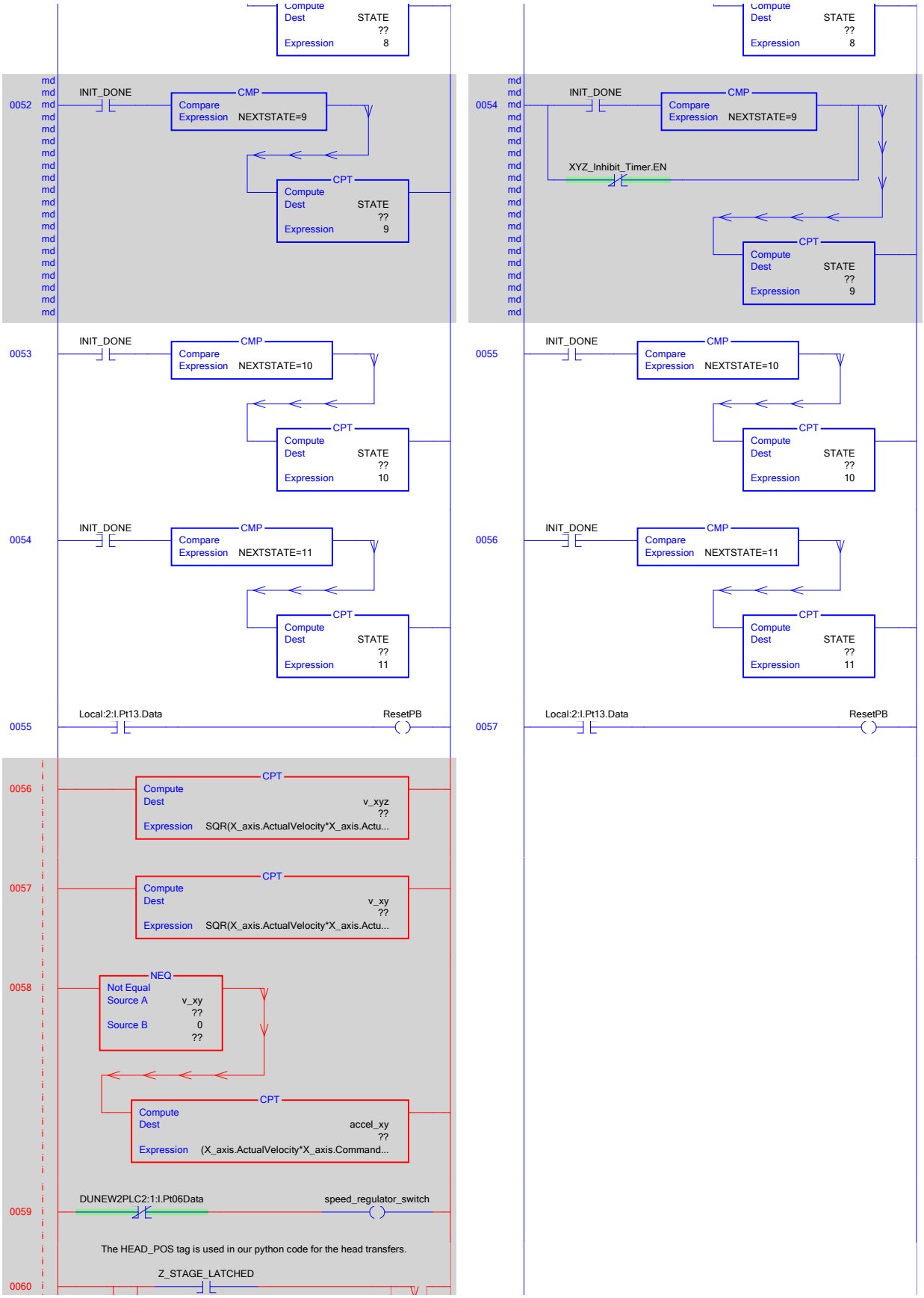
0045 INIT_DONE CMP Compare Expression NEXTSTATE=0

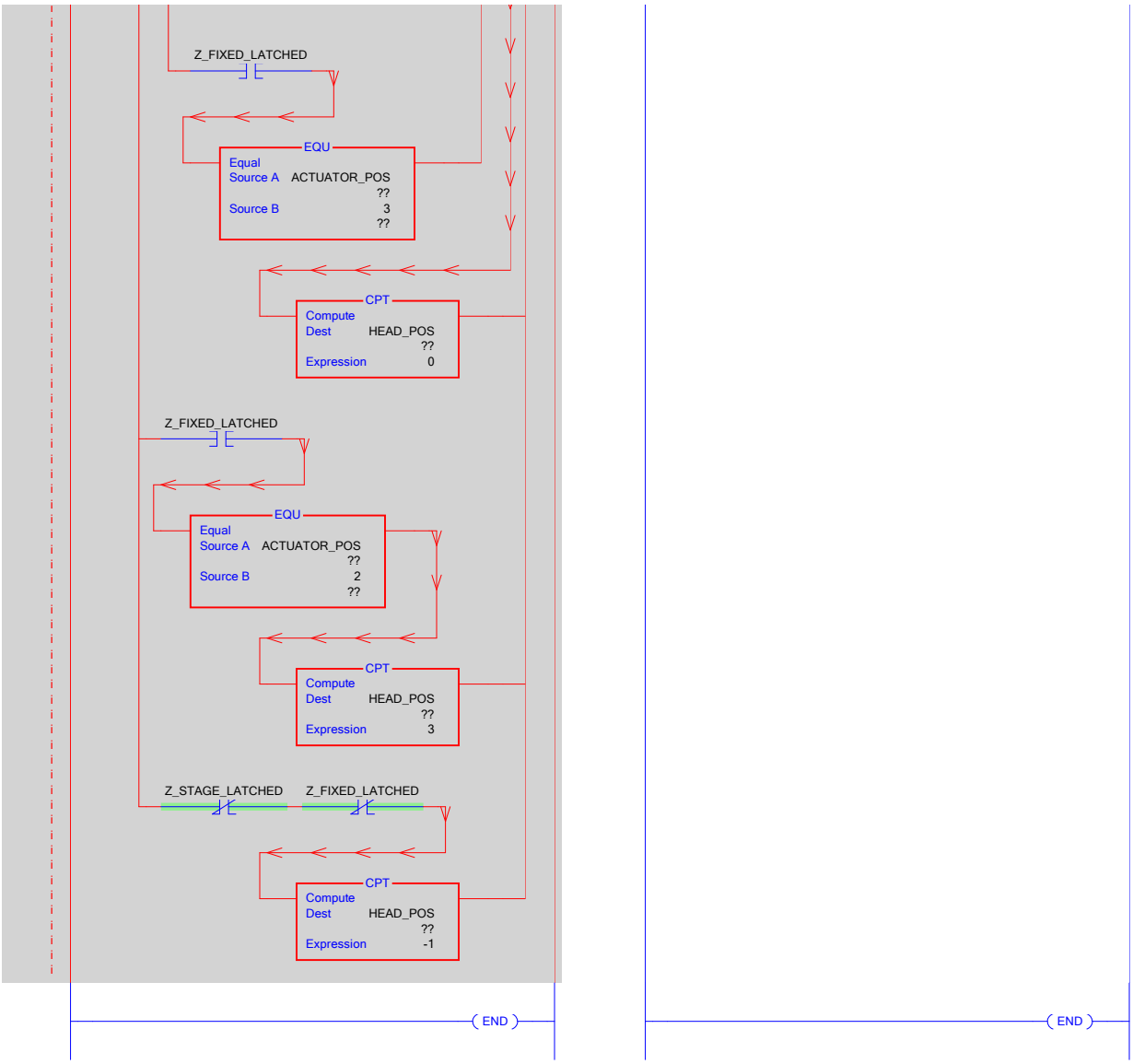
CPT Compute Dest STATE ?? Expression 0

0046 INIT_DONE CMP Compare Expression NEXTSTATE=1

CPT Compute Dest STATE


Dest STATE
??
Expression 1
0045
INIT_DONE
CMP
Compare
Expression NEXTSTATE=2
CPT
Compute
Dest STATE
??
Expression 2
0046
INIT_DONE
CMP
Compare
Expression NEXTSTATE=3
CPT
Compute
Dest STATE
??
Expression 3
0047
INIT_DONE
CMP
Compare
Expression NEXTSTATE=4
CPT
Compute
Dest STATE
??
Expression 4
0048
INIT_DONE
CMP
Compare
Expression NEXTSTATE=5
CPT
Compute
Dest STATE
??
Expression 5
0049
INIT_DONE
CMP
Compare
Expression NEXTSTATE=6
CPT
Compute
Dest STATE
??
Expression 6
0050
INIT_DONE
CMP
Compare
Expression NEXTSTATE=7
CPT
Compute
Dest STATE
??
Expression 7
0051
INIT_DONE
CMP
Compare
Expression NEXTSTATE=8
CPT
Dest STATE
??
Expression 1
0047
INIT_DONE
CMP
Compare
Expression NEXTSTATE=2
CPT
Compute
Dest STATE
??
Expression 2
0048
INIT_DONE
CMP
Compare
Expression NEXTSTATE=3
CPT
Compute
Dest STATE
??
Expression 3
0049
INIT_DONE
CMP
Compare
Expression NEXTSTATE=4
CPT
Compute
Dest STATE
??
Expression 4
0050
INIT_DONE
CMP
Compare
Expression NEXTSTATE=5
CPT
Compute
Dest STATE
??
Expression 5
0051
INIT_DONE
CMP
Compare
Expression NEXTSTATE=6
CPT
Compute
Dest STATE
??
Expression 6
0052
INIT_DONE
CMP
Compare
Expression NEXTSTATE=7
CPT
Compute
Dest STATE
??
Expression 7
0053
INIT_DONE
CMP
Compare
Expression NEXTSTATE=8
CPT


Compute
Dest STATE
??
Expression 8
0052
INIT_DONE
CMP
Compare
Expression NEXTSTATE=9
CPT
Compute
Dest STATE
??
Expression 9
0053
INIT_DONE
CMP
Compare
Expression NEXTSTATE=10
CPT
Compute
Dest STATE
??
Expression 10
0054
INIT_DONE
CMP
Compare
Expression NEXTSTATE=11
CPT
Compute
Dest STATE
??
Expression 11
0055
Local:2:I.Pt13.Data
ResetPB
0056
CPT
Compute
Dest v_xyz
??
Expression SQR(X_axis.ActualVelocity*X_axis.Actu...
0057
CPT
Compute
Dest v_xy
??
Expression SQR(X_axis.ActualVelocity*X_axis.Actu...
0058
NEQ
Not Equal
Source A v_xy
??
Source B 0
??
CPT
Compute
Dest accel_xy
??
Expression (X_axis.ActualVelocity*X_axis.Command...
0059
DUNEW2PLC2:1:I.Pt06Data
speed_regulator_switch
The HEAD_POS tag is used in our python code for the head transfers.
0060
Z_STAGE_LATCHED
Compute
Dest STATE
??
Expression 8
0054
INIT_DONE
CMP
Compare
Expression NEXTSTATE=9
XYZ_Inhibit_Timer.EN
CPT
Compute
Dest STATE
??
Expression 9
0055
INIT_DONE
CMP
Compare
Expression NEXTSTATE=10
CPT
Compute
Dest STATE
??
Expression 10
0056
INIT_DONE
CMP
Compare
Expression NEXTSTATE=11
CPT
Compute
Dest STATE
??
Expression 11
0057
Local:2:I.Pt13.Data
ResetPB


Z_FIXED_LATCHED
EQU
Equal
Source A ACTUATOR_POS
??
Source B 3
??
CPT
Compute
Dest HEAD_POS
??
Expression 0
Z_FIXED_LATCHED
EQU
Equal
Source A ACTUATOR_POS
??
Source B 2
??
CPT
Compute
Dest HEAD_POS
??
Expression 3
Z_STAGE_LATCHED
Z_FIXED_LATCHED
CPT
Compute
Dest HEAD_POS
??
Expression -1
END
END

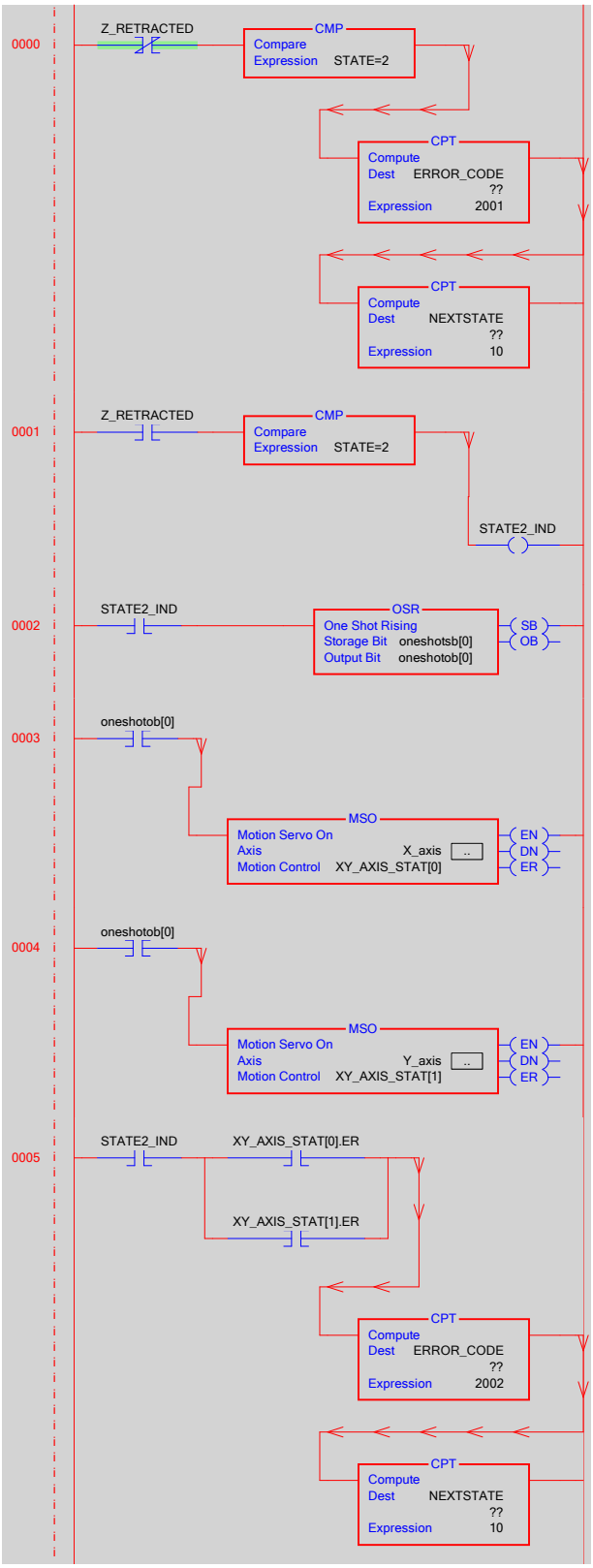
MoveXY_State_2_3 Parameters and Local Tags

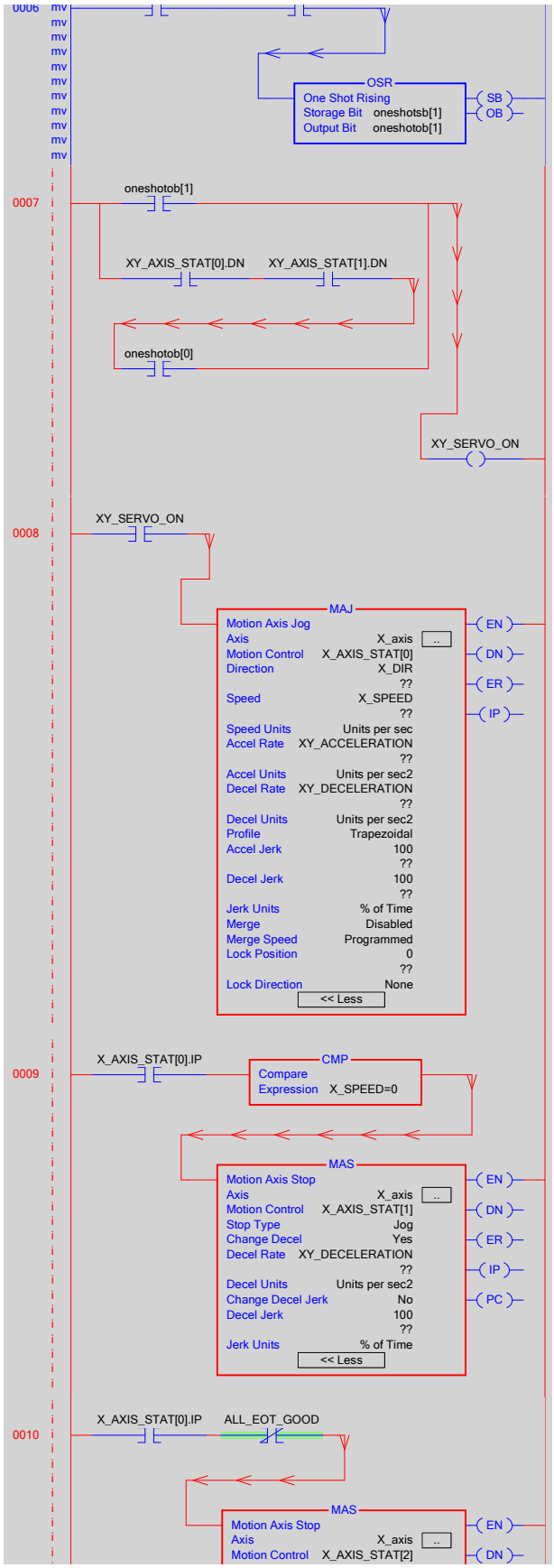
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
AXIS	---	---	---	---		INT		0
Current_x	---	---	---	---		REAL		0.0057
Current_y	---	---	---	---		DINT		0
d_raw		REAL		0.0	---	---	---	---
dx		REAL		0.0039	---	---	---	---
dy		REAL		0.0017	---	---	---	---
eot_triggered		BOOL		0	---	---	---	---
ERROR	---	---	---	---		BOOL[32]		
k		REAL		211538.47	---	---	---	---
MANUAL_SW	---	---	---	---		BOOL		0
MANUAL_SW1	---	---	---	---		BOOL		0
MANUAL_SW2	---	---	---	---		BOOL		0
max_integral		DINT		1000	---	---	---	---
mike	---	---	---	---		BOOL		0
min_integral		DINT		-1000	---	---	---	---
min_regulated_speed		DINT		100	---	---	---	---
MOTOR_SERVO_ON	---	---	---	---		BOOL		0
MountHeight	---	---	---	---		REAL		1330.0
MOVING	---	---	---	---		BOOL		0
MOVINGSTATE	---	---	---	---		DINT		0
near_ending		BOOL	True if distance to xy-move target is less than the required stopping distance	0	---	---	---	---
near_ending_trigger		BOOL		0	---	---	---	---
NOTMOVING	---	---	---	---		BOOL		0
oneshottestob	---	---	---	---		BOOL[32]		
oneshottestsb	---	---	---	---		BOOL[32]		
OUTPUTBIT	---	---	---	---		BOOL[32]		
PB0	---	---	---	---		BOOL		0
PB1	---	---	---	---		BOOL		0
regulated_speed		REAL		899.9713	---	---	---	---
regulator_loop_timer		TIMER			---	---	---	---
set	---	---	---	---		BOOL		0
speed_set_to_zero		BOOL		0	---	---	---	---
speed_tension_setpoint		REAL		8.5	---	---	---	---
starting_x		REAL		-0.0039	---	---	---	---
starting_y		REAL		-0.0017	---	---	---	---
STATE2_IND	---	---	---	---		BOOL		0

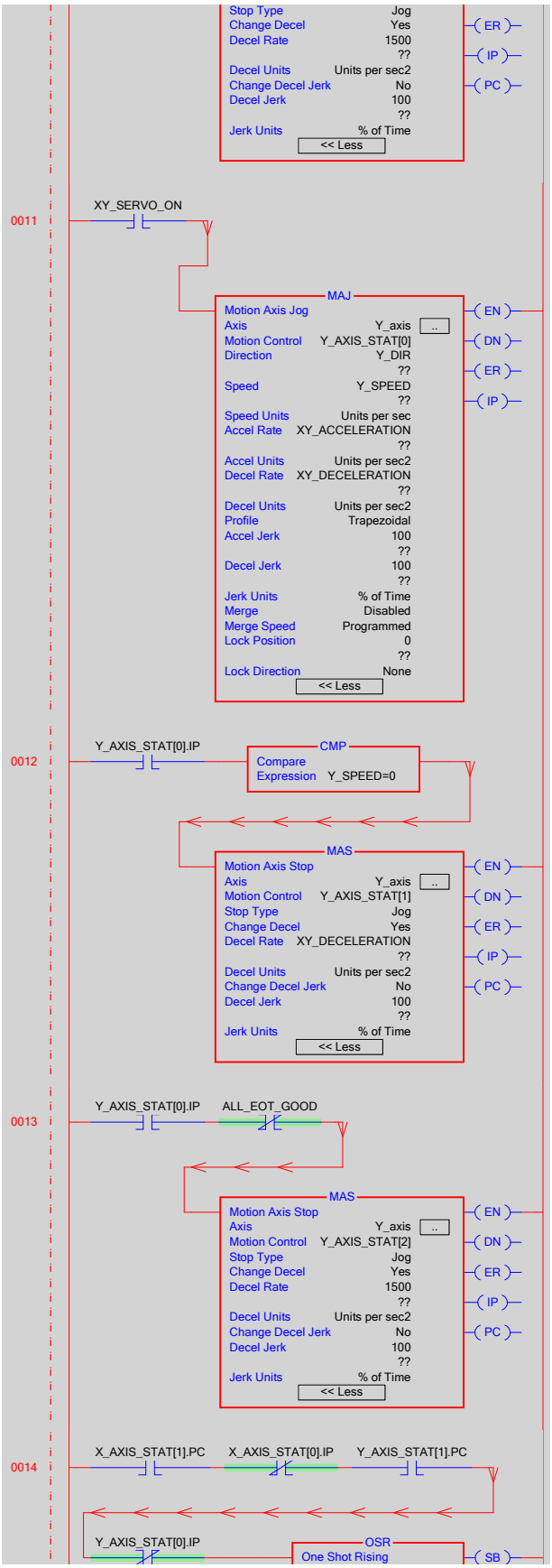
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
stopping_distance		REAL	Calculate the distance required to perform a jerk-limited stop at the gurrent jerk, acceleration, and speed	0.0	---	---	---	---
stopping_ratio		REAL		0.2532453	---	---	---	---
STOPPING_TIME	---	---	---	---		TIMER		
STORAGEBIT	---	---	---	---		BOOL[32]		
SW0	---	---	---	---		BOOL		0
trigger_xy_move		BOOL		0	---	---	---	---
v_max		REAL		899.9713	---	---	---	---
X_AXIS_STAT	---	---	---	---		MOTION_INSTRUCTION [32]		
x_dist_to_target		REAL		0.6653	---	---	---	---
X_max_speed		REAL		825.0	---	---	---	---
x_time		REAL		211538.47	---	---	---	---
xy_accel_jerk		DINT		1500	---	---	---	---
xy_d_alpha		REAL		1.0	---	---	---	---
xy_d_dt		DINT		50	---	---	---	---
xy_d_term		REAL		0.0	---	---	---	---
xy_d_term_prev		REAL		0.0	---	---	---	---
xy_d_timer		TIMER			---	---	---	---
xy_decel_jerk		DINT		2000	---	---	---	---
xy_default_speed		DINT		0	---	---	---	---
xy_dist_to_target		REAL		0.66530246	---	---	---	---
xy_dt		DINT		1	---	---	---	---
xy_error		REAL		6.0954127	---	---	---	---
xy_error_prev		REAL		6.0985594	---	---	---	---
xy_i_term		REAL		900.0352	---	---	---	---
xy_kd		REAL		0.0	---	---	---	---
xy_ki		REAL		300.0	---	---	---	---
xy_kp		REAL		0.0	---	---	---	---
xy_p_term		REAL		0.0	---	---	---	---
XY_SERVO_ON	---	---	---	---		BOOL		0
Y_AXIS_STAT	---	---	---	---		MOTION_INSTRUCTION [32]		
y_dist_to_target		REAL		-0.0018	---	---	---	---
Y_estop_active_pos	---	---	---	---		REAL		0.0
Y_max_speed		REAL		600.0	---	---	---	---

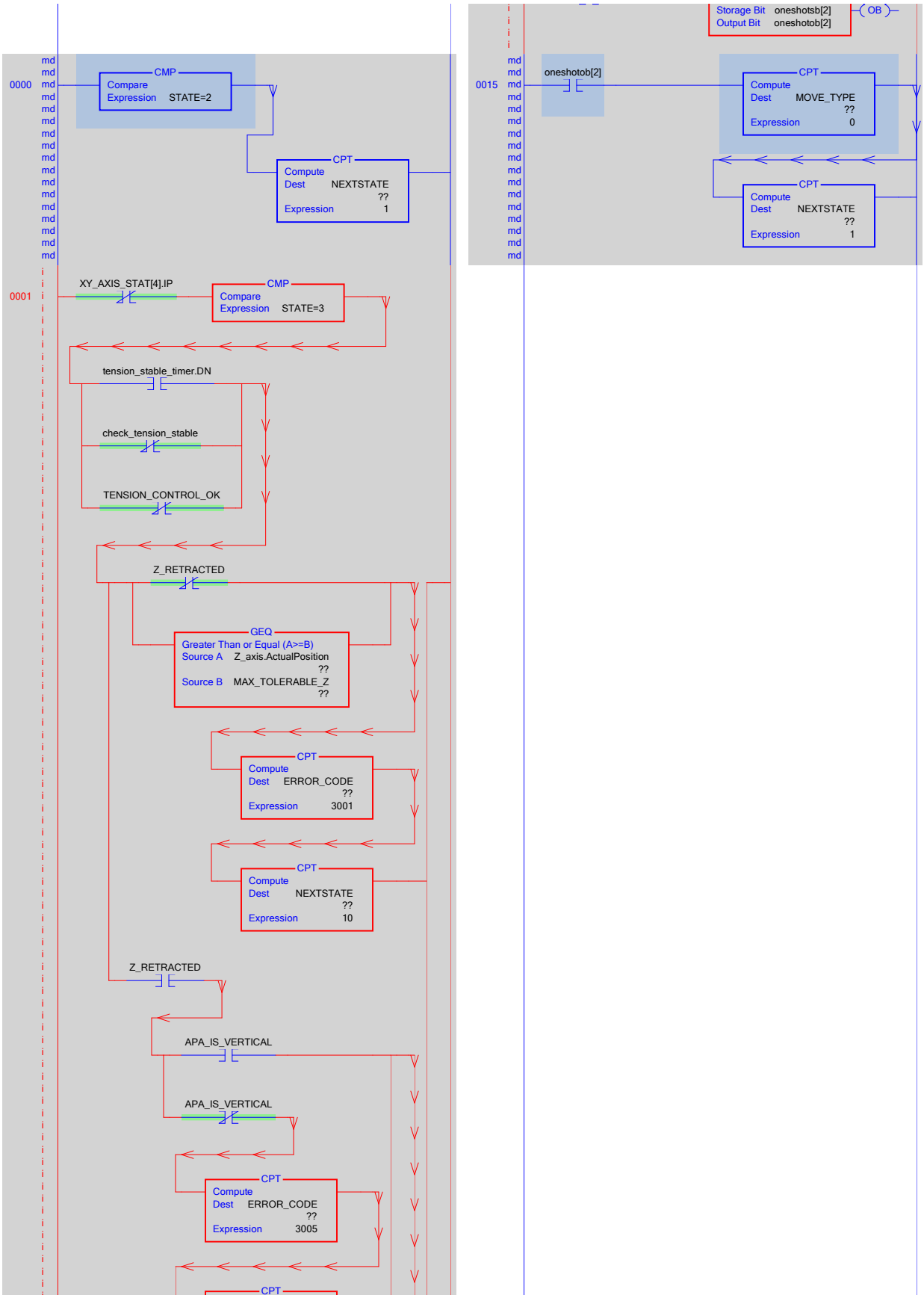
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
y_time		REAL		352941.16	---	---	---	---
YOHOHO	---	---	---	---		BOOL		0

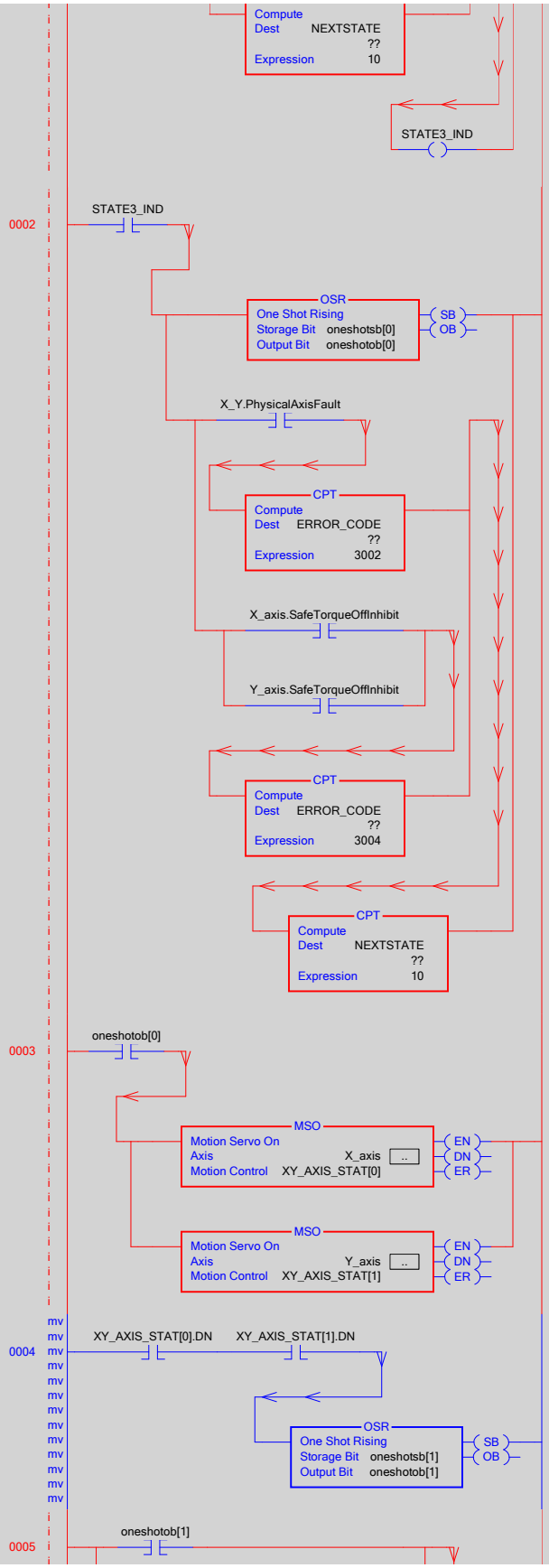
Program: MoveXY_State_2_3
Routine: MoveXY

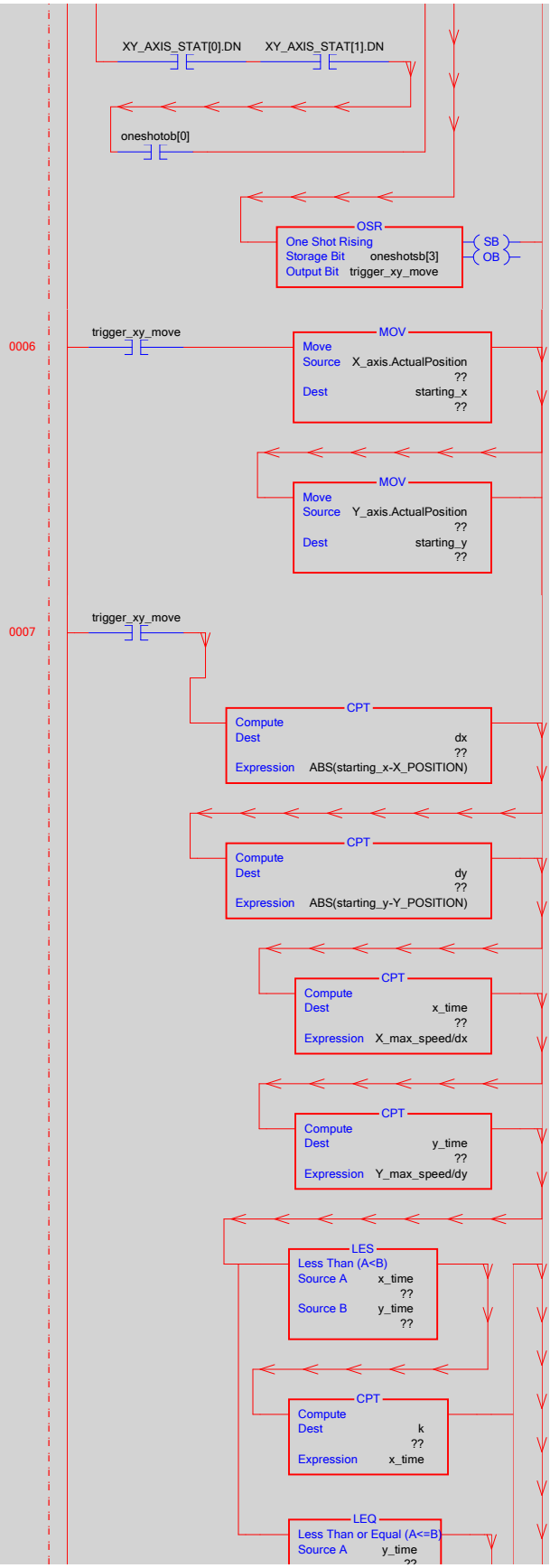


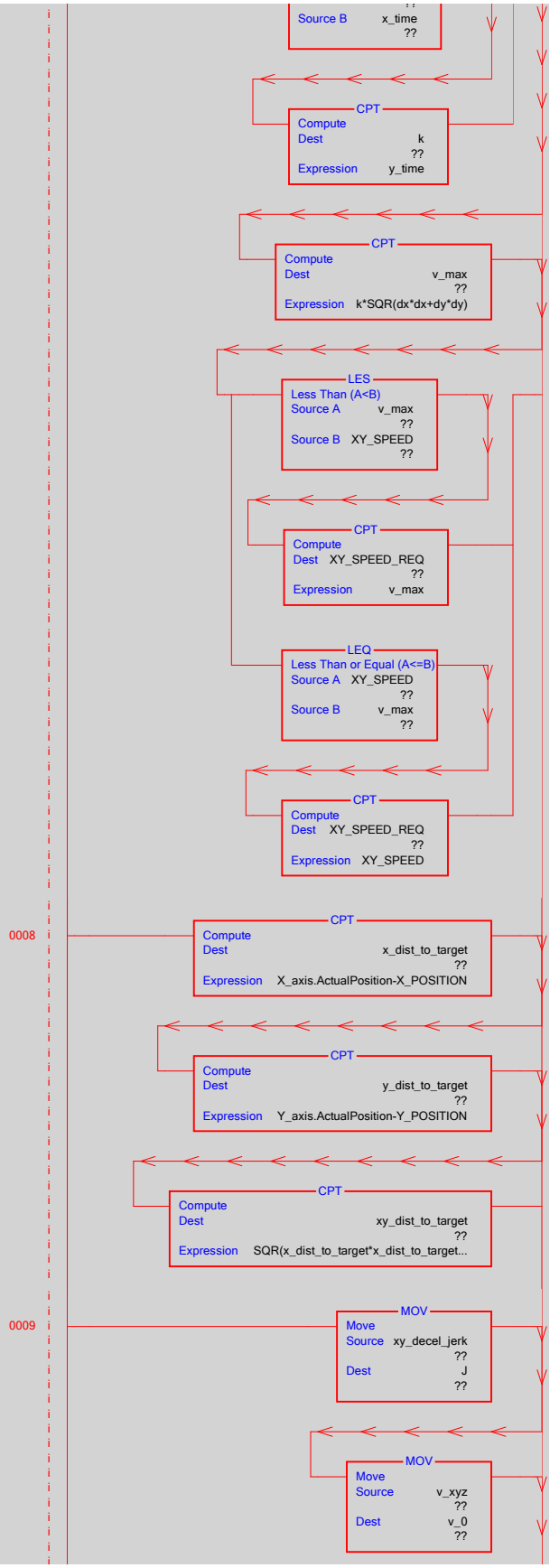

0006
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB
0007
oneshotob[1]
XY_AXIS_STAT[0].DN
XY_AXIS_STAT[1].DN
oneshotob[0]
XY_SERVO_ON
0008
XY_SERVO_ON
MAJ
Motion Axis Jog
Axis X_axis
Motion Control X_AXIS_STAT[0]
Direction X_DIR
??
Speed X_SPEED
??
Speed Units Units per sec
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
<< Less
EN
DN
ER
IP
0009
X_AXIS_STAT[0].IP
CMP
Compare
Expression X_SPEED=0
MAS
Motion Axis Stop
Axis X_axis
Motion Control X_AXIS_STAT[1]
Stop Type Jog
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0010
X_AXIS_STAT[0].IP
ALL_EOT_GOOD
MAS
Motion Axis Stop
Axis X_axis
Motion Control X_AXIS_STAT[2]
EN
DN

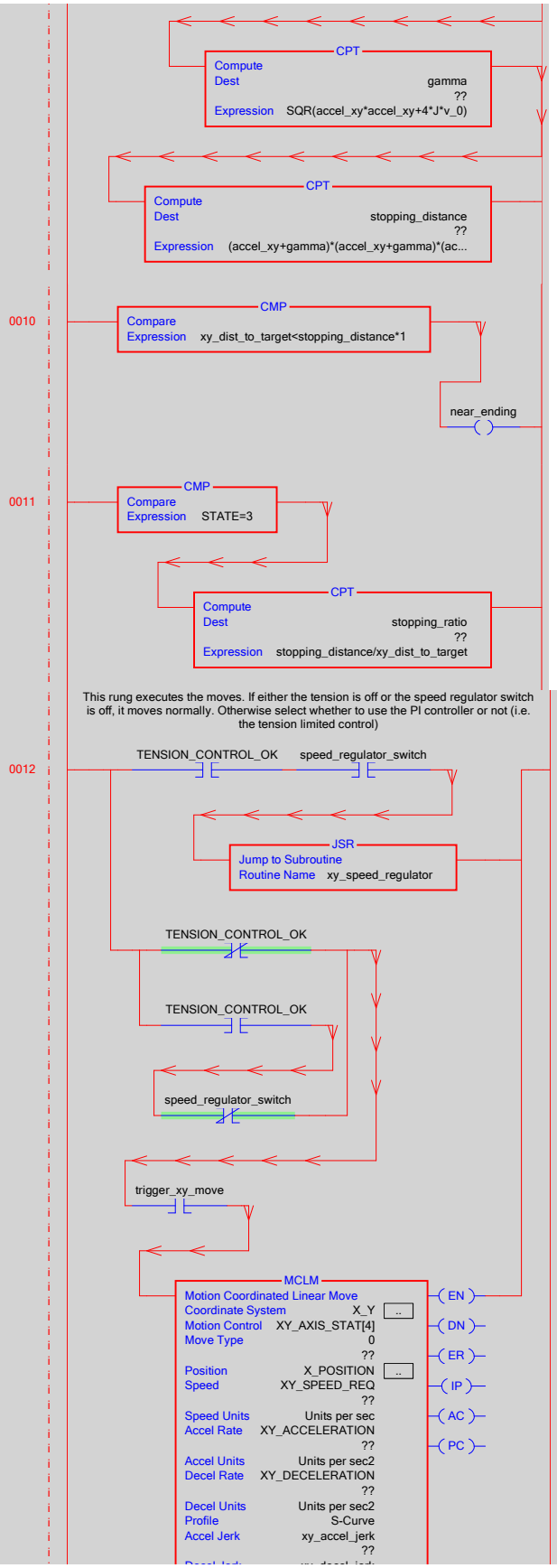

Stop Type Jog
Change Decel Yes
Decel Rate 1500
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
ER
IP
PC
0011
XY_SERVO_ON
MAJ
Motion Axis Jog
Axis Y_axis
Motion Control Y_AXIS_STAT[0]
Direction Y_DIR
??
Speed Y_SPEED
??
Speed Units Units per sec
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
<< Less
EN
DN
ER
IP
0012
Y_AXIS_STAT[0].IP
CMP
Compare
Expression Y_SPEED=0
MAS
Motion Axis Stop
Axis Y_axis
Motion Control Y_AXIS_STAT[1]
Stop Type Jog
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0013
Y_AXIS_STAT[0].IP
ALL_EOT_GOOD
MAS
Motion Axis Stop
Axis Y_axis
Motion Control Y_AXIS_STAT[2]
Stop Type Jog
Change Decel Yes
Decel Rate 1500
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0014
X_AXIS_STAT[1].PC
X_AXIS_STAT[0].IP
Y_AXIS_STAT[1].PC
Y_AXIS_STAT[0].IP
OSR
One Shot Rising
SB

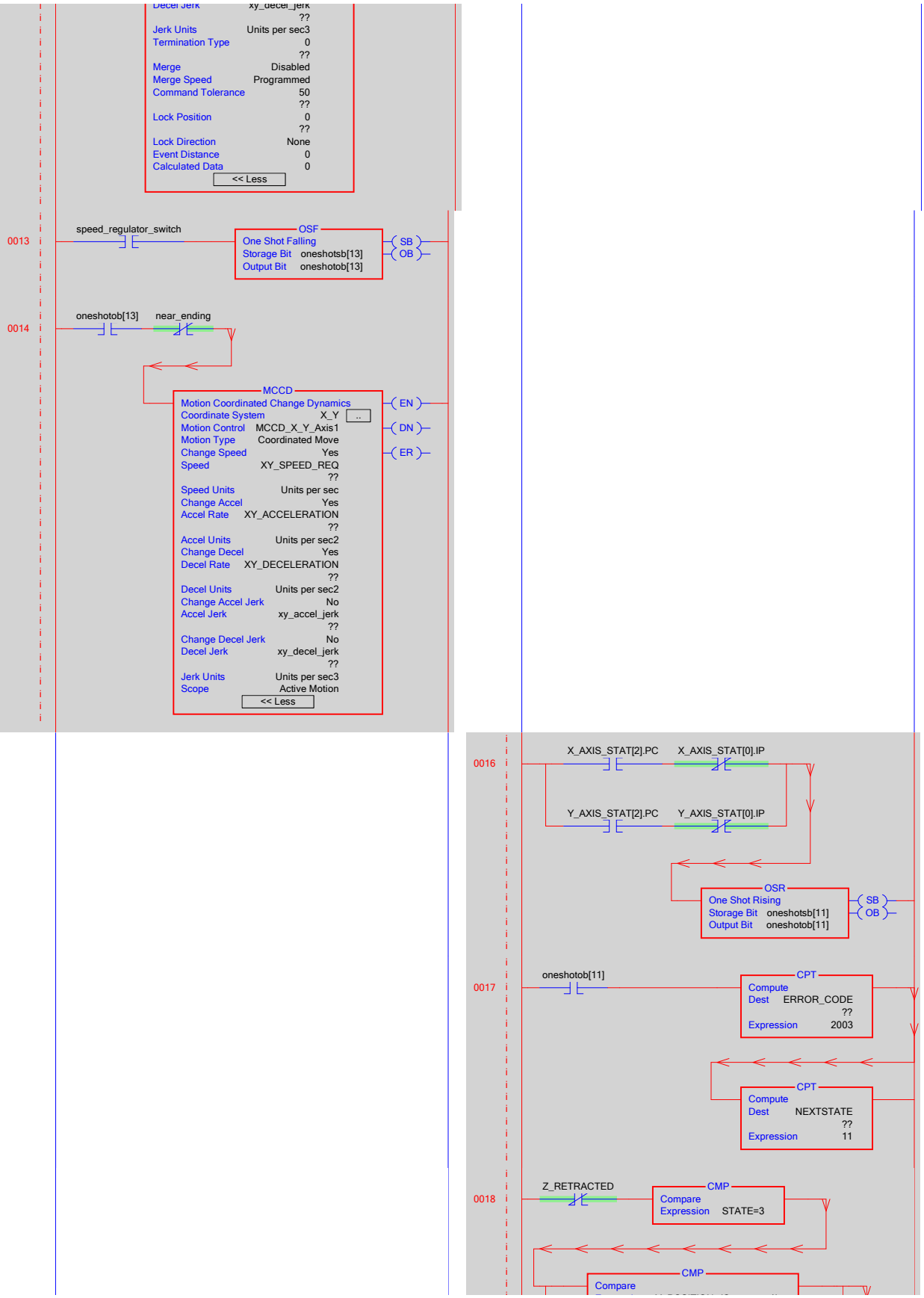

0000
CMP
Compare
Expression STATE=2
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0001
XY_AXIS_STAT[4].IP
CMP
Compare
Expression STATE=3
tension_stable_timer.DN
check_tension_stable
TENSION_CONTROL_OK
Z_RETRACTED
GEQ
Greater Than or Equal (A>=B)
Source A Z_axis.ActualPosition
??
Source B MAX_TOLERABLE_Z
??
CPT
Compute
Dest ERROR_CODE
??
Expression 3001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
Z_RETRACTED
APA_IS_VERTICAL
APA_IS_VERTICAL
CPT
Compute
Dest ERROR_CODE
??
Expression 3005
CPT
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
OB
0015
oneshotob[2]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1

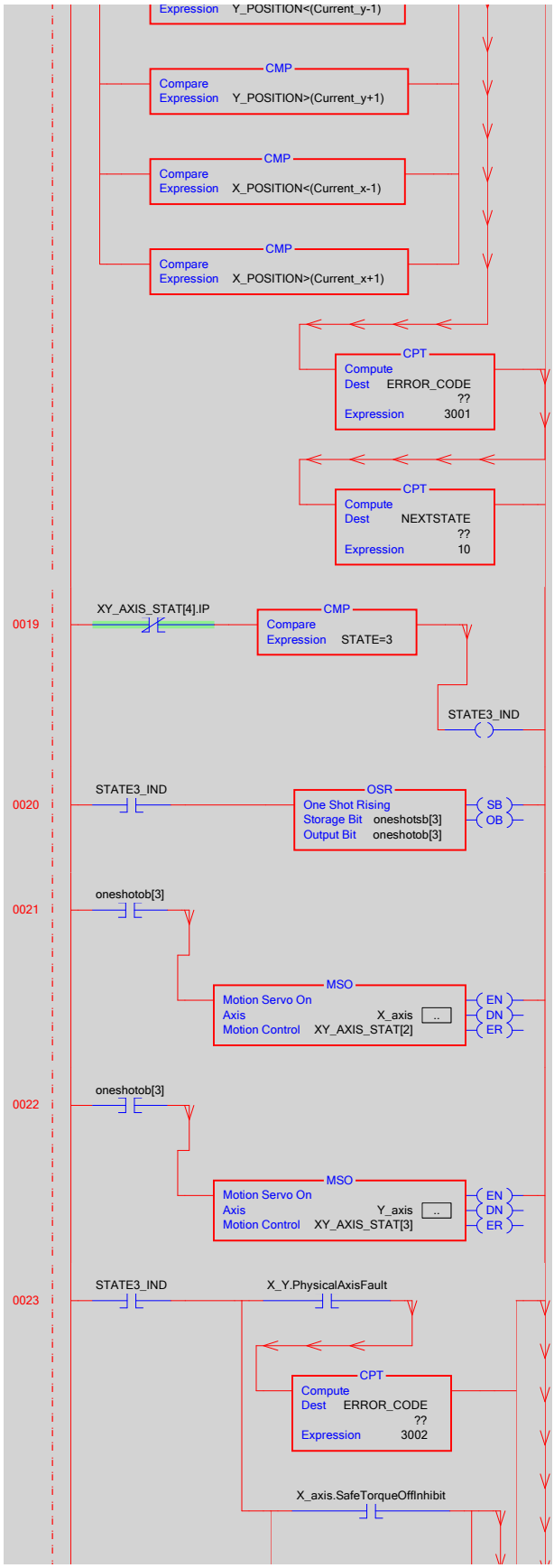

Compute
Dest NEXTSTATE
??
Expression 10
STATE3_IND
0002
STATE3_IND
OSR
One Shot Rising
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB
X_Y.PhysicalAxisFault
CPT
Compute
Dest ERROR_CODE
??
Expression 3002
X_axis.SafeTorqueOffInhibit
Y_axis.SafeTorqueOffInhibit
CPT
Compute
Dest ERROR_CODE
??
Expression 3004
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0003
oneshotob[0]
MSO
Motion Servo On
Axis X_axis
Motion Control XY_AXIS_STAT[0]
EN
DN
ER
MSO
Motion Servo On
Axis Y_axis
Motion Control XY_AXIS_STAT[1]
EN
DN
ER
0004
XY_AXIS_STAT[0].DN
XY_AXIS_STAT[1].DN
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB
0005
oneshotob[1]

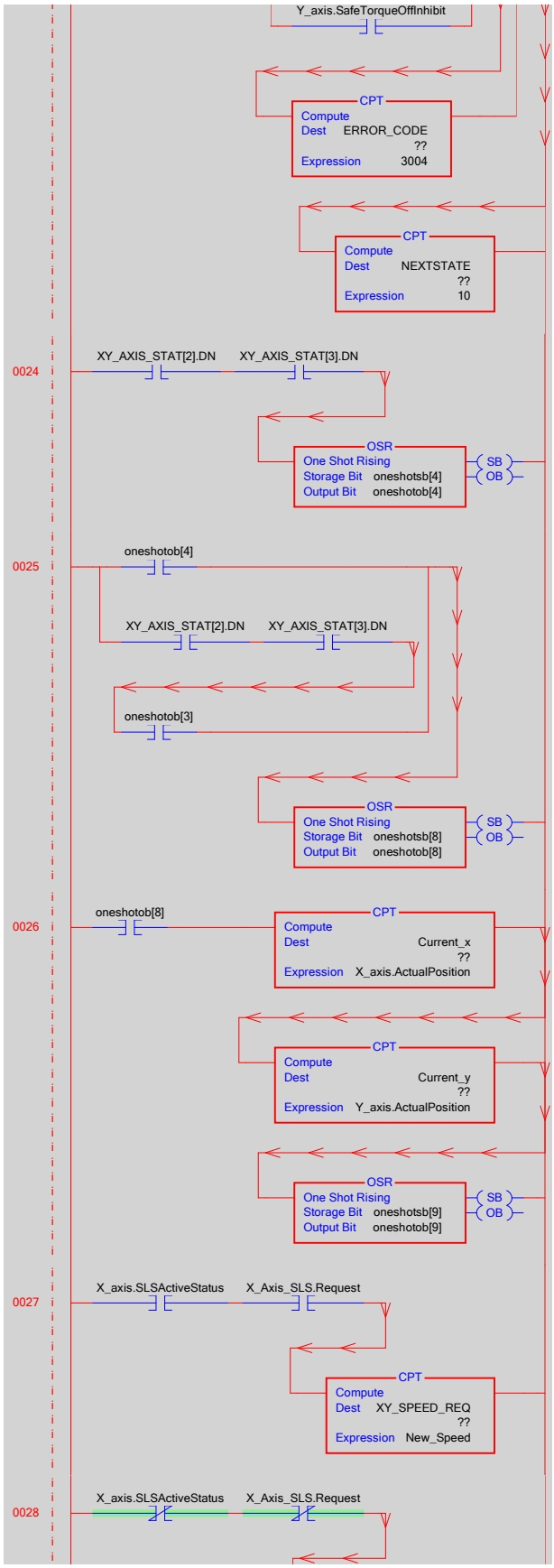

XY_AXIS_STAT[0].DN
XY_AXIS_STAT[1].DN
oneshotob[0]
OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit trigger_xy_move
SB
OB
0006
trigger_xy_move
MOV
Move
Source X_axis.ActualPosition
??
Dest starting_x
??
MOV
Move
Source Y_axis.ActualPosition
??
Dest starting_y
??
0007
trigger_xy_move
CPT
Compute
Dest dx
??
Expression ABS(starting_x-X_POSITION)
CPT
Compute
Dest dy
??
Expression ABS(starting_y-Y_POSITION)
CPT
Compute
Dest x_time
??
Expression X_max_speed/dx
CPT
Compute
Dest y_time
??
Expression Y_max_speed/dy
LES
Less Than (A<B)
Source A x_time
??
Source B y_time
??
CPT
Compute
Dest k
??
Expression x_time
LEQ
Less Than or Equal (A<=B)
Source A y_time

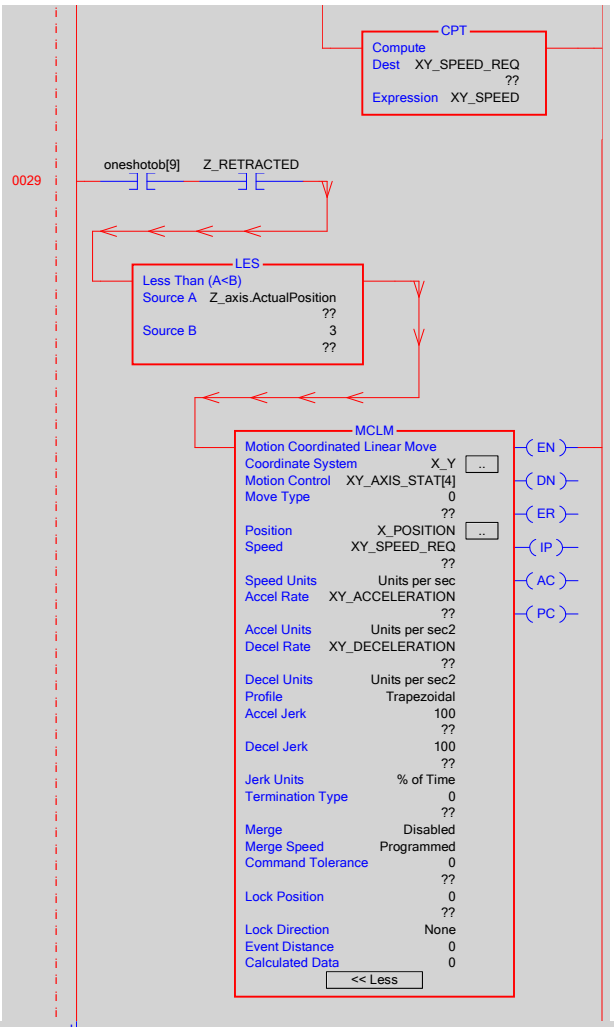

Source B x_time
??
CPT
Compute
Dest k
??
Expression y_time
CPT
Compute
Dest v_max
??
Expression k*SQR(dx*dx+dy*dy)
LES
Less Than (A<B)
Source A v_max
??
Source B XY_SPEED
??
CPT
Compute
Dest XY_SPEED_REQ
??
Expression v_max
LEQ
Less Than or Equal (A<=B)
Source A XY_SPEED
??
Source B v_max
??
CPT
Compute
Dest XY_SPEED_REQ
??
Expression XY_SPEED
0008
CPT
Compute
Dest x_dist_to_target
??
Expression X_axis.ActualPosition-X_POSITION
CPT
Compute
Dest y_dist_to_target
??
Expression Y_axis.ActualPosition-Y_POSITION
CPT
Compute
Dest xy_dist_to_target
??
Expression SQR(x_dist_to_target*x_dist_to_target...
0009
MOV
Move
Source xy_decel_jerk
??
Dest J
??
MOV
Move
Source v_xyz
??
Dest v_0
??

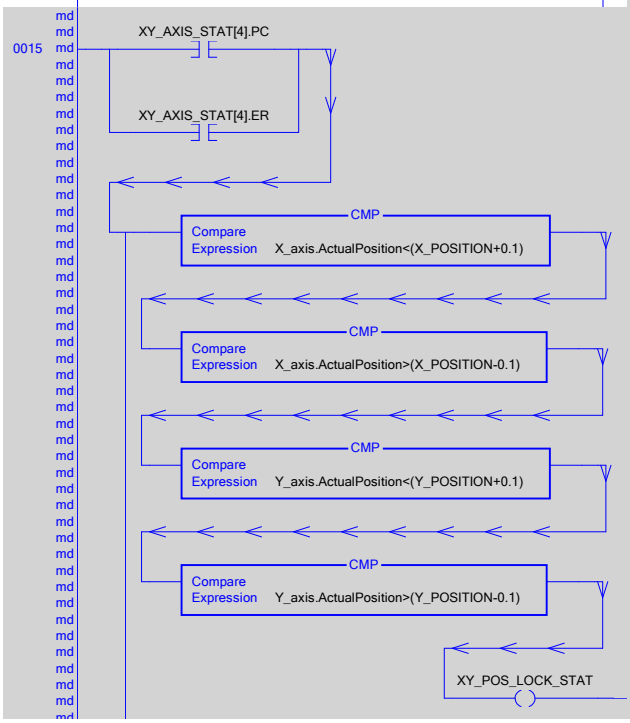

CPT
Compute
Dest gamma
??
Expression SQR(accel_xy*accel_xy+4*J*v_0)
CPT
Compute
Dest stopping_distance
??
Expression (accel_xy+gamma)*(accel_xy+gamma)*(ac...
0010
CMP
Compare
Expression xy_dist_to_target<stopping_distance*1
near_ending
0011
CMP
Compare
Expression STATE=3
CPT
Compute
Dest stopping_ratio
??
Expression stopping_distance/xy_dist_to_target
This rung executes the moves. If either the tension is off or the speed regulator switch is off, it moves normally. Otherwise select whether to use the PI controller or not (i.e. the tension limited control)
0012
TENSION_CONTROL_OK
speed_regulator_switch
JSR
Jump to Subroutine
Routine Name xy_speed_regulator
TENSION_CONTROL_OK
TENSION_CONTROL_OK
speed_regulator_switch
trigger_xy_move
MCLM
Motion Coordinated Linear Move
Coordinate System X_Y
Motion Control XY_AXIS_STAT[4]
Move Type 0
??
Position X_POSITION
Speed XY_SPEED_REQ
??
Speed Units Units per sec
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Profile S-Curve
Accel Jerk xy_accel_jerk
??
EN
DN
ER
IP
AC
PC

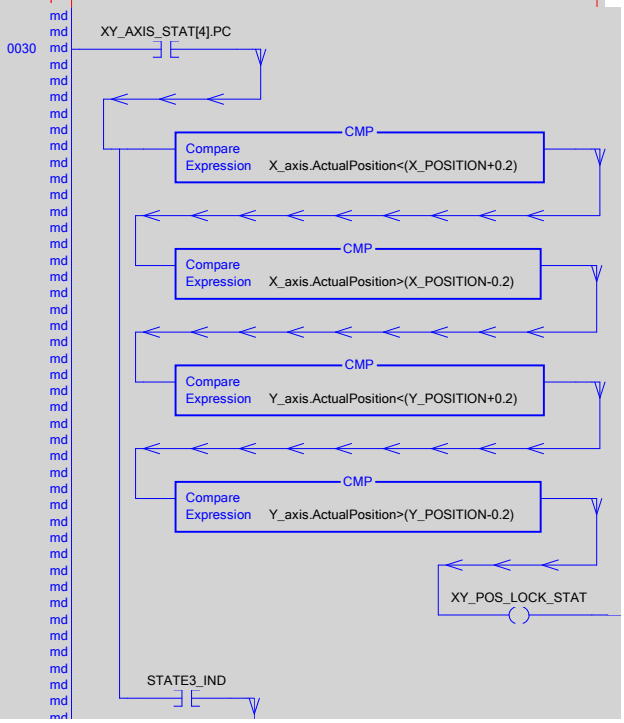

Decel Jerk xy_decel_jerk
??
Jerk Units Units per sec3
Termination Type 0
??
Merge Disabled
Merge Speed Programmed
Command Tolerance 50
??
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
0013
speed_regulator_switch
OSF
One Shot Falling
Storage Bit oneshotsb[13]
Output Bit oneshotob[13]
SB
OB
0014
oneshotob[13]
near_ending
MCCD
Motion Coordinated Change Dynamics
Coordinate System X_Y
Motion Control MCCD_X_Y_Axis1
Motion Type Coordinated Move
Change Speed Yes
Speed XY_SPEED_REQ
??
Speed Units Units per sec
Change Accel Yes
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Accel Jerk No
Accel Jerk xy_accel_jerk
??
Change Decel Jerk No
Decel Jerk xy_decel_jerk
??
Jerk Units Units per sec3
Scope Active Motion
<< Less
EN
DN
ER
0016
X_AXIS_STAT[2].PC
X_AXIS_STAT[0].IP
Y_AXIS_STAT[2].PC
Y_AXIS_STAT[0].IP
OSR
One Shot Rising
Storage Bit oneshotsb[11]
Output Bit oneshotob[11]
SB
OB
0017
oneshotob[11]
CPT
Compute
Dest ERROR_CODE
??
Expression 2003
CPT
Compute
Dest NEXTSTATE
??
Expression 11
0018
Z_RETRACTED
CMP
Compare
Expression STATE=3
CMP
Compare

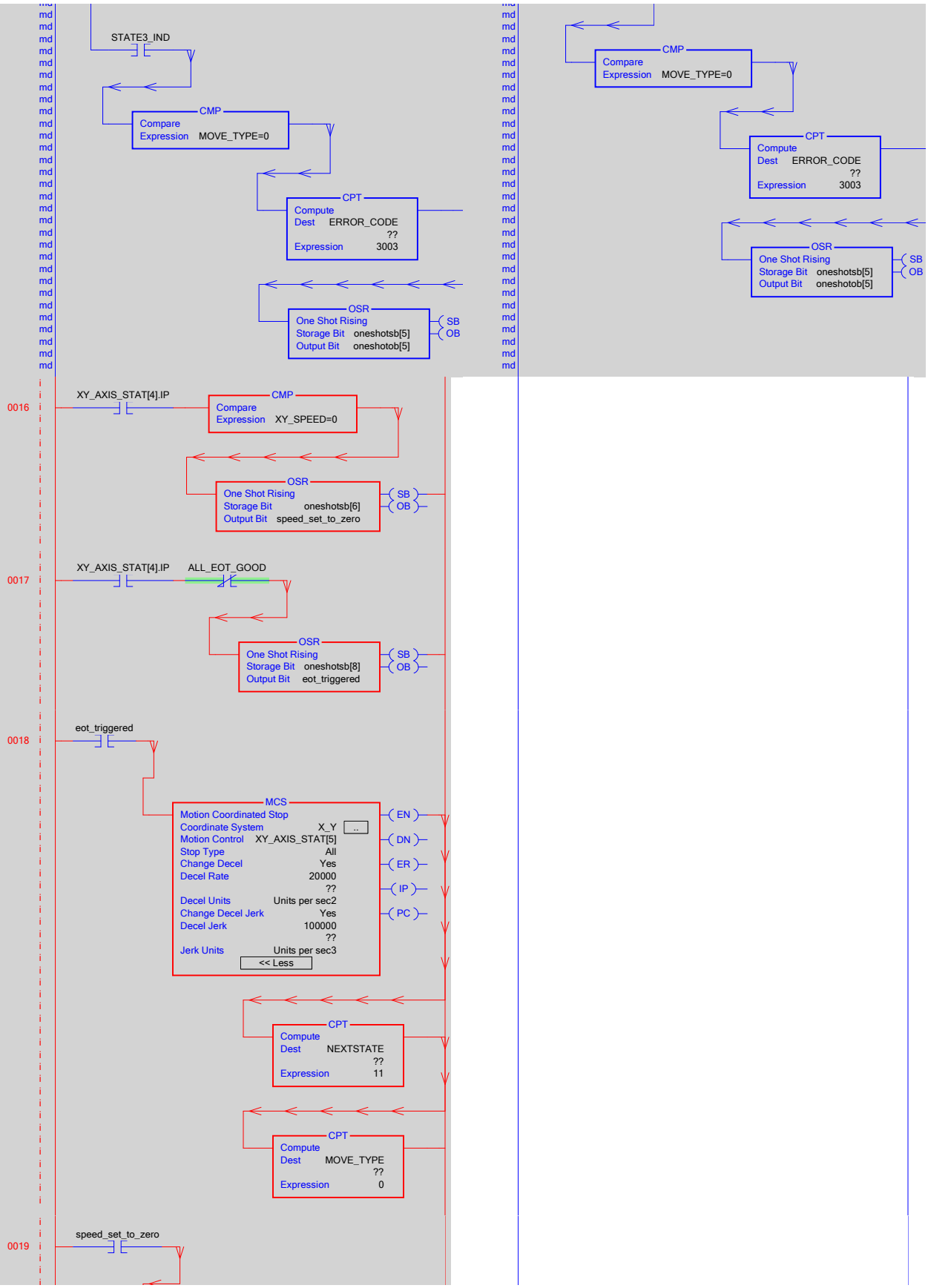

Expression Y_POSITION<(Current_y-1)
CMP
Compare
Expression Y_POSITION>(Current_y+1)
CMP
Compare
Expression X_POSITION<(Current_x-1)
CMP
Compare
Expression X_POSITION>(Current_x+1)
CPT
Compute
Dest ERROR_CODE
??
Expression 3001
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0019
XY_AXIS_STAT[4].IP
CMP
Compare
Expression STATE=3
STATE3_IND
0020
STATE3_IND
OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB
0021
oneshotob[3]
MSO
Motion Servo On
Axis X_axis
Motion Control XY_AXIS_STAT[2]
EN
DN
ER
0022
oneshotob[3]
MSO
Motion Servo On
Axis Y_axis
Motion Control XY_AXIS_STAT[3]
EN
DN
ER
0023
STATE3_IND
X_Y.PhysicalAxisFault
CPT
Compute
Dest ERROR_CODE
??
Expression 3002
X_axis.SafeTorqueOffInhibit

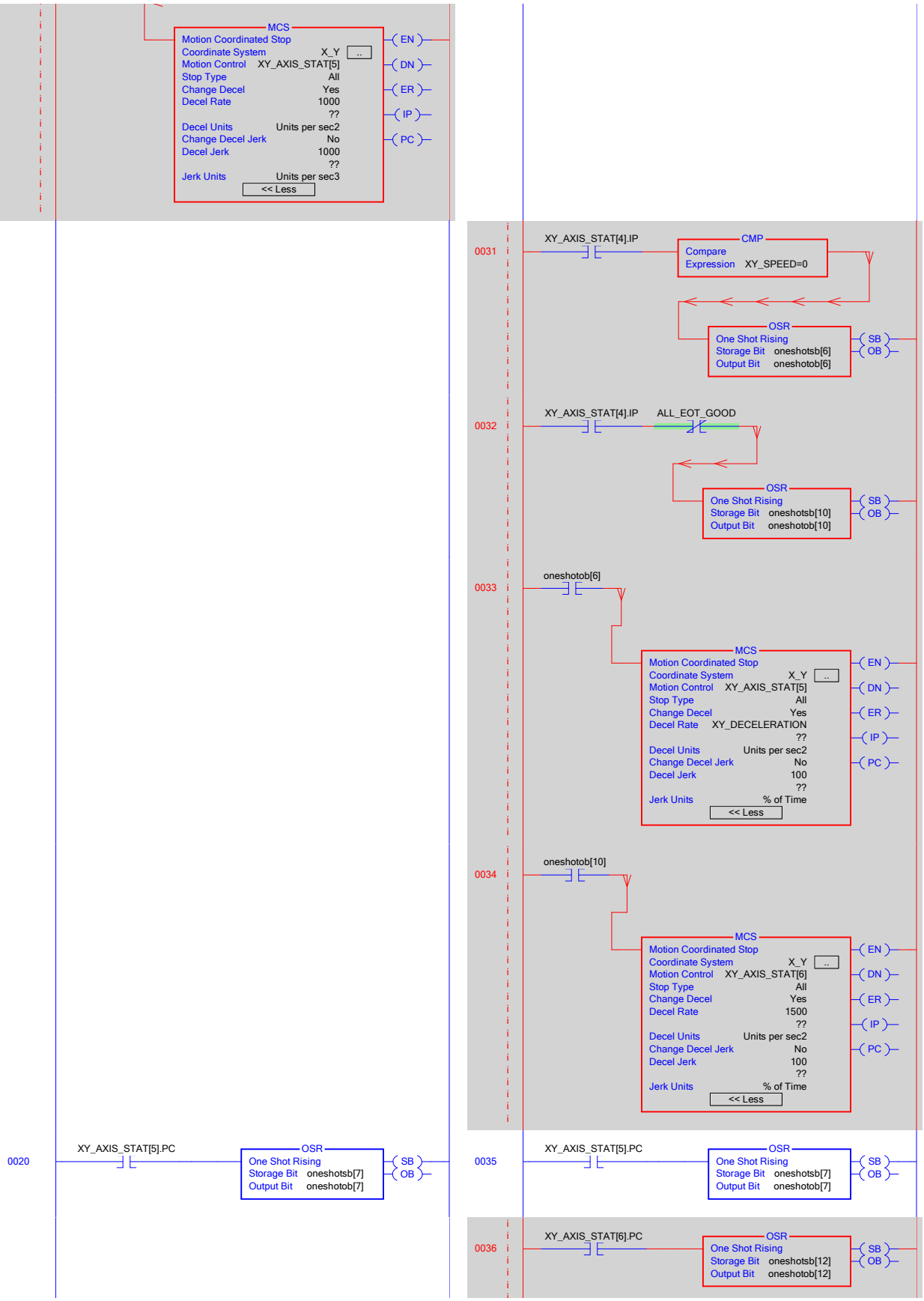

Y_axis.SafeTorqueOffInhibit
CPT
Compute
Dest ERROR_CODE
??
Expression 3004
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0024
XY_AXIS_STAT[2].DN
XY_AXIS_STAT[3].DN
OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB
0025
oneshotob[4]
XY_AXIS_STAT[2].DN
XY_AXIS_STAT[3].DN
oneshotob[3]
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0026
oneshotob[8]
CPT
Compute
Dest Current_x
??
Expression X_axis.ActualPosition
CPT
Compute
Dest Current_y
??
Expression Y_axis.ActualPosition
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB
0027
X_axis.SLSActiveStatus
X_Axis_SLS.Request
CPT
Compute
Dest XY_SPEED_REQ
??
Expression New_Speed
0028
X_axis.SLSActiveStatus
X_Axis_SLS.Request

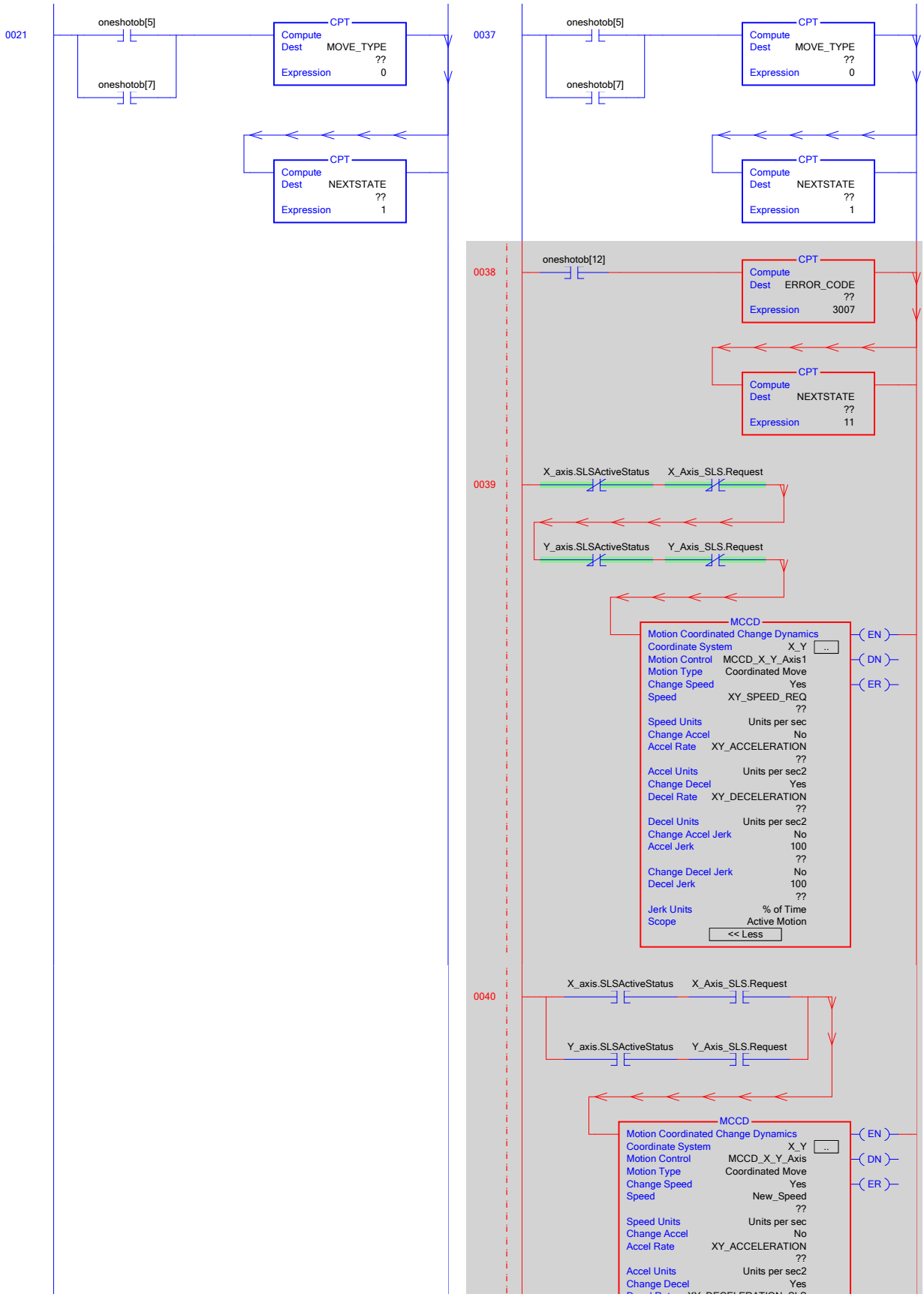

CPT
Compute
Dest XY_SPEED_REQ
??
Expression XY_SPEED
0029
oneshotob[9]
Z_RETRACTED
LES
Less Than (A<B)
Source A Z_axis.ActualPosition
??
Source B 3
??
MCLM
Motion Coordinated Linear Move
Coordinate System X_Y
Motion Control XY_AXIS_STAT[4]
Move Type 0
??
Position X_POSITION
Speed XY_SPEED_REQ
??
Speed Units Units per sec
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Termination Type 0
??
Merge Disabled
Merge Speed Programmed
Command Tolerance 0
??
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
AC
PC

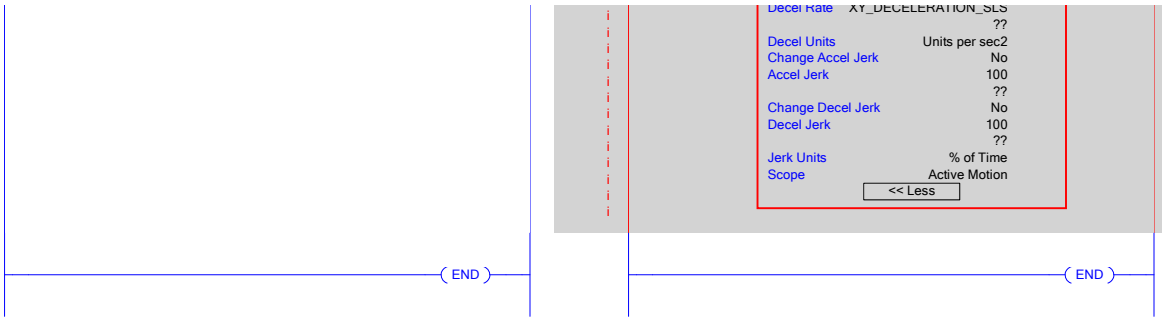

0015
XY_AXIS_STAT[4].PC
XY_AXIS_STAT[4].ER
CMP
Compare
Expression X_axis.ActualPosition<(X_POSITION+0.1)
CMP
Compare
Expression X_axis.ActualPosition>(X_POSITION-0.1)
CMP
Compare
Expression Y_axis.ActualPosition<(Y_POSITION+0.1)
CMP
Compare
Expression Y_axis.ActualPosition>(Y_POSITION-0.1)
XY_POS_LOCK_STAT


0030
XY_AXIS_STAT[4].PC
CMP
Compare
Expression X_axis.ActualPosition<(X_POSITION+0.2)
CMP
Compare
Expression X_axis.ActualPosition>(X_POSITION-0.2)
CMP
Compare
Expression Y_axis.ActualPosition<(Y_POSITION+0.2)
CMP
Compare
Expression Y_axis.ActualPosition>(Y_POSITION-0.2)
XY_POS_LOCK_STAT
STATE3_IND

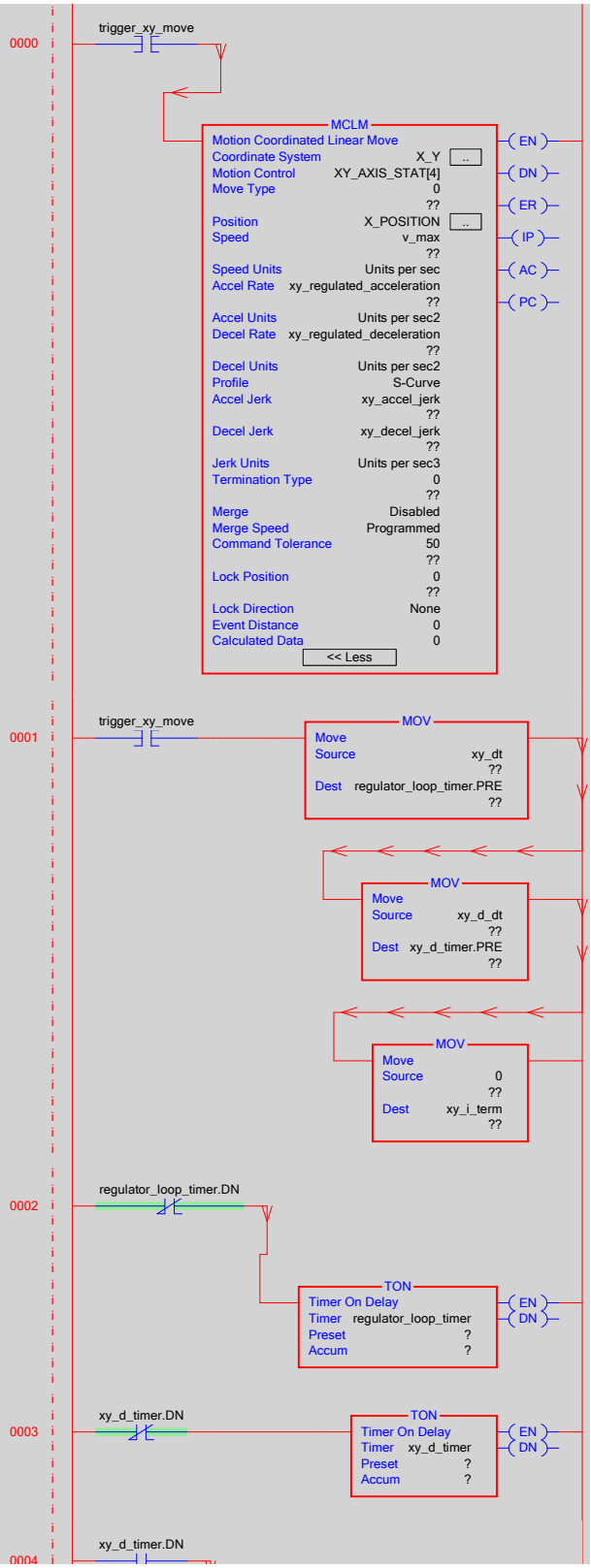

STATE3_IND
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 3003
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 3003
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
0016
XY_AXIS_STAT[4].IP
CMP
Compare
Expression XY_SPEED=0
OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit speed_set_to_zero
SB
OB
0017
XY_AXIS_STAT[4].IP
ALL_EOT_GOOD
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit eot_triggered
SB
OB
0018
eot_triggered
MCS
Motion Coordinated Stop
Coordinate System X_Y
Motion Control XY_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate 20000
??
Decel Units Units per sec2
Change Decel Jerk Yes
Decel Jerk 100000
??
Jerk Units Units per sec3
<< Less
EN
DN
ER
IP
PC
CPT
Compute
Dest NEXTSTATE
??
Expression 11
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
0019
speed_set_to_zero

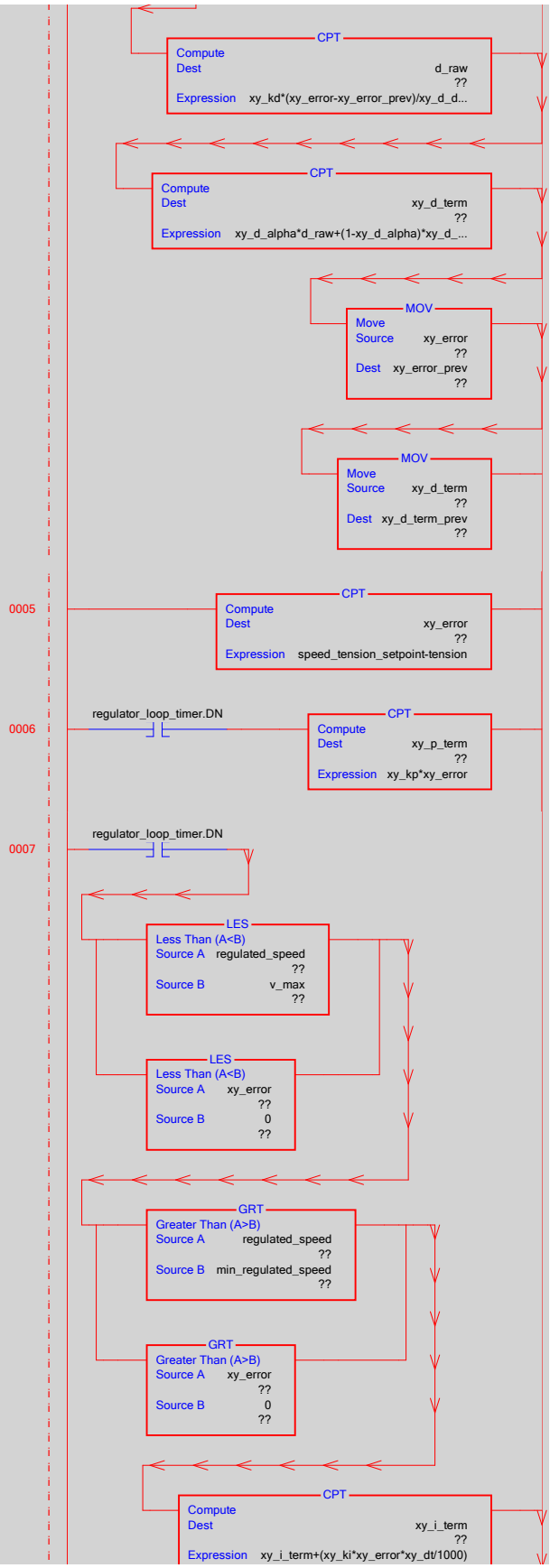

MCS
Motion Coordinated Stop
Coordinate System X_Y
Motion Control XY_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate 1000
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 1000
??
Jerk Units Units per sec3
<< Less
EN
DN
ER
IP
PC
0031
XY_AXIS_STAT[4].IP
CMP
Compare
Expression XY_SPEED=0
OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB
0032
XY_AXIS_STAT[4].IP
ALL_EOT_GOOD
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
0033
oneshotob[6]
MCS
Motion Coordinated Stop
Coordinate System X_Y
Motion Control XY_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
0034
oneshotob[10]
MCS
Motion Coordinated Stop
Coordinate System X_Y
Motion Control XY_AXIS_STAT[6]
Stop Type All
Change Decel Yes
Decel Rate 1500
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
0020
XY_AXIS_STAT[5].PC
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
0035
XY_AXIS_STAT[5].PC
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
0036
XY_AXIS_STAT[6].PC
OSR
One Shot Rising
Storage Bit oneshotsb[12]
Output Bit oneshotob[12]

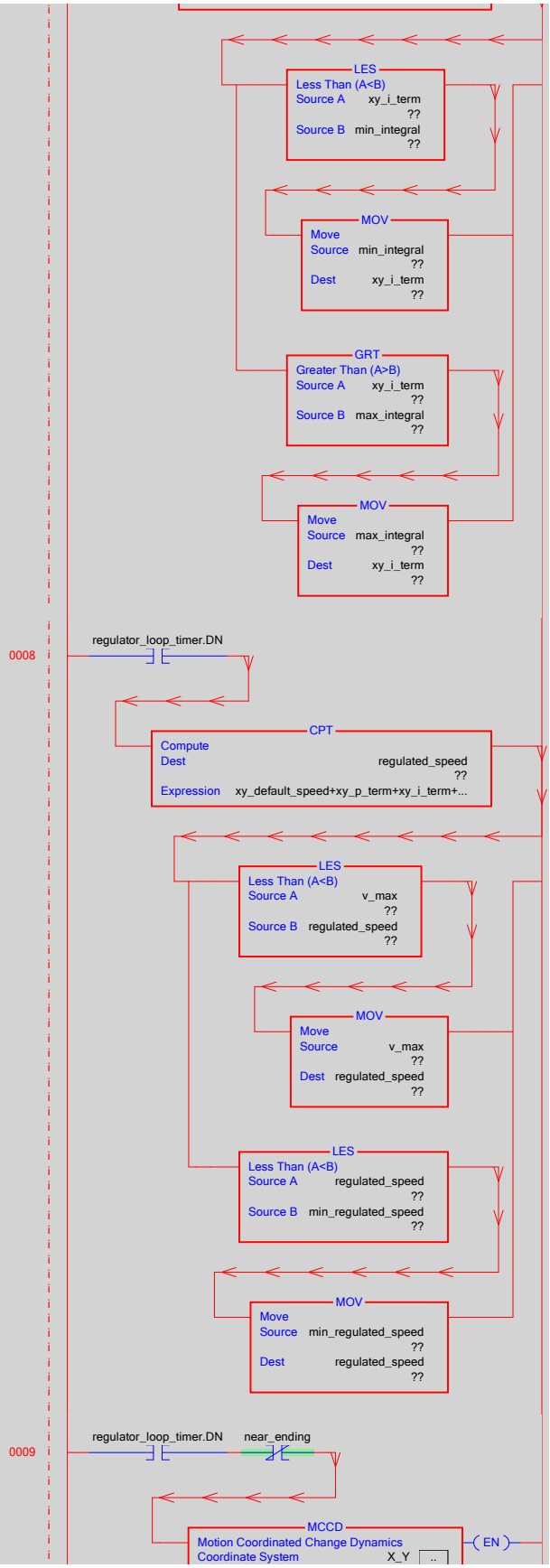

0021
oneshotob[5]
oneshotob[7]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0037
oneshotob[5]
oneshotob[7]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0038
oneshotob[12]
CPT
Compute
Dest ERROR_CODE
??
Expression 3007
CPT
Compute
Dest NEXTSTATE
??
Expression 11
0039
X_axis.SLSActiveStatus
X_Axis_SLS.Request
Y_axis.SLSActiveStatus
Y_Axis_SLS.Request
MCCD
Motion Coordinated Change Dynamics
Coordinate System X_Y
Motion Control MCCD_X_Y_Axis1
Motion Type Coordinated Move
Change Speed Yes
Speed XY_SPEED_REQ
??
Speed Units Units per sec
Change Accel No
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Change Decel Yes
Decel Rate XY_DECELERATION
??
Decel Units Units per sec2
Change Accel Jerk No
Accel Jerk 100
??
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
Scope Active Motion
<< Less
EN
DN
ER
0040
X_axis.SLSActiveStatus
X_Axis_SLS.Request
Y_axis.SLSActiveStatus
Y_Axis_SLS.Request
MCCD
Motion Coordinated Change Dynamics
Coordinate System X_Y
Motion Control MCCD_X_Y_Axis
Motion Type Coordinated Move
Change Speed Yes
Speed New_Speed
??
Speed Units Units per sec
Change Accel No
Accel Rate XY_ACCELERATION
??
Accel Units Units per sec2
Change Decel Yes
EN
DN
ER

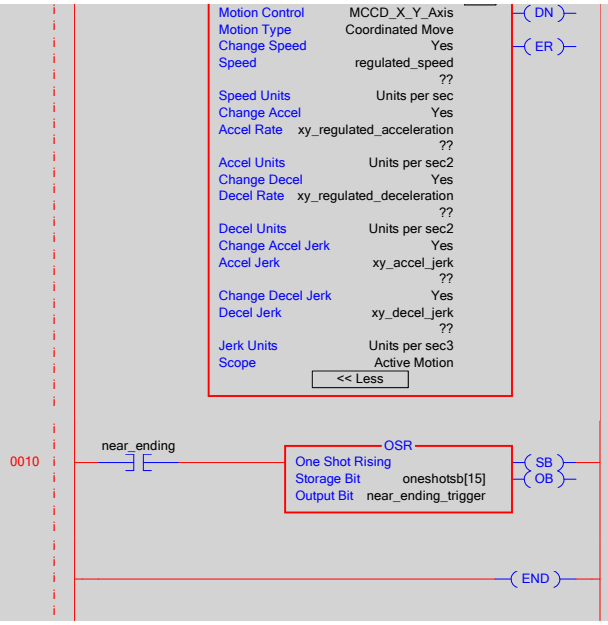

END
Decel Rate XY_DECELERATION_SLS
??
Decel Units Units per sec2
Change Accel Jerk No
Accel Jerk 100
??
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
Scope Active Motion
<< Less
END

Program: MoveXY_State_2_3
Routine: xy_speed_regulator




CPT
Compute
Dest d_raw
??
Expression xy_kd*(xy_error-xy_error_prev)/xy_d_d...
CPT
Compute
Dest xy_d_term
??
Expression xy_d_alpha*d_raw+(1-xy_d_alpha)*xy_d_...
MOV
Move
Source xy_error
??
Dest xy_error_prev
??
MOV
Move
Source xy_d_term
??
Dest xy_d_term_prev
??
0005
CPT
Compute
Dest xy_error
??
Expression speed_tension_setpoint-tension
0006
regulator_loop_timer.DN
CPT
Compute
Dest xy_p_term
??
Expression xy_kp*xy_error
0007
regulator_loop_timer.DN
LES
Less Than (A<B)
Source A regulated_speed
??
Source B v_max
??
LES
Less Than (A<B)
Source A xy_error
??
Source B 0
??
GRT
Greater Than (A>B)
Source A regulated_speed
??
Source B min_regulated_speed
??
GRT
Greater Than (A>B)
Source A xy_error
??
Source B 0
??
CPT
Compute
Dest xy_i_term
??
Expression xy_i_term+(xy_ki*xy_error*xy_dt/1000)

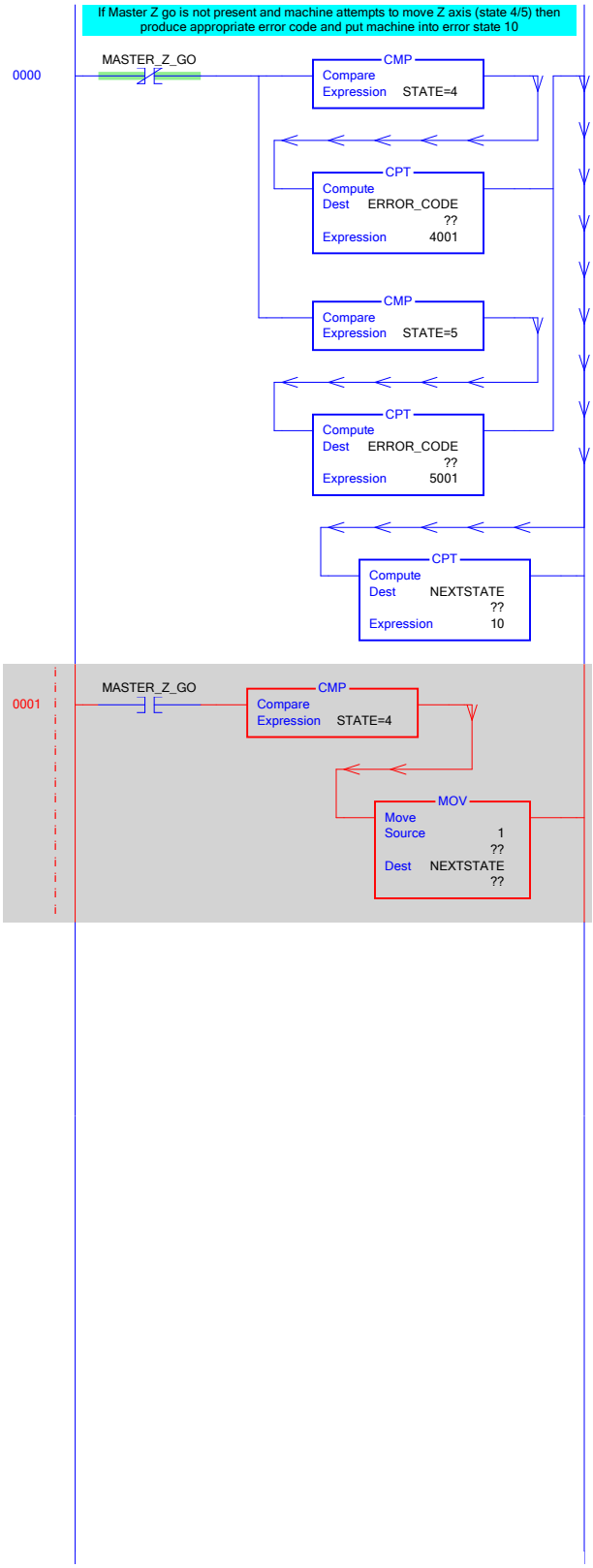

LES
Less Than (A<B)
Source A xy_i_term
??
Source B min_integral
??
MOV
Move
Source min_integral
??
Dest xy_i_term
??
GRT
Greater Than (A>B)
Source A xy_i_term
??
Source B max_integral
??
MOV
Move
Source max_integral
??
Dest xy_i_term
??
0008
regulator_loop_timer.DN
CPT
Compute
Dest regulated_speed
??
Expression xy_default_speed+xy_p_term+xy_i_term+...
LES
Less Than (A<B)
Source A v_max
??
Source B regulated_speed
??
MOV
Move
Source v_max
??
Dest regulated_speed
??
LES
Less Than (A<B)
Source A regulated_speed
??
Source B min_regulated_speed
??
MOV
Move
Source min_regulated_speed
??
Dest regulated_speed
??
0009
regulator_loop_timer.DN
near_ending
MCCD
Motion Coordinated Change Dynamics
Coordinate System X_Y
EN

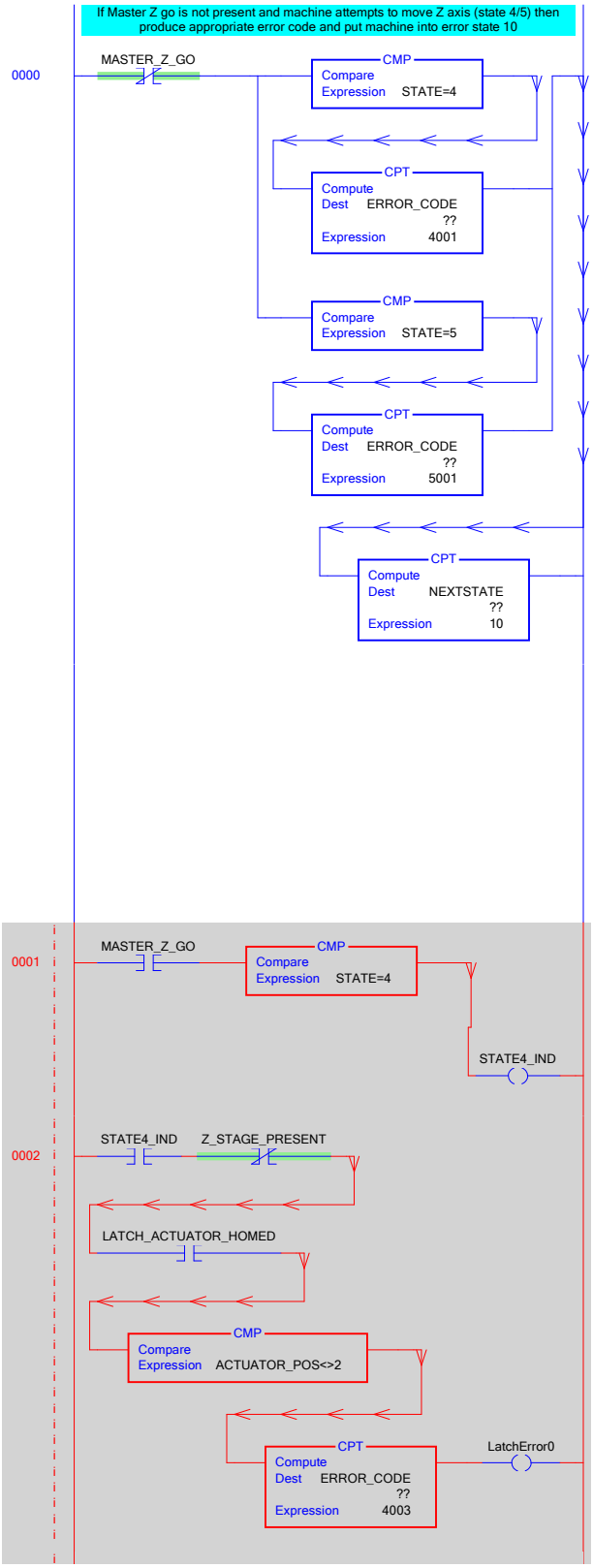

Motion Control MCCD_X_Y_Axis
Motion Type Coordinated Move
Change Speed Yes
Speed regulated_speed
??
Speed Units Units per sec
Change Accel Yes
Accel Rate xy_regulated_acceleration
??
Accel Units Units per sec2
Change Decel Yes
Decel Rate xy_regulated_deceleration
??
Decel Units Units per sec2
Change Accel Jerk Yes
Accel Jerk xy_accel_jerk
??
Change Decel Jerk Yes
Decel Jerk xy_decel_jerk
??
Jerk Units Units per sec3
Scope Active Motion
<< Less
DN
ER
0010
near_ending
OSR
One Shot Rising
Storage Bit oneshotsb[15]
Output Bit near_ending_trigger
SB
OB
END

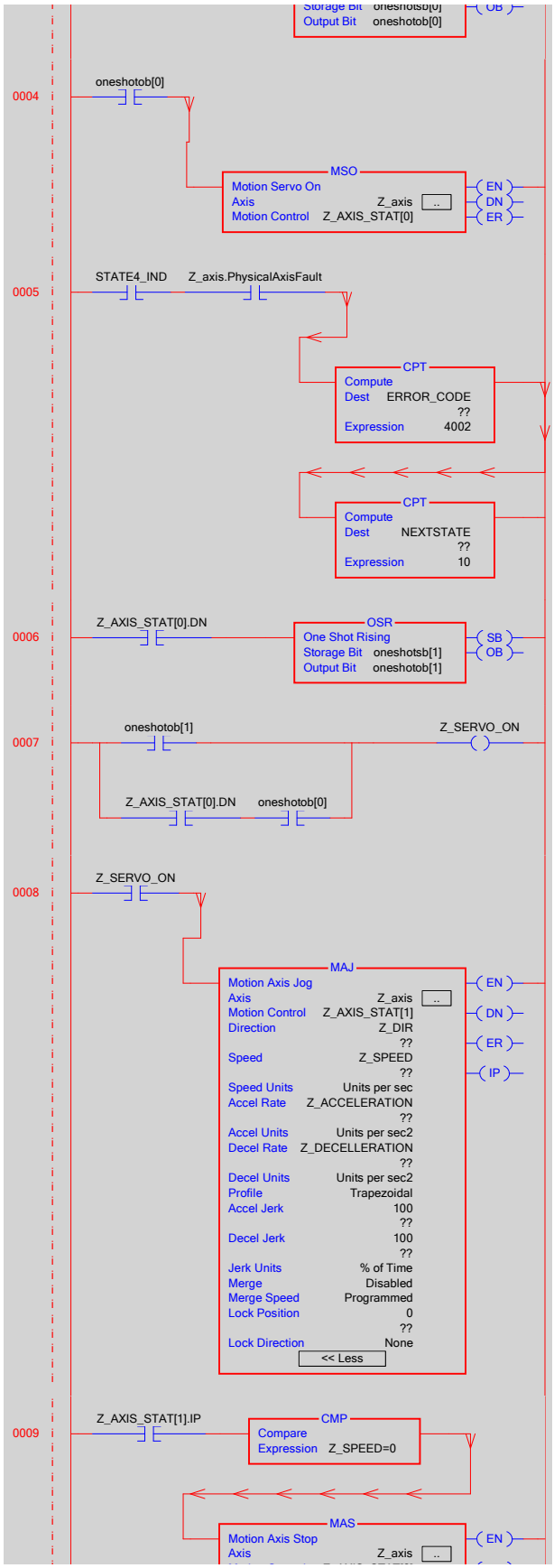
MoveZ_State_4_5 Parameters and Local Tags

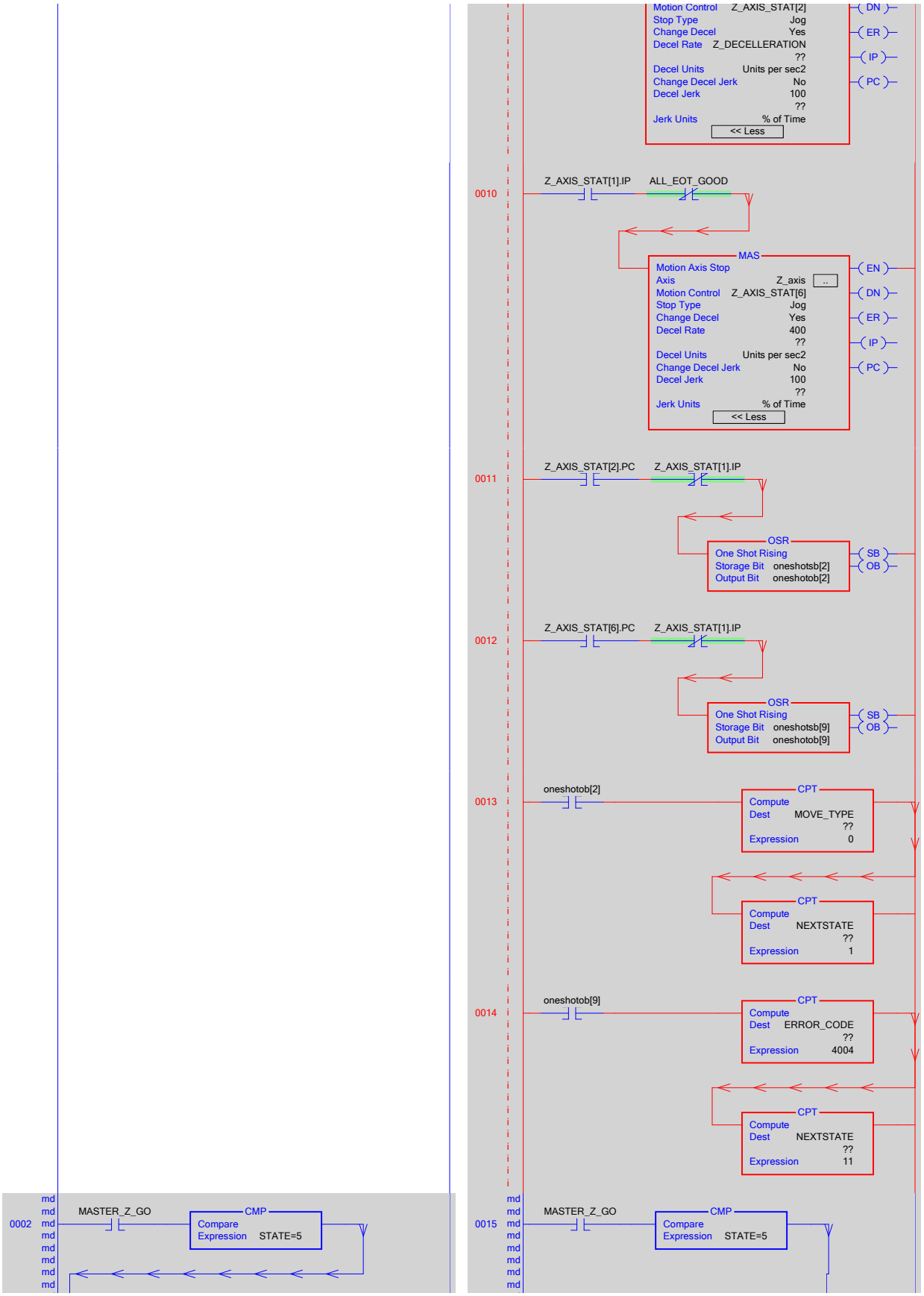
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
error_suppress		BOOL		0	---	---	---	---
ian	---	---	---	---		BOOL		0
LatchError0	---	---	---	---		BOOL		0
LatchError1	---	---	---	---		BOOL		0
STATE4_IND	---	---	---	---		BOOL		0
STOPPING_TIME	---	---	---	---		TIMER		
z_accel_jerk		DINT		800	---	---	---	---
z_decel_jerk		DINT		1600	---	---	---	---
Z_EOT_active	---	---	---	---		REAL		0.0
Z_fixedPresent_active	---	---	---	---		REAL		427.4913
Z_Retracted1A_active	---	---	---	---		REAL		1.6748
Z_Retracted1B_active	---	---	---	---		REAL		1.6748
Z_Retracted2A_active	---	---	---	---		REAL		1.4798
Z_Retracted2B_active	---	---	---	---		REAL		1.4798
Z_SERVO_ON	---	---	---	---		BOOL		0
Z_stagePresent_active	---	---	---	---		REAL		428.8098

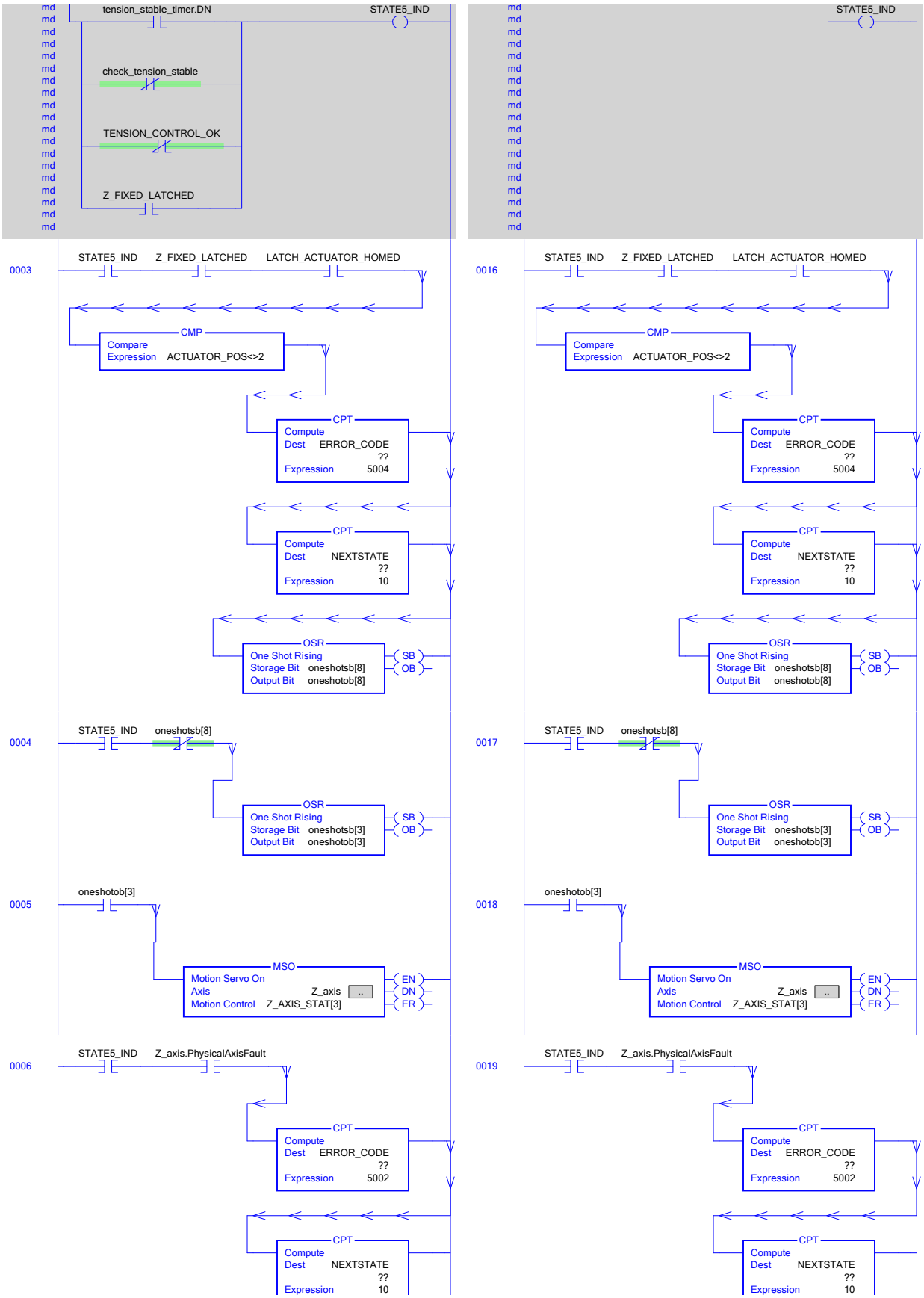
Program: MoveZ_State_4_5
Routine: MoveZ






Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
OB
0004
oneshotob[0]
MSO
Motion Servo On
Axis Z_axis
Motion Control Z_AXIS_STAT[0]
EN
DN
ER
0005
STATE4_IND
Z_axis.PhysicalAxisFault
CPT
Compute
Dest ERROR_CODE
??
Expression 4002
CPT
Compute
Dest NEXTSTATE
??
Expression 10
0006
Z_AXIS_STAT[0].DN
OSR
One Shot Rising
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB
0007
oneshotob[1]
Z_SERVO_ON
Z_AXIS_STAT[0].DN
oneshotob[0]
0008
Z_SERVO_ON
MAJ
Motion Axis Jog
Axis Z_axis
Motion Control Z_AXIS_STAT[1]
Direction Z_DIR
??
Speed Z_SPEED
??
Speed Units Units per sec
Accel Rate Z_ACCELERATION
??
Accel Units Units per sec2
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
<< Less
EN
DN
ER
IP
0009
Z_AXIS_STAT[1].IP
CMP
Compare
Expression Z_SPEED=0
MAS
Motion Axis Stop
Axis Z_axis
EN


Motion Control Z_AXIS_STAT[2]
Stop Type Jog
Change Decel Yes
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
DN
ER
IP
PC
0010
Z_AXIS_STAT[1].IP
ALL_EOT_GOOD
MAS
Motion Axis Stop
Axis Z_axis
Motion Control Z_AXIS_STAT[6]
Stop Type Jog
Change Decel Yes
Decel Rate 400
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0011
Z_AXIS_STAT[2].PC
Z_AXIS_STAT[1].IP
OSR
One Shot Rising
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
SB
OB
0012
Z_AXIS_STAT[6].PC
Z_AXIS_STAT[1].IP
OSR
One Shot Rising
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB
0013
oneshotob[2]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0014
oneshotob[9]
CPT
Compute
Dest ERROR_CODE
??
Expression 4004
CPT
Compute
Dest NEXTSTATE
??
Expression 11
0002
MASTER_Z_GO
CMP
Compare
Expression STATE=5
0015
MASTER_Z_GO
CMP
Compare
Expression STATE=5


tension_stable_timer.DN
STATE5_IND
check_tension_stable
TENSION_CONTROL_OK
Z_FIXED_LATCHED
0003
STATE5_IND Z_FIXED_LATCHED LATCH_ACTUATOR_HOMED
CMP
Compare
Expression ACTUATOR_POS<>2
CPT
Compute
Dest ERROR_CODE
??
Expression 5004
CPT
Compute
Dest NEXTSTATE
??
Expression 10
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0004
STATE5_IND oneshotsb[8]
OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB
0005
oneshotob[3]
MSO
Motion Servo On
Axis Z_axis
Motion Control Z_AXIS_STAT[3]
EN
DN
ER
0006
STATE5_IND Z_axis.PhysicalAxisFault
CPT
Compute
Dest ERROR_CODE
??
Expression 5002
CPT
Compute
Dest NEXTSTATE
??
Expression 10
STATE5_IND
0016
STATE5_IND Z_FIXED_LATCHED LATCH_ACTUATOR_HOMED
CMP
Compare
Expression ACTUATOR_POS<>2
CPT
Compute
Dest ERROR_CODE
??
Expression 5004
CPT
Compute
Dest NEXTSTATE
??
Expression 10
OSR
One Shot Rising
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB
0017
STATE5_IND oneshotsb[8]
OSR
One Shot Rising
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB
0018
oneshotob[3]
MSO
Motion Servo On
Axis Z_axis
Motion Control Z_AXIS_STAT[3]
EN
DN
ER
0019
STATE5_IND Z_axis.PhysicalAxisFault
CPT
Compute
Dest ERROR_CODE
??
Expression 5002
CPT
Compute
Dest NEXTSTATE
??
Expression 10

0007 Z_AXIS_STAT[3].DN

OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

Move more abruptly when the winding head isn't on the z-arm

0008 oneshotob[4]

Z_AXIS_STAT[3].DN oneshotob[3]

Z_FIXED_LATCHED

MAM
Motion Axis Move
Axis Z_axis ..
Motion Control Z_AXIS_STAT[4]
Move Type 0
??
Position Z_POSITION
??
Speed Z_SPEED
??
Speed Units Units per sec
Accel Rate Z_ACCELERATION
??
Accel Units Units per sec2
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Profile S-Curve
Accel Jerk z_accel_jerk
??
Decel Jerk z_decel_jerk
??
Jerk Units Units per sec3
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC

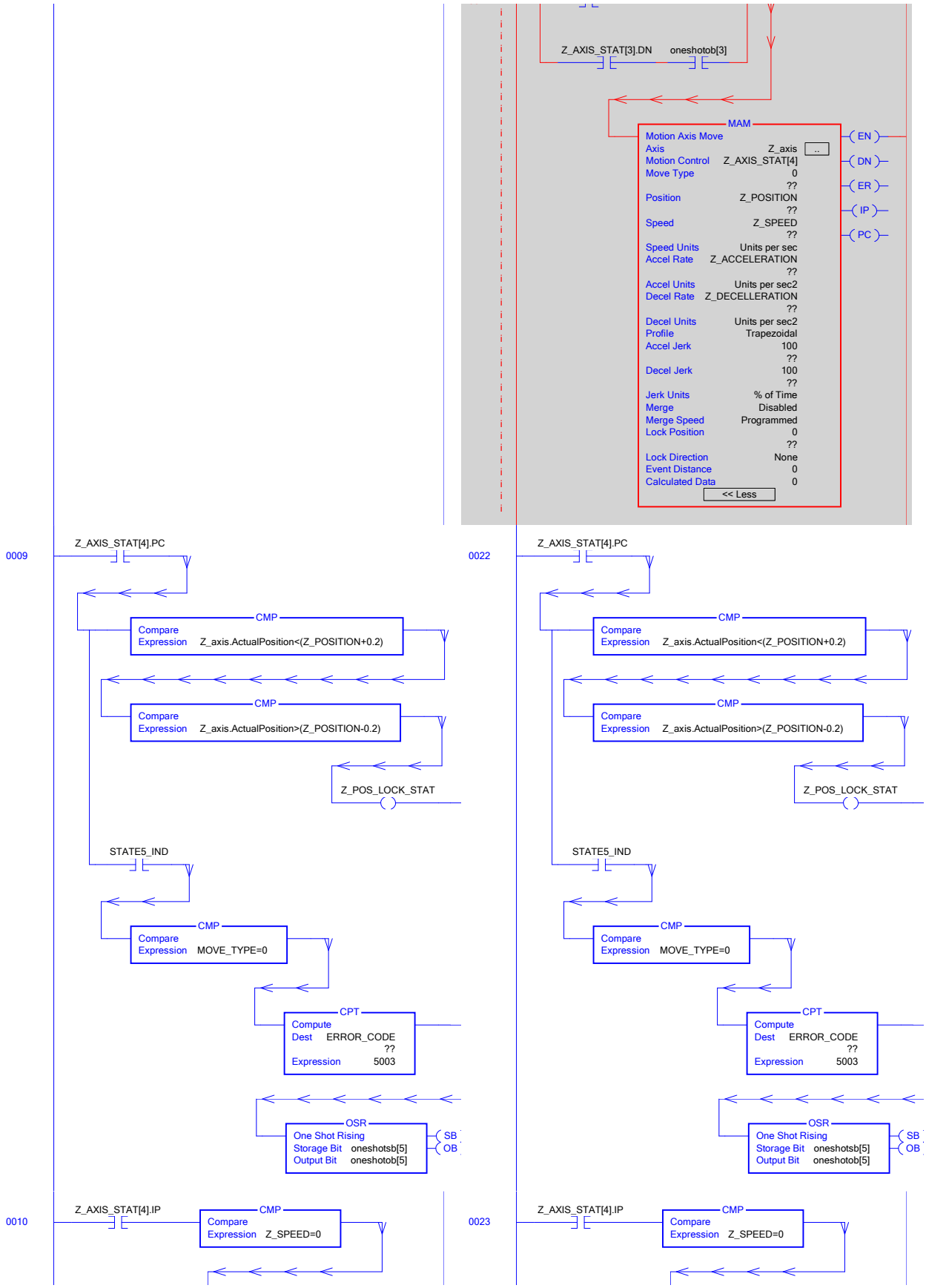
Z_FIXED_LATCHED

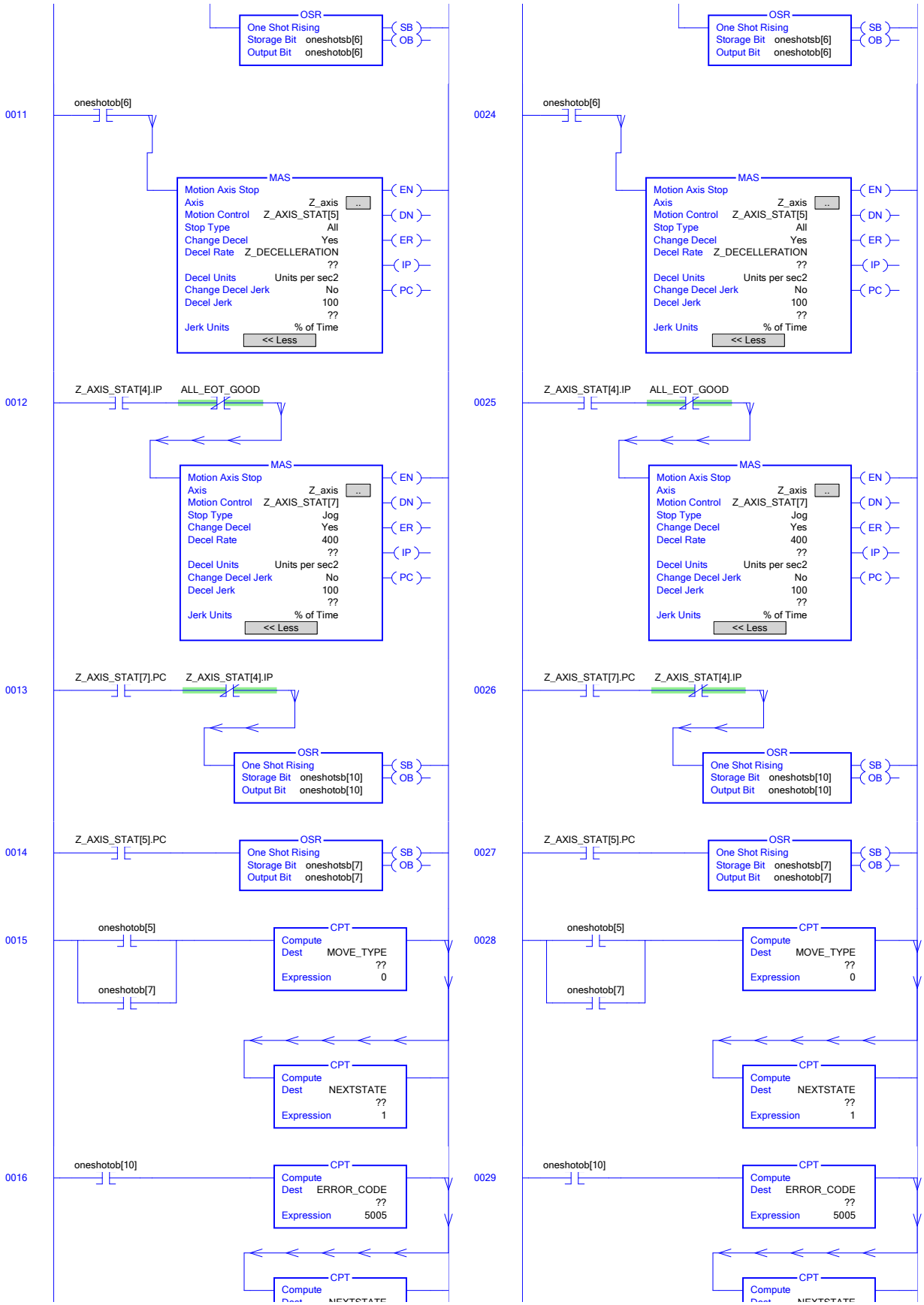
MAM
Motion Axis Move
Axis Z_axis ..
Motion Control Z_AXIS_STAT[4]
Move Type 0
??
Position Z_POSITION
??
Speed 1000
??
Speed Units Units per sec
Accel Rate 5000
??
Accel Units Units per sec2
Decel Rate 5000
??
Decel Units Units per sec2
Profile S-Curve
Accel Jerk 20000
??
Decel Jerk 10000
??
Jerk Units Units per sec3
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC

0020 Z_AXIS_STAT[3].DN

OSR
One Shot Rising
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

0021 oneshotob[4]


Z_AXIS_STAT[3].DN
oneshotob[3]
MAM
Motion Axis Move
Axis Z_axis
Motion Control Z_AXIS_STAT[4]
Move Type 0
??
Position Z_POSITION
??
Speed Z_SPEED
??
Speed Units Units per sec
Accel Rate Z_ACCELERATION
??
Accel Units Units per sec2
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
??
Decel Jerk 100
??
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
??
Lock Direction None
Event Distance 0
Calculated Data 0
<< Less
EN
DN
ER
IP
PC
0009
Z_AXIS_STAT[4].PC
CMP
Compare
Expression Z_axis.ActualPosition<(Z_POSITION+0.2)
CMP
Compare
Expression Z_axis.ActualPosition>(Z_POSITION-0.2)
Z_POS_LOCK_STAT
STATE5_IND
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 5003
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
0010
Z_AXIS_STAT[4].IP
CMP
Compare
Expression Z_SPEED=0
0022
Z_AXIS_STAT[4].PC
CMP
Compare
Expression Z_axis.ActualPosition<(Z_POSITION+0.2)
CMP
Compare
Expression Z_axis.ActualPosition>(Z_POSITION-0.2)
Z_POS_LOCK_STAT
STATE5_IND
CMP
Compare
Expression MOVE_TYPE=0
CPT
Compute
Dest ERROR_CODE
??
Expression 5003
OSR
One Shot Rising
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB
0023
Z_AXIS_STAT[4].IP
CMP
Compare
Expression Z_SPEED=0


OSR
One Shot Rising
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB
0011
oneshotob[6]
MAS
Motion Axis Stop
Axis Z_axis
Motion Control Z_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate Z_DECELLERATION
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
EN
DN
ER
IP
PC
0012
Z_AXIS_STAT[4].IP
ALL_EOT_GOOD
MAS
Motion Axis Stop
Axis Z_axis
Motion Control Z_AXIS_STAT[7]
Stop Type Jog
Change Decel Yes
Decel Rate 400
??
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
??
Jerk Units % of Time
<< Less
0013
Z_AXIS_STAT[7].PC
Z_AXIS_STAT[4].IP
OSR
One Shot Rising
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
0014
Z_AXIS_STAT[5].PC
OSR
One Shot Rising
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
0015
oneshotob[5]
oneshotob[7]
CPT
Compute
Dest MOVE_TYPE
??
Expression 0
CPT
Compute
Dest NEXTSTATE
??
Expression 1
0016
oneshotob[10]
CPT
Compute
Dest ERROR_CODE
??
Expression 5005
CPT
Compute
0024
0025
0026
0027
0028
0029

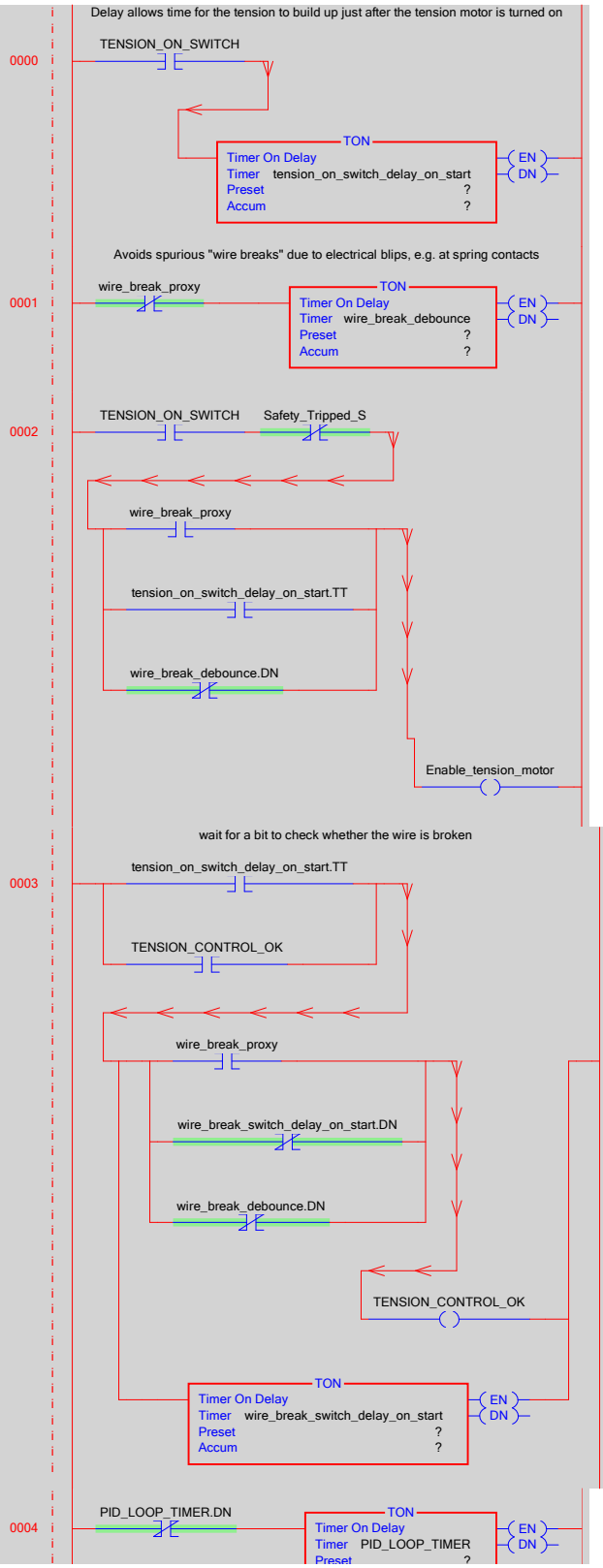
Dest NEXTSTATE
??
Expression 11
END

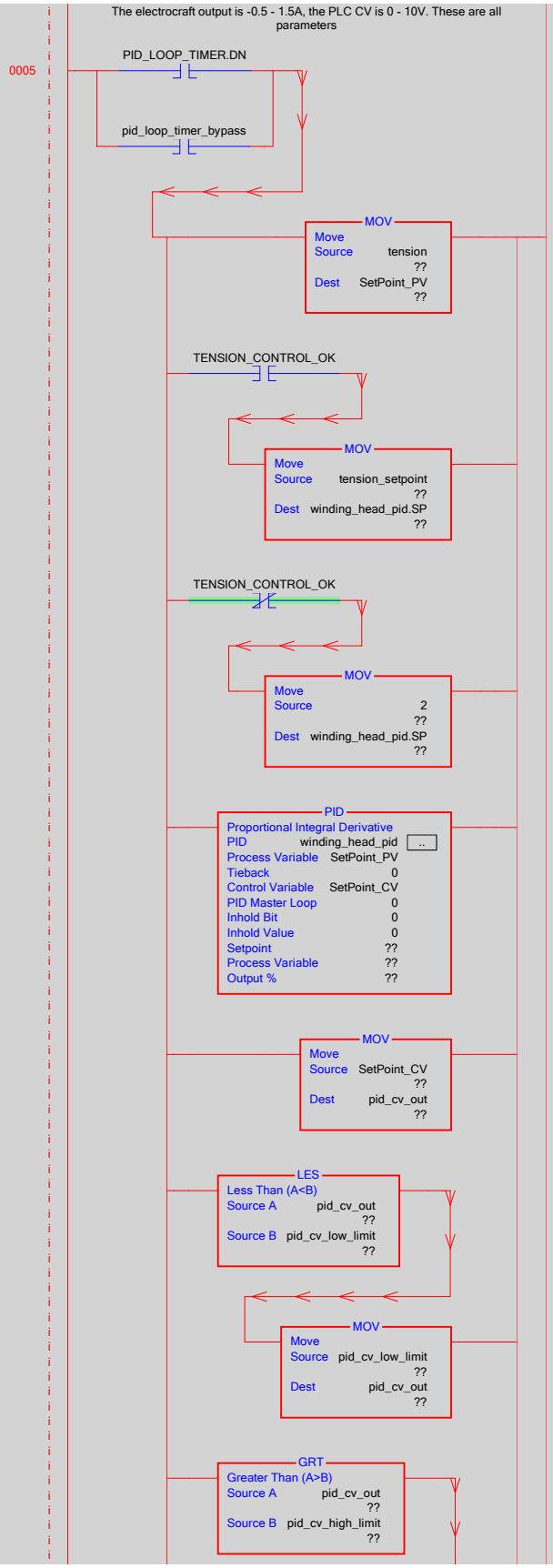
Dest NEXTSTATE
??
Expression 11
END

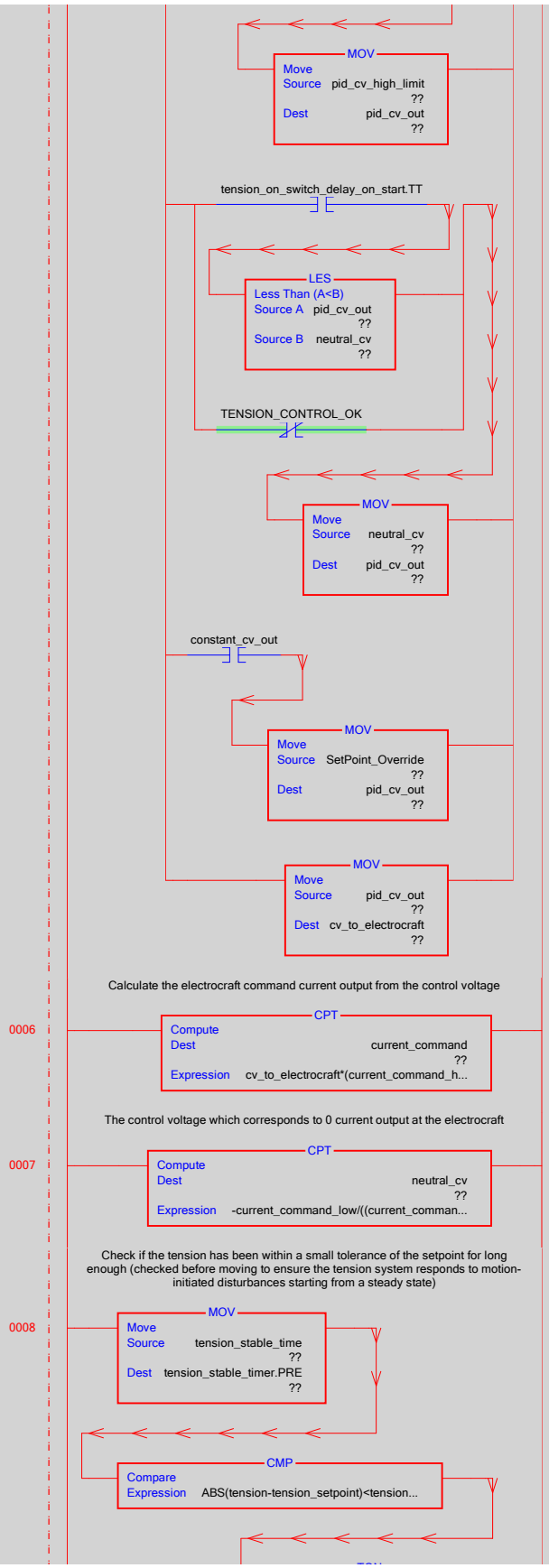
PID_Tension_Servo Parameters and Local Tags

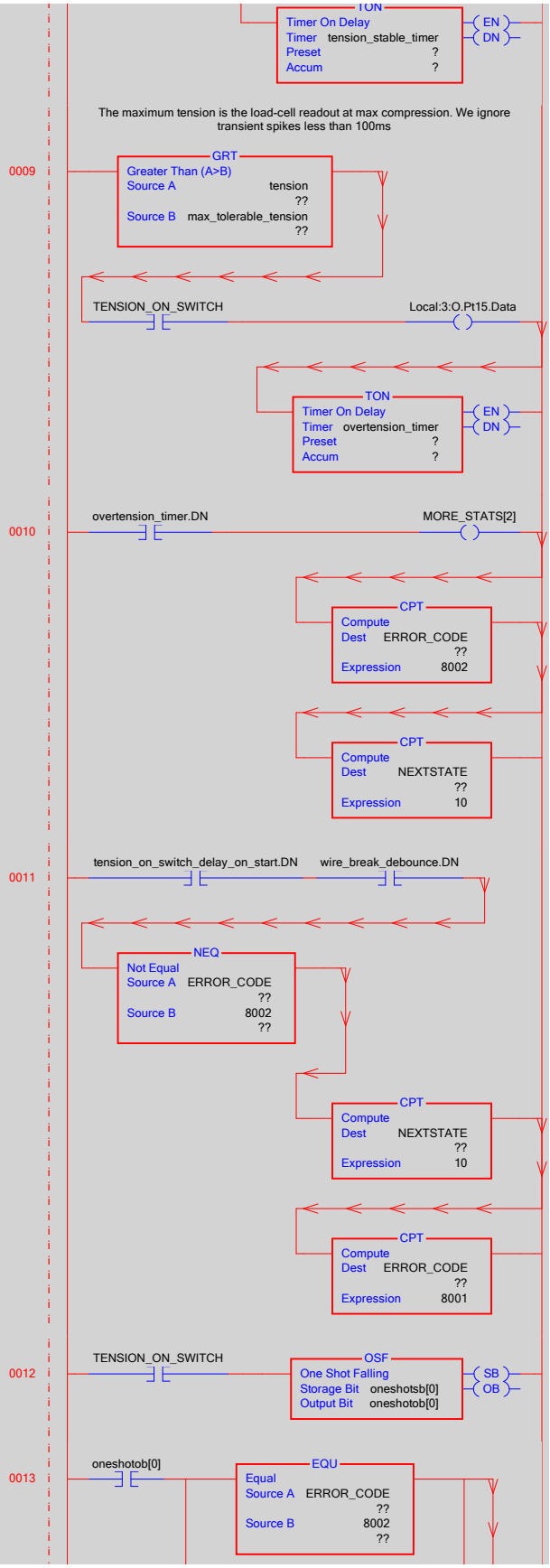
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
constant_cv_out		BOOL		0	---	---	---	---
cv_to_electrocraft	Local:5:O.Ch00.Data			2.5	---	---	---	---
Enable_tension_motor	Local:3:O.Pt03.Data			0	---	---	---	---
oneshotob		BOOL[32]			---	---	---	---
oneshotsb		BOOL[32]			---	---	---	---
TENSION_CNTL_OK	---	---	---	---		BOOL		0
TENSION_CNTL_REQUEST	---	---	---	---		BOOL		0
tension_stable_time		DINT		200	---	---	---	---
tension_stable_tolerance		REAL		0.15	---	---	---	---
TestBit	---	---	---	---		BOOL		0
wire_break_debounce		TIMER			---	---	---	---
wire_break_switch_delay_on_start		TIMER			---	---	---	---

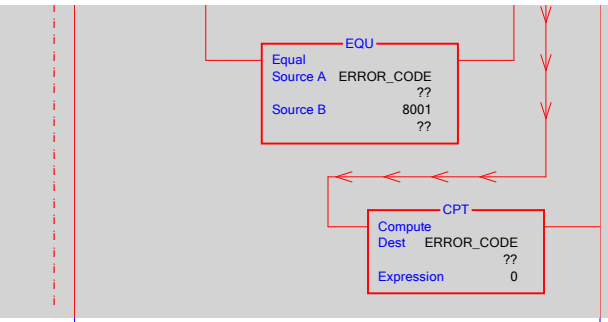
Program: PID_Tension_Servo
Routine: PID_Tension

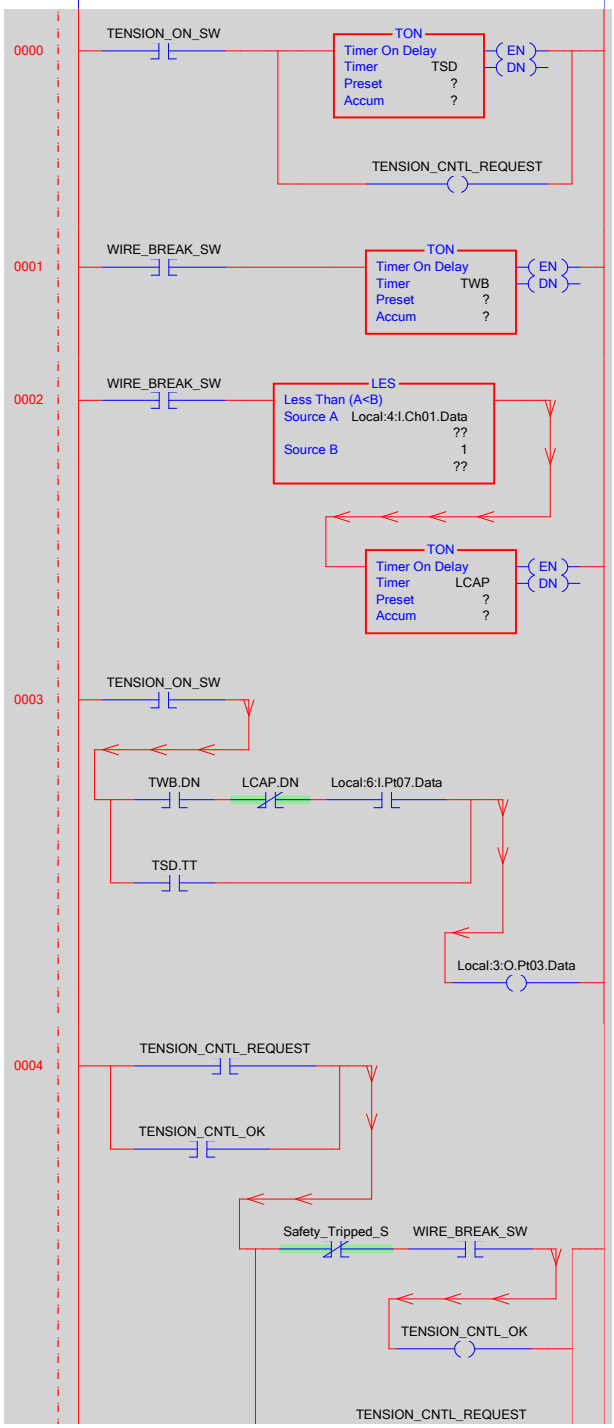


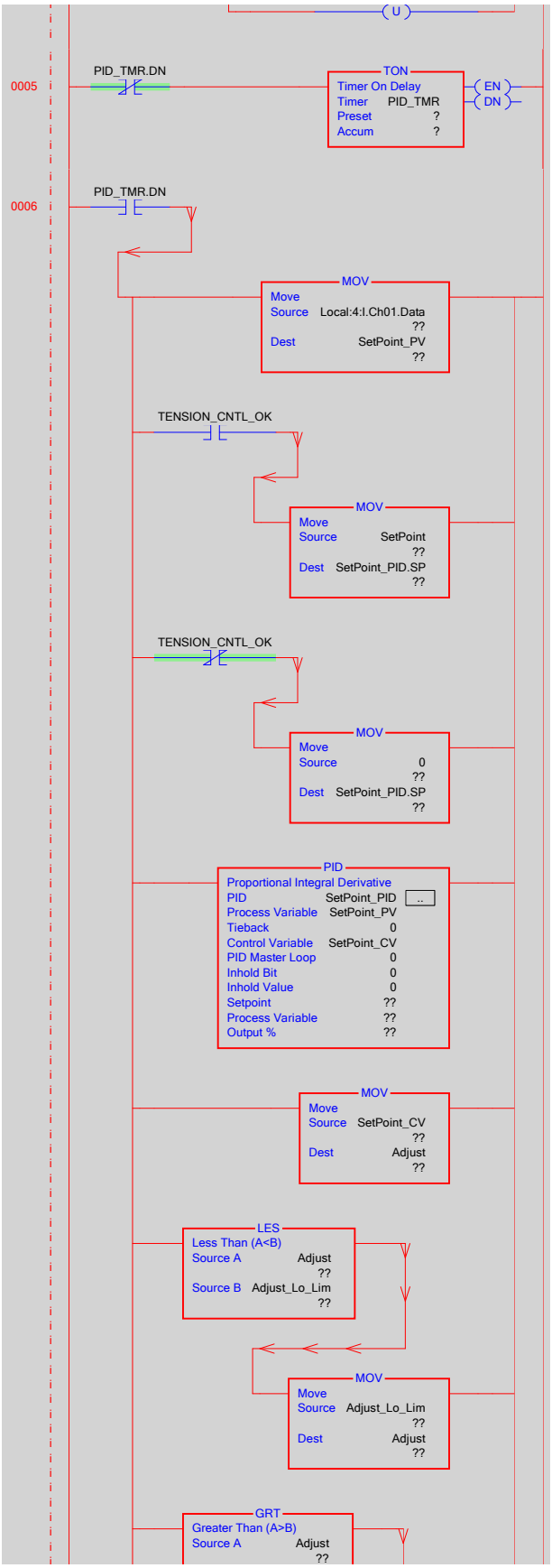

The electrocraft output is -0.5 - 1.5A, the PLC CV is 0 - 10V. These are all parameters
0005
PID_LOOP_TIMER.DN
pid_loop_timer_bypass
MOV
Move
Source tension
??
Dest SetPoint_PV
??
TENSION_CONTROL_OK
MOV
Move
Source tension_setpoint
??
Dest winding_head_pid.SP
??
TENSION_CONTROL_OK
MOV
Move
Source 2
??
Dest winding_head_pid.SP
??
PID
Proportional Integral Derivative
PID winding_head_pid
Process Variable SetPoint_PV
Tieback 0
Control Variable SetPoint_CV
PID Master Loop 0
Inhold Bit 0
Inhold Value 0
Setpoint ??
Process Variable ??
Output % ??
MOV
Move
Source SetPoint_CV
??
Dest pid_cv_out
??
LES
Less Than (A<B)
Source A pid_cv_out
??
Source B pid_cv_low_limit
??
MOV
Move
Source pid_cv_low_limit
??
Dest pid_cv_out
??
GRT
Greater Than (A>B)
Source A pid_cv_out
??
Source B pid_cv_high_limit
??

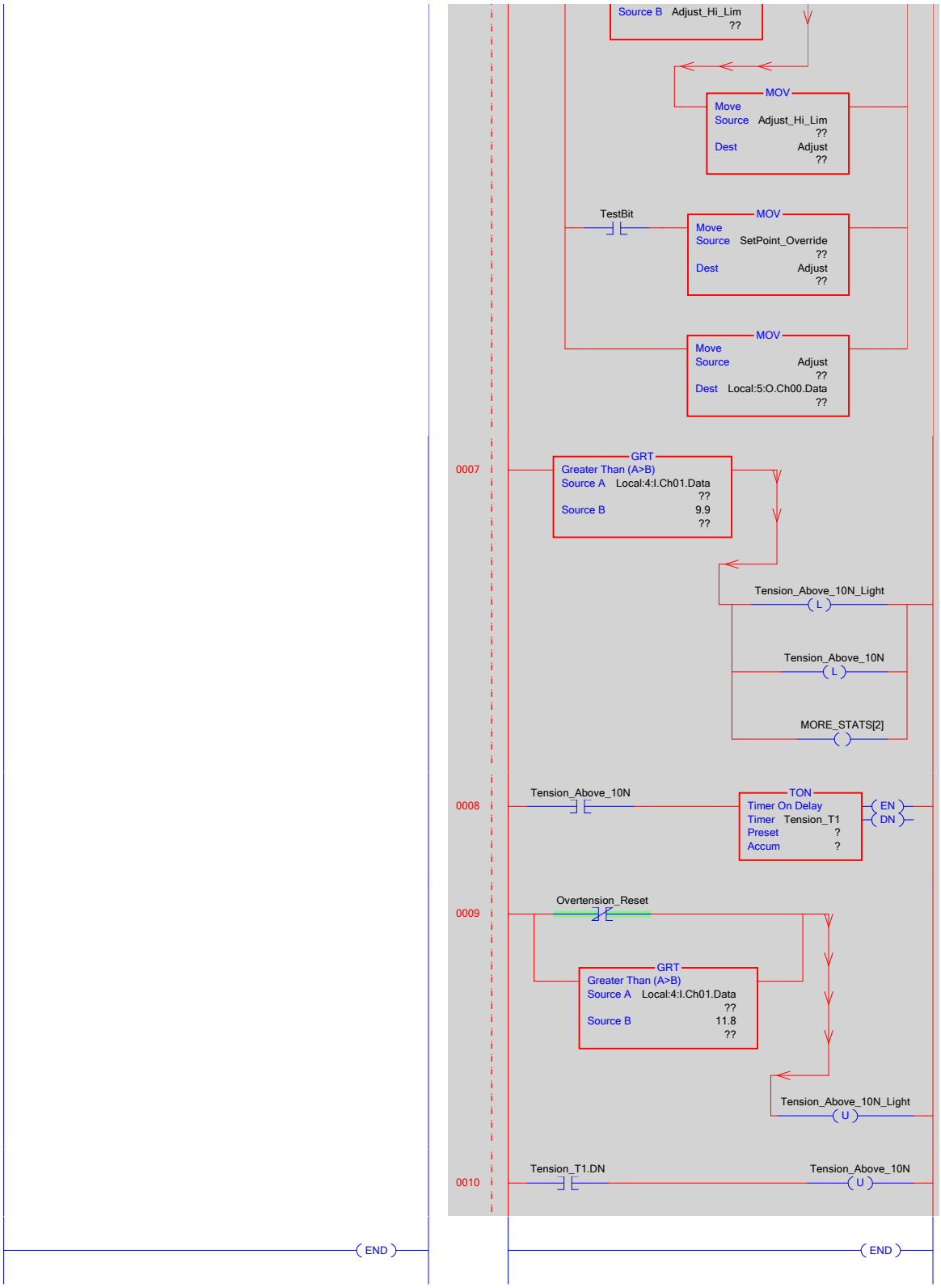

MOV
Move
Source pid_cv_high_limit
??
Dest pid_cv_out
??
tension_on_switch_delay_on_start.TT
LES
Less Than (A<B)
Source A pid_cv_out
??
Source B neutral_cv
??
TENSION_CONTROL_OK
MOV
Move
Source neutral_cv
??
Dest pid_cv_out
??
constant_cv_out
MOV
Move
Source SetPoint_Override
??
Dest pid_cv_out
??
MOV
Move
Source pid_cv_out
??
Dest cv_to_electrocraft
??
Calculate the electrocraft command current output from the control voltage
0006
CPT
Compute
Dest current_command
??
Expression cv_to_electrocraft*(current_command_h...
The control voltage which corresponds to 0 current output at the electrocraft
0007
CPT
Compute
Dest neutral_cv
??
Expression -current_command_low/((current_comman...
Check if the tension has been within a small tolerance of the setpoint for long enough (checked before moving to ensure the tension system responds to motion-initiated disturbances starting from a steady state)
0008
MOV
Move
Source tension_stable_time
??
Dest tension_stable_timer.PRE
??
CMP
Compare
Expression ABS(tension-tension_setpoint)<tension...


TON
Timer On Delay
Timer tension_stable_timer
Preset ?
Accum ?
EN
DN
The maximum tension is the load-cell readout at max compression. We ignore transient spikes less than 100ms
0009
GRT
Greater Than (A>B)
Source A tension ??
Source B max_tolerable_tension ??
TENSION_ON_SWITCH
Local:3:O.Pt15.Data
TON
Timer On Delay
Timer overtension_timer
Preset ?
Accum ?
EN
DN
0010
overtension_timer.DN
MORE_STATS[2]
CPT
Compute
Dest ERROR_CODE ??
Expression 8002
CPT
Compute
Dest NEXTSTATE ??
Expression 10
0011
tension_on_switch_delay_on_start.DN
wire_break_debounce.DN
NEQ
Not Equal
Source A ERROR_CODE ??
Source B 8002 ??
CPT
Compute
Dest NEXTSTATE ??
Expression 10
CPT
Compute
Dest ERROR_CODE ??
Expression 8001
0012
TENSION_ON_SWITCH
OSF
One Shot Falling
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB
0013
oneshotob[0]
EQU
Equal
Source A ERROR_CODE ??
Source B 8002 ??


EQU
Equal
Source A ERROR_CODE
??
Source B 8001
??
CPT
Compute
Dest ERROR_CODE
??
Expression 0


0000
TENSION_ON_SW
TON
Timer On Delay
Timer TSD
Preset ?
Accum ?
EN
DN
TENSION_CNTL_REQUEST
0001
WIRE_BREAK_SW
TON
Timer On Delay
Timer TWB
Preset ?
Accum ?
EN
DN
0002
WIRE_BREAK_SW
LES
Less Than (A<B)
Source A Local:4:I.Ch01.Data
??
Source B 1
??
TON
Timer On Delay
Timer LCAP
Preset ?
Accum ?
EN
DN
0003
TENSION_ON_SW
TWB.DN
LCAP.DN
Local:6:I.Pt07.Data
TSD.TT
Local:3:O.Pt03.Data
0004
TENSION_CNTL_REQUEST
TENSION_CNTL_OK
Safety_Tripped_S
WIRE_BREAK_SW
TENSION_CNTL_OK
TENSION_CNTL_REQUEST

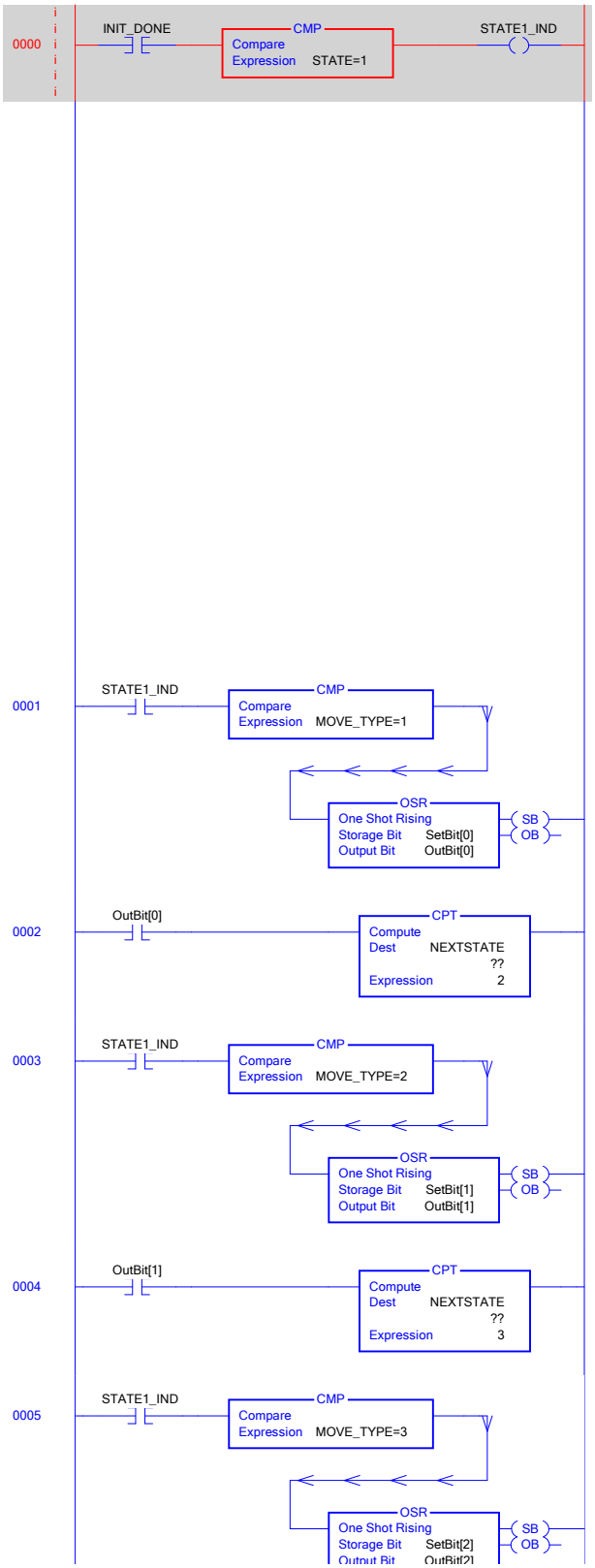

U
0005
PID_TMR.DN
TON
Timer On Delay
Timer PID_TMR
Preset ?
Accum ?
EN
DN
0006
PID_TMR.DN
MOV
Move
Source Local:4:I.Ch01.Data
??
Dest SetPoint_PV
??
TENSION_CNTL_OK
MOV
Move
Source SetPoint
??
Dest SetPoint_PID.SP
??
TENSION_CNTL_OK
MOV
Move
Source 0
??
Dest SetPoint_PID.SP
??
PID
Proportional Integral Derivative
PID SetPoint_PID
Process Variable SetPoint_PV
Tieback 0
Control Variable SetPoint_CV
PID Master Loop 0
Inhold Bit 0
Inhold Value 0
Setpoint ??
Process Variable ??
Output % ??
MOV
Move
Source SetPoint_CV
??
Dest Adjust
??
LES
Less Than (A<B)
Source A Adjust
??
Source B Adjust_Lo_Lim
??
MOV
Move
Source Adjust_Lo_Lim
??
Dest Adjust
??
GRT
Greater Than (A>B)
Source A Adjust
??

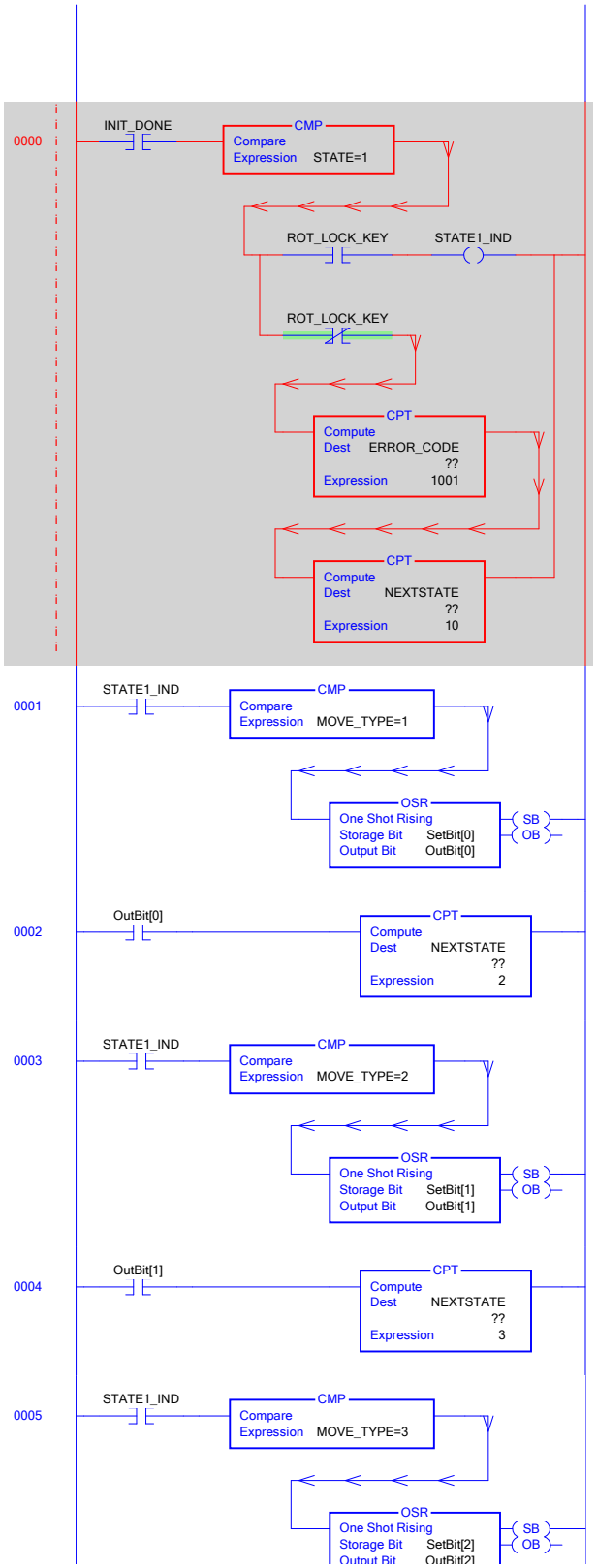

Source B Adjust_Hi_Lim
??
MOV
Move
Source Adjust_Hi_Lim
??
Dest Adjust
??
TestBit
MOV
Move
Source SetPoint_Override
??
Dest Adjust
??
MOV
Move
Source Adjust
??
Dest Local:5:O.Ch00.Data
??
0007
GRT
Greater Than (A>B)
Source A Local:4:I.Ch01.Data
??
Source B 9.9
??
Tension_Above_10N_Light
L
Tension_Above_10N
L
MORE_STATS[2]
0008
Tension_Above_10N
TON
Timer On Delay
Timer Tension_T1
Preset ?
Accum ?
EN
DN
0009
Overtension_Reset
GRT
Greater Than (A>B)
Source A Local:4:I.Ch01.Data
??
Source B 11.8
??
Tension_Above_10N_Light
U
0010
Tension_T1.DN
Tension_Above_10N
U
END
END

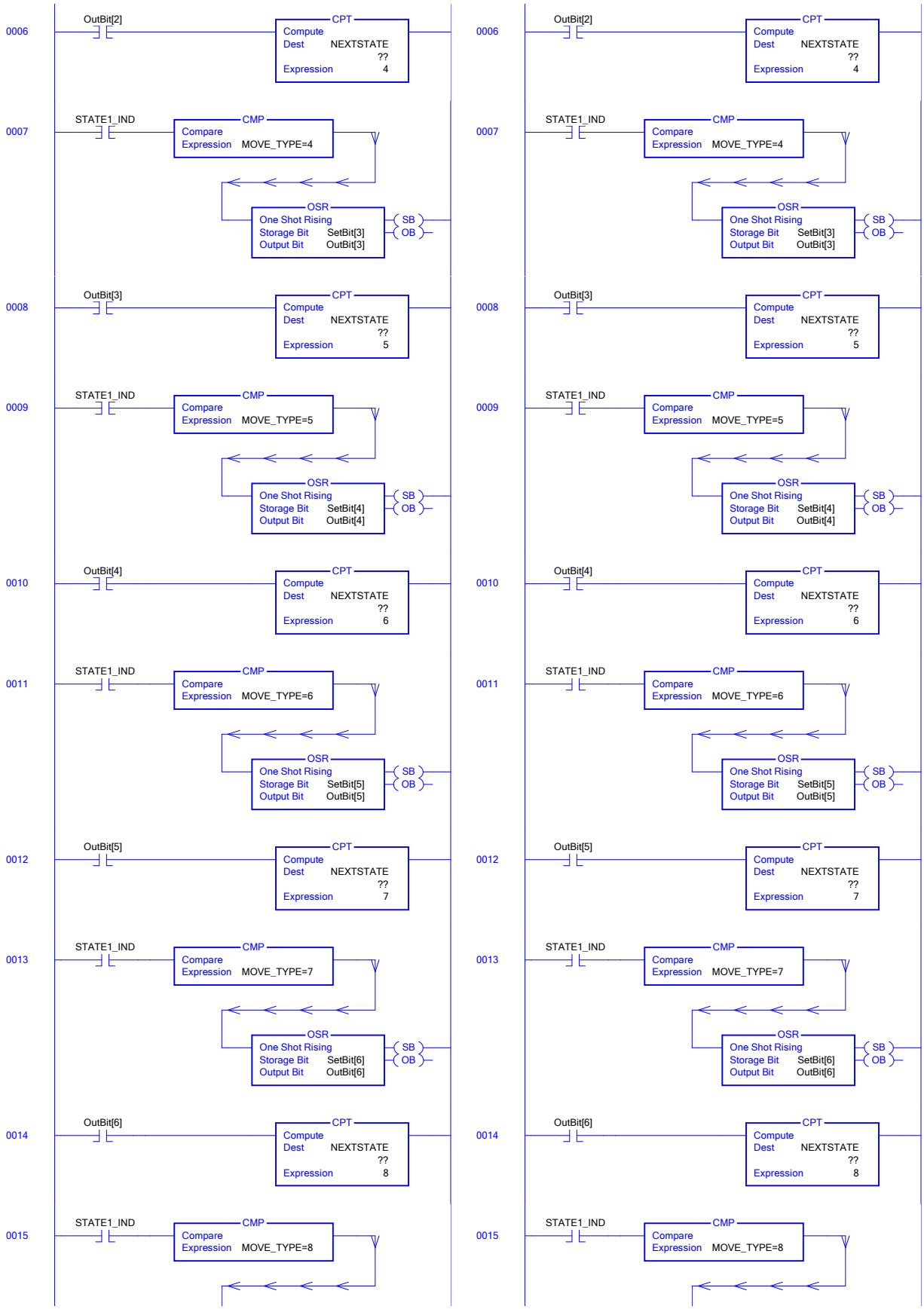
Ready_State_1 Parameters and Local Tags

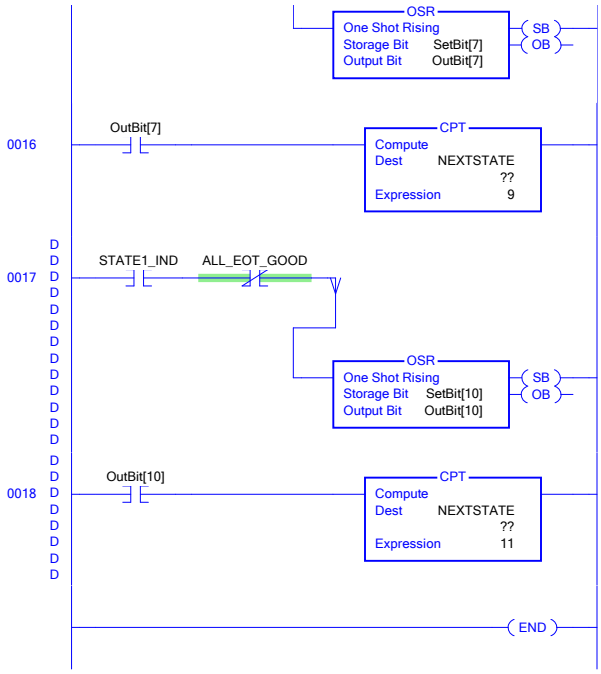
	Left Content				Right Content			
Tag Name	Alias For	Type	Description	Data	AliasFor	Type	Description	Data
MANUAL_SW	---	---	---	---		BOOL		0
MANUAL_SW1	---	---	---	---		BOOL		0
PB0	---	---	---	---		BOOL		0
PB1	---	---	---	---		BOOL		0
RX_MSG	---	---	---	---		BOOL		0
TIMER	---	---	---	---		TIMER		

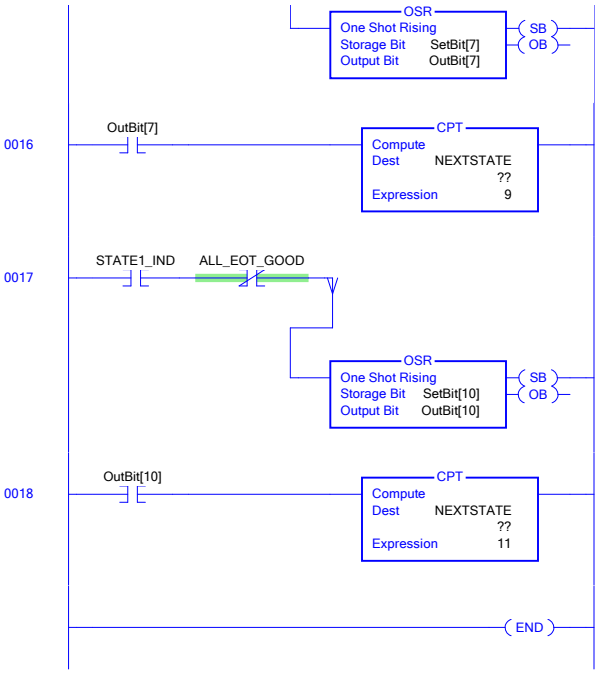
Program: Ready_State_1
Routine: Ready



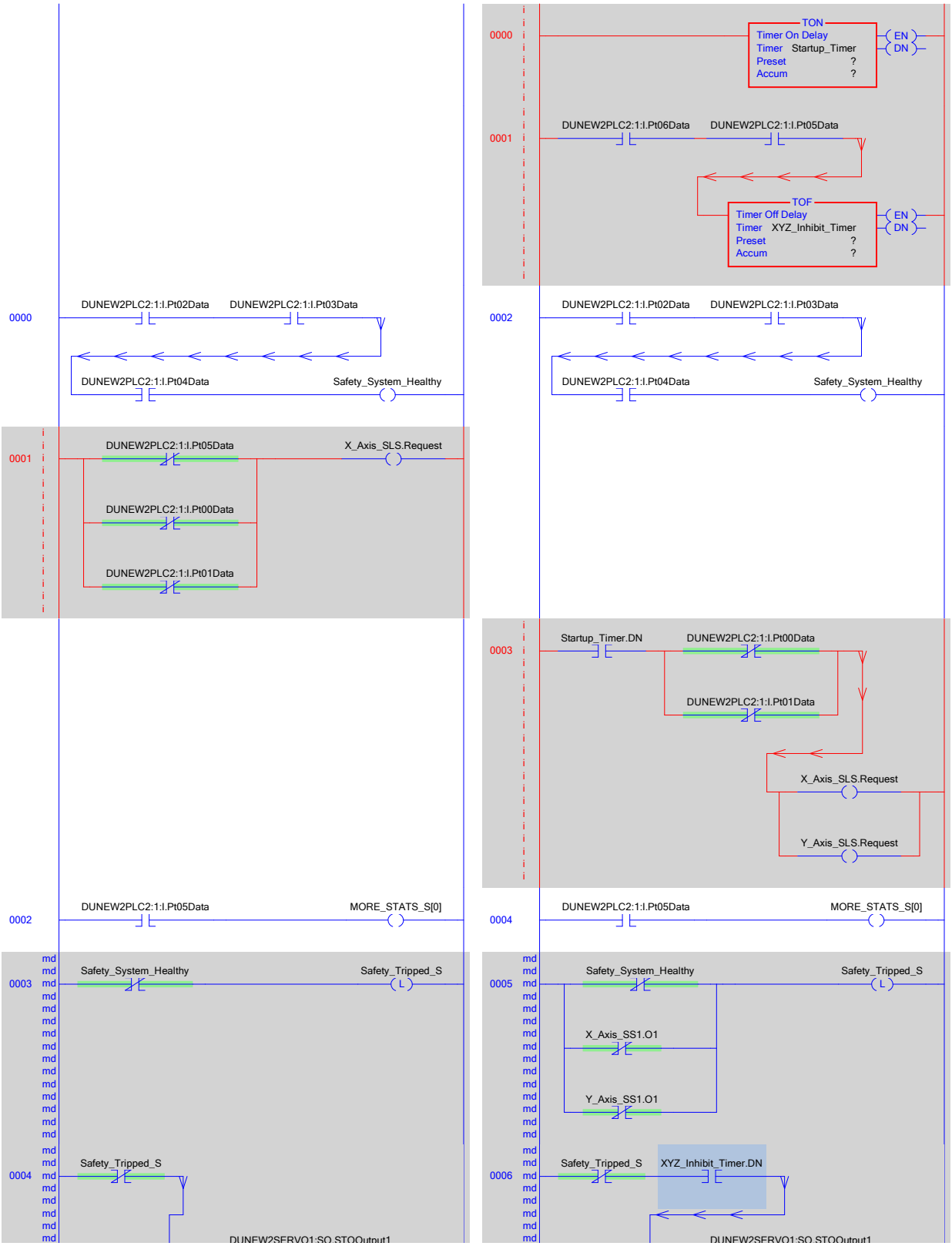


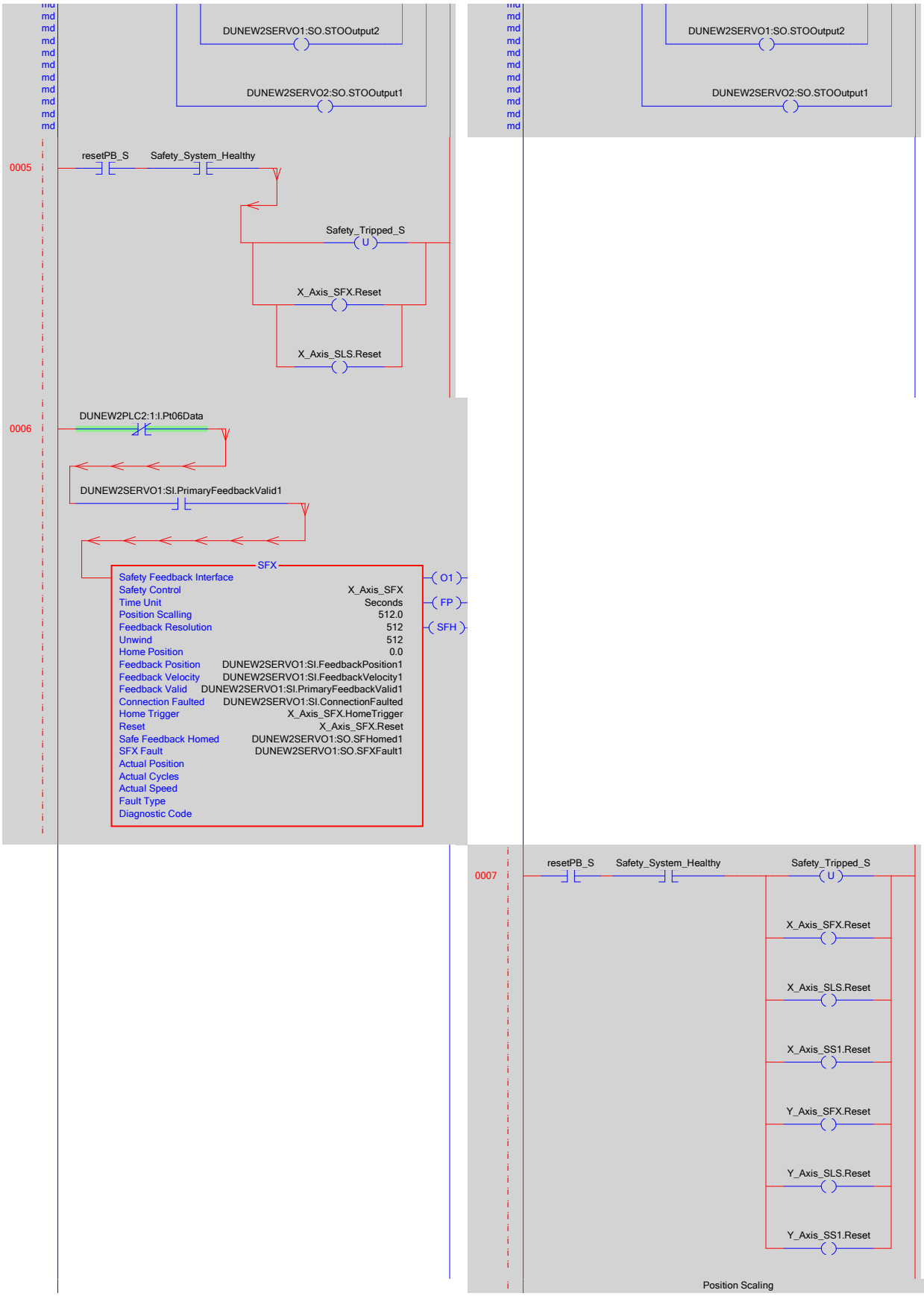

0006
OutBit[2]
CPT
Compute
Dest NEXTSTATE
??
Expression 4
0007
STATE1_IND
CMP
Compare
Expression MOVE_TYPE=4
OSR
One Shot Rising
Storage Bit SetBit[3]
Output Bit OutBit[3]
SB
OB
0008
OutBit[3]
CPT
Compute
Dest NEXTSTATE
??
Expression 5
0009
STATE1_IND
CMP
Compare
Expression MOVE_TYPE=5
OSR
One Shot Rising
Storage Bit SetBit[4]
Output Bit OutBit[4]
SB
OB
0010
OutBit[4]
CPT
Compute
Dest NEXTSTATE
??
Expression 6
0011
STATE1_IND
CMP
Compare
Expression MOVE_TYPE=6
OSR
One Shot Rising
Storage Bit SetBit[5]
Output Bit OutBit[5]
SB
OB
0012
OutBit[5]
CPT
Compute
Dest NEXTSTATE
??
Expression 7
0013
STATE1_IND
CMP
Compare
Expression MOVE_TYPE=7
OSR
One Shot Rising
Storage Bit SetBit[6]
Output Bit OutBit[6]
SB
OB
0014
OutBit[6]
CPT
Compute
Dest NEXTSTATE
??
Expression 8
0015
STATE1_IND
CMP
Compare
Expression MOVE_TYPE=8


OSR
One Shot Rising
Storage Bit SetBit[7]
Output Bit OutBit[7]
SB
OB
0016
OutBit[7]
CPT
Compute
Dest NEXTSTATE
??
Expression 9
0017
STATE1_IND
ALL_EOT_GOOD
OSR
One Shot Rising
Storage Bit SetBit[10]
Output Bit OutBit[10]
SB
OB
0018
OutBit[10]
CPT
Compute
Dest NEXTSTATE
??
Expression 11
END


OSR
One Shot Rising
Storage Bit SetBit[7]
Output Bit OutBit[7]
SB
OB
0016
OutBit[7]
CPT
Compute
Dest NEXTSTATE
??
Expression 9
0017
STATE1_IND
ALL_EOT_GOOD
OSR
One Shot Rising
Storage Bit SetBit[10]
Output Bit OutBit[10]
SB
OB
0018
OutBit[10]
CPT
Compute
Dest NEXTSTATE
??
Expression 11
END

Program: Safety
Routine: Safety




DUNEW2SERVO1:SO.STOOutput2
DUNEW2SERVO2:SO.STOOutput1
0005
resetPB_S
Safety_System_Healthy
Safety_Tripped_S
U
X_Axis_SFX.Reset
X_Axis_SLS.Reset
0006
DUNEW2PLC2:1:I.Pt06Data
DUNEW2SERVO1:SI.PrimaryFeedbackValid1
SFX
Safety Feedback Interface
Safety Control X_Axis_SFX
Time Unit Seconds
Position Scalling 512.0
Feedback Resolution 512
Unwind 512
Home Position 0.0
Feedback Position DUNEW2SERVO1:SI.FeedbackPosition1
Feedback Velocity DUNEW2SERVO1:SI.FeedbackVelocity1
Feedback Valid DUNEW2SERVO1:SI.PrimaryFeedbackValid1
Connection Faulted DUNEW2SERVO1:SI.ConnectionFaulted
Home Trigger X_Axis_SFX.HomeTrigger
Reset X_Axis_SFX.Reset
Safe Feedback Homed DUNEW2SERVO1:SO.SFHomed1
SFX Fault DUNEW2SERVO1:SO.SFXFault1
Actual Position
Actual Cycles
Actual Speed
Fault Type
Diagnostic Code
O1
FP
SFH
0007
resetPB_S
Safety_System_Healthy
Safety_Tripped_S
U
X_Axis_SFX.Reset
X_Axis_SLS.Reset
X_Axis_SS1.Reset
Y_Axis_SFX.Reset
Y_Axis_SLS.Reset
Y_Axis_SS1.Reset
Position Scaling

md
md
0007 md DUNEW2PLC2:1:I.Pt06Data
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md
md

SLS
Safely-limited Speed
Safety Control X_Axis_SLS
Restart Type AUTOMATIC
Cold Start Type AUTOMATIC
Check Delay 3000
Active Limit X_Axis_SLS.ActiveLimit
Feedback SFX X_Axis_SFX
Request X_Axis_SLS.Request
Reset X_Axis_SLS.Reset
SLS Active DUNEW2SERVO1:SO.SLSActive1
SLS Limit DUNEW2SERVO1:SO.SLSLimit1
SLS Fault DUNEW2SERVO1:SO.SLSFault1
Fault Type
Diagnostic Code
O1
RR
FP

0008 DUNEW2SERVO1:SO.SLSActive1 MORE_STATS_S[1]

Encoder Pulses Per Revolution = 512
Gearbox Ratio = 35:1
Actuator Diameter = 95.5mm

= (PPR * Ratio) / (Pi * Dia)
= 59.72893

0008 DUNEW2SERVO1:SI.PrimaryFeedbackValid1

SFX
Safety Feedback Interface
Safety Control X_Axis_SFX
Time Unit Seconds
Position Scalling 59.723
Feedback Resolution 512
Unwind 0
Home Position 0.0
Feedback Position DUNEW2SERVO1:SI.FeedbackPosition1
Feedback Velocity DUNEW2SERVO1:SI.FeedbackVelocity1
Feedback Valid DUNEW2SERVO1:SI.PrimaryFeedbackValid1
Connection Faulted DUNEW2SERVO1:SI.ConnectionFaulted
Home Trigger X_Axis_SFX.HomeTrigger
Reset X_Axis_SFX.Reset
Safe Feedback Homed DUNEW2SERVO1:SO.SFHomed1
SFX Fault DUNEW2SERVO1:SO.SFXFault1
Actual Position
Actual Cycles
Actual Speed
Fault Type
Diagnostic Code
O1
FP
SFH

0009

MOV
Move
Source 28
??
Dest X_Axis_SLS.ActiveLimit
??

md
md
0010 md
md
md
md
md
md
md
md
md
md
md
md
md
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md

SLS
Safely-limited Speed
Safety Control X_Axis_SLS
Restart Type AUTOMATIC
Cold Start Type AUTOMATIC
Check Delay 3000
Active Limit X_Axis_SLS.ActiveLimit
Feedback SFX X_Axis_SFX
Request X_Axis_SLS.Request
Reset X_Axis_SLS.Reset
SLS Active DUNEW2SERVO1:SO.SLSActive1
SLS Limit DUNEW2SERVO1:SO.SLSLimit1
SLS Fault DUNEW2SERVO1:SO.SLSFault1
Fault Type
Diagnostic Code
O1
RR
FP

0011 DUNEW2SERVO1:SO.SLSActive1 MORE_STATS_S[1]

Safe Stop Modification 13.06.23

0012 DUNEW2SERVO1:SO.SLSLimit1 X_Axis_SS1.Request

0013

SS1
Safe Stop 1
Safety Control X_Axis_SS1
Restart Type AUTOMATIC
Cold Start Type AUTOMATIC
Stop Monitor Delay 250
Stop Delay 1000
Standstill Speed 1.0
Decel Ref Speed 50.0
Decel Speed Tolerance 5.0
Feedback SFX X_Axis_SFX
Request X_Axis_SS1.Request
Reset X_Axis_SS1.Reset
SS1 Active DUNEW2SERVO1:SO.SS1Active1
SS1 Fault DUNEW2SERVO1:SO.SS1Fault1
Fault Type
Diagnostic Code
O1
RR
FP

Position Scaling

Encoder Pulses Per Revolution = 512
Gearbox Ratio = 35:1
Actuator Diameter = 79.6mm

= (PPR * Ratio) / (Pi * Dia)
= 71.65971

0014 DUNEW2SERVO1:SI.PrimaryFeedbackValid2

SFX

Safety Feedback Interface	
Safety Control	Y_Axis_SFX
Time Unit	Seconds
Position Scalling	71.6597
Feedback Resolution	512
Unwind	0
Home Position	0.0
Feedback Position	DUNEW2SERVO1:SI.FeedbackPosition2
Feedback Velocity	DUNEW2SERVO1:SI.FeedbackVelocity2
Feedback Valid	DUNEW2SERVO1:SI.PrimaryFeedbackValid2
Connection Faulted	DUNEW2SERVO1:SI.ConnectionFaulted
Home Trigger	Y_Axis_SFX.HomeTrigger
Reset	Y_Axis_SFX.Reset
Safe Feedback Homed	DUNEW2SERVO1:SO.SFHomed2
SFX Fault	DUNEW2SERVO1:SO.SFXFault2
Actual Position	
Actual Cycles	
Actual Speed	
Fault Type	
Diagnostic Code	

O1 FP SFH

0015

MOV

Move	
Source	28
	??
Dest	Y_Axis_SLS.ActiveLimit
	??

0016

SLS

Safely-limited Speed	
Safety Control	Y_Axis_SLS
Restart Type	AUTOMATIC
Cold Start Type	AUTOMATIC
Check Delay	3000
Active Limit	Y_Axis_SLS.ActiveLimit
Feedback SFX	Y_Axis_SFX
Request	Y_Axis_SLS.Request
Reset	Y_Axis_SLS.Reset
SLS Active	DUNEW2SERVO1:SO.SLSActive2
SLS Limit	DUNEW2SERVO1:SO.SLSLimit2
SLS Fault	DUNEW2SERVO1:SO.SLSFault2
Fault Type	
Diagnostic Code	

O1 RR FP

0017 DUNEW2SERVO1:SO.SLSActive2 — MORE_STATS_S[2]

Safe Stop Modification 13.06.23

0018 DUNEW2SERVO1:SO.SLSLimit2 — Y_Axis_SS1.Request

0019

SS1

Safe Stop 1	
Safety Control	Y_Axis_SS1
Restart Type	AUTOMATIC
Cold Start Type	AUTOMATIC
Stop Monitor Delay	250
Stop Delay	1000
Standstill Speed	1.0
Decel Ref Speed	50.0
Decel Speed Tolerance	5.0
Feedback SFX	Y_Axis_SFX
Request	Y_Axis_SS1.Request
Reset	Y_Axis_SS1.Reset
SS1 Active	DUNEW2SERVO1:SO.SS1Active2
SS1 Fault	DUNEW2SERVO1:SO.SS1Fault2
Fault Type	
Diagnostic Code	

O1 RR FP

END END

Alarm Condition

Alarm Condition	Left Content	Right Content
Tension_Above_10N:Tension_Above_10N	---	Right Content Only